

**Design and Implementation of an Interoperable Risk based Food
Safety Inspection Information Management System (IIMS) for the
FCAU of Ministry of Health**

SOFTWARE DESIGN SPECIFICATION

Prepared By



PROCONS INFOTECH (PVT) LTD

171 2/1, NAWALA ROAD, NUGEGODA, SRI LANKA.

T: +94 11 2199555 F: _94 11 4209555

E: info@procons.lk W: www.proconsinfotech.com

Document Version: V1.0

Last Edited: 05th May 2026

Key Summary

Project Name	Design and Implementation of an Interoperable Risk based Food Safety Inspection Information Management System (IIMS) for the FCAU of Ministry of Health
Tender No	2026/FASRL/FASRL/135600
Contract No	304/2026/UNJP/SRL/076/EC
Sponsor	UNFAO
Approver	Food Control and Administration Unit (FCAU) of Ministry of Health (MOH)
Author	Procons Infotech (Pvt) Ltd

Table of Contents

Introduction	8
Document Purpose	8
1. System Workflows	10
1.1 Premises Registration and Lifecycle Management Module.....	11
1.2 Inspection and Grading Module	11
1.3 Laboratory Coordination Module	12
1.4 Legal, Prosecution and Enforcement Management Module	13
1.4.1 Summarized	13
1.4.2 Seized	14
1.4.3 Prosecution	15
1.4.4 Arrested.....	16
1.5 Import Control Module	17
1.6 Export Control Module	18
1.7 Registration of Common Salt Manufacturing and Transports Management Module	19
1.7.1 Salt Factory Registration	19
1.7.2 Salt Factory Annual Renewal.....	20
1.7.3 Salt Lagoon Registration.....	21
1.7.4 Salt Lagoon Annual Renewal.....	22
1.7.5 Salt Factory/Lagoon Registration Amendment.....	23
1.8 Bottled Water Manufacturing and Registration Management Module	24
1.8.1 Bottled Water Registration.....	24
1.8.2 Bottled Water Registration Renewal.....	25
1.8.3 Bottled Water Registration Amendment	26
1.8.4 Bottled Water Multiple Brands	27
2. Use Case Modeling.....	28
2.1 Premises Registration and Lifecycle Management Module.....	29
Use Case Narratives	30
2.2 Inspection and Grading Module	36

Use Case Narratives	37
2.3 Laboratory Coordination Module	43
Use Case Narratives	44
2.4 Legal, Prosecution and Enforcement Management Module	49
Use Case Narratives	50
2.5 Import Control Module	54
Use Case Narratives	55
2.6 Export Control Module	61
Use Case Narratives	62
2.7 Registration of Common Salt Manufacturing and Transports Management Module	67
Use Case Narratives	68
2.8 Bottled Water Manufacturing and Registration Management Module	72
Use Case Narratives	73
3. Sequence Modeling	78
3.1 Premises Registration and Lifecycle Management Module.....	79
3.2 Inspection and Grading Module	80
3.3 Laboratory Coordination Module	81
3.4 Legal, Prosecution and Enforcement Management Module	82
3.5 Import Control Module	83
3.6 Export Control Module	84
3.7 Registration of Common Salt Manufacturing and Transports Management Module	85
3.8 Bottled Water Manufacturing and Registration Management Module	86
4. Activity Modeling	87
4.1 Premises Registration and Lifecycle Management Module.....	88
4.2 Inspection and Grading Module	89
4.3 Laboratory Coordination Module	90
4.4 Legal, Prosecution and Enforcement Management Module	91
4.5 Import Control Module	92
4.6 Export Control Module	93

4.7 Registration of Common Salt Manufacturing and Transports Management Module	94
4.8 Bottled Water Manufacturing and Registration Management Module	95
5. System Architecture	97
5.1 Comprehensive Explanation of the Proposed Architecture	98
5.1.1 Client and Access Layer	98
5.1.2 Load Balancer Layer	98
5.1.3 Web Server Layer	99
5.1.4 API Gateway Layer.....	99
5.1.5 Microservices Layer (Core System)	100
5.1.6 Database Layer	101
5.1.7 File Storage Layer	101
5.1.8 Messaging & Event Processing Layer	102
5.1.9 External Services Layer	103
6. CI/CD Pipeline Design.....	104
6.1 High-Level CI/CD Flow	104
6.2 Pipeline Architecture Overview	105
6.3 Pipeline Stages	105
6.3.1 Source Code Management (SCM)	105
6.3.2 Build Stage.....	105
6.3.3 Code Quality & Static Analysis	106
6.3.4 Testing Stage	106
6.3.5 Security Scanning	106
6.3.6 Docker Build Stage and Push to Container Registry	106
6.3.7 Deployment Stage.....	107
6.3.8 Post-Deployment Verification	107
6.3.9 Rollback Mechanism	107
7. OpenAPI 3.0 Design.....	108
7.1 AUTH SERVICE	108
7.2 USER PROFILE SERVICE.....	110

7.3 PREMISES SERVICE	111
7.4 WORKFLOW SERVICE	112
7.5 INSPECTION SERVICE.....	113
7.6 RISK ENGINE SERVICE.....	115
7.7 SCHEDULING SERVICE	115
7.8 LAB SERVICE	116
7.9 APPEALS SERVICE	117
7.10 DOCUMENT SERVICE.....	118
7.11 NOTIFICATION SERVICE	119
7.12 CONFIG SERVICE.....	120
7.13 REPORTING SERVICE	120
7.14 IMPORT CONTROL SERVICE	121
7.15 EXPORT CONTROL SERVICE	123
7.16 SALT REG SERVICE	126
7.18 BOTTLED WATER REG SERVICE.....	128
8. Tech Stack.....	131
8.1 Development.....	131
8.2 Deployment.....	131
9. UI Prototype Samples.....	132
9.1 New System User Creation.....	132
9.2 FBO Registration.....	133
9.3 Premises List.....	136
9.4 Grading Form Configuration	137
9.5 Inspection Calendar	138
9.6 Inspection.....	139
9.7 FCAU User Dashboard	140
9.8 Import Control	141
9.9 Sample Management.....	142
10. Error Handling and Logging Strategy	143

10.1 Error Handling Strategy	143
10.1.1 Standardized Error Handling Approach	143
10.1.2 Error Categorization	144
10.1.3 Graceful Degradation	144
10.1.4 Retry & Recovery Mechanisms	145
10.1.5 Exception Handling in Microservices	145
10.1.6 User-Friendly Error Messages	145
10.2 Logging Strategy	146
10.2.1 Centralized Logging	146
10.2.2 Log Levels	146
10.2.3 Structured Logging Format	146
10.2.4 Correlation & Traceability	146
10.2.5 Audit Logging	147
10.2.6 Logging Scope	147
11. Design Constraints and Assumptions.....	148
11.1 Constraints	148
11.1.1 Technical Constraints	148
11.1.2 Architecture Constraints	148
11.1.3 Integration Constraints	148
11.1.4 Data Constraints.....	149
11.2 Design Assumptions.....	149
11.2.1 Technical Assumptions.....	149
11.2.2 Integration Assumptions.....	149
11.2.3 Data Assumptions	149
11.2.4 User Assumptions	150
11.2.5 Operational Assumptions.....	150

Introduction

This Software Design Specification (SDS) defines the technical design and architectural blueprint for the **Interoperable Risk-Based Food Safety Inspection Information Management System (IIMS)** developed for the Food Control Administration Unit (FCAU) of the Ministry of Health.

Building upon the requirements outlined in the Software Requirements Specification (SRS), this document translates business, functional, and non-functional requirements into a structured system design. The SDS provides a detailed representation of how the system will be constructed, including system architecture, module decomposition, data models, and interaction flows.

The IIMS is envisioned as a centralized, web-based, and interoperable platform that supports the complete lifecycle of food safety regulation. This includes premises registration, inspection and grading (H-800), laboratory coordination, enforcement actions, import/export control, complaint management, notifications, and reporting.

This document serves as a technical foundation for implementing a scalable, secure, and workflow-driven system that integrates multiple stakeholders such as Food Business Operators (FBOs), Public Health Inspectors (PHIs), Medical Officers of Health (MOHs), laboratories, and regulatory authorities. It ensures that the system design aligns with regulatory frameworks, operational workflows, and interoperability requirements defined in the SRS.

Document Purpose

The purpose of this Software Design Specification is to provide a comprehensive and detailed technical guide for the development and implementation of the IIMS. It acts as a bridge between the system requirements and the actual system construction, ensuring that all specified functionalities are accurately realized through a well-defined design.

Specifically, this document aims to:

- Define the overall system architecture, including high-level design patterns, component interactions, and deployment considerations
- Describe the internal design of all major modules identified in the SRS, such as:
 - Premises Registration and Lifecycle Management Module
 - Inspection and Grading Module
 - Laboratory Coordination Module
 - Legal, Prosecution and Enforcement Management Module

- Import Control Module
 - Export Control Module
 - Registration of Common Salt Manufacturing and Transports Management Module
 - Bottled Water Manufacturing and Registration Management
-
- Module Specify database design, entity relationships, and data flow mechanisms based on the defined data architecture
 - Detail APIs, and external integrations
 - Provide design specifications for security, performance, scalability, and interoperability in alignment with non-functional requirements
 - UI/UX samples
 - Support developers, architects, and QA teams by offering clear guidance for implementation, testing, and validation
 - Ensure traceability from requirements (SRS) to design and eventual system implementation

This document also serves as a reference for future system maintenance, enhancements, and scalability planning, ensuring that the system remains adaptable to evolving regulatory and technological needs.

1. System Workflows

System workflows describe the end-to-end operational processes that govern how different components of the system interact to deliver business functionality. Within the context of this Software Design Document (SDD), workflows provide a structured representation of how user actions, system services, and data flows are coordinated across the microservices based architecture.

These workflows serve as a bridge between the functional requirements defined in the Software Requirements Specification (SRS) and the technical implementation outlined in the SDD. They illustrate how discrete modules collaborate to execute business processes in a cohesive and efficient manner.

Each workflow captures:

- The sequence of interactions between users and system components
- The orchestration
- Data exchange patterns across services and databases
- Trigger points for events such as notifications, validations, and escalations

Given the adoption of a microservices architecture, workflows are particularly critical in defining service boundaries, asynchronous communication mechanisms (e.g., event-driven messaging), and inter-service dependencies. They ensure that the system maintains consistency, scalability, and fault tolerance while handling complex real-world scenarios.

The workflows documented in this section will be represented using standardized process flow diagrams, enabling clear visualization of system behavior for both technical and non-technical stakeholders.

Overall, system workflows provide a comprehensive view of how the system operates dynamically, ensuring alignment between business objectives and technical design while supporting maintainability and future scalability.

1.1 Premises Registration and Lifecycle Management Module

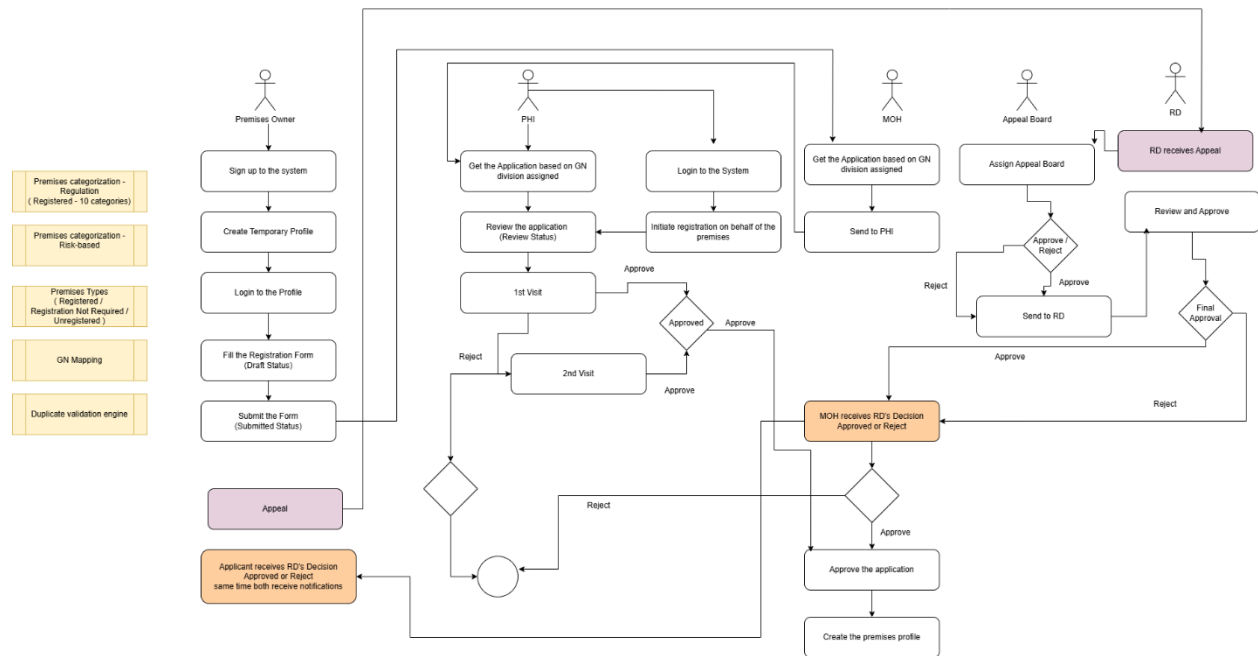


Figure 1.0

1.2 Inspection and Grading Module

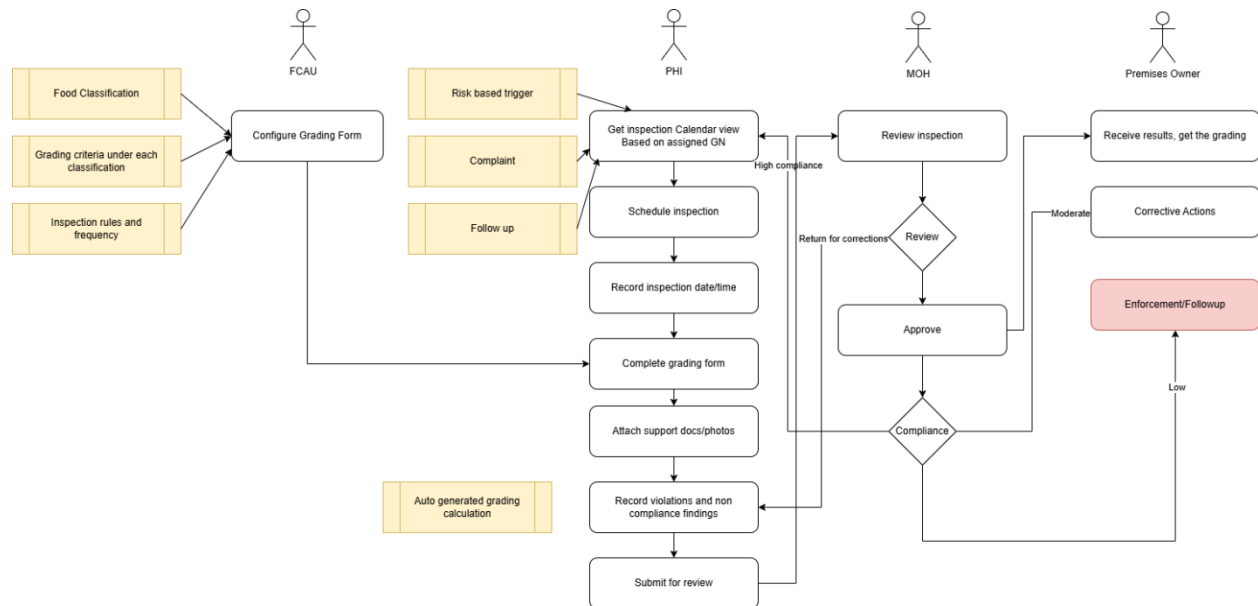


Figure 1.1

1.3 Laboratory Coordination Module

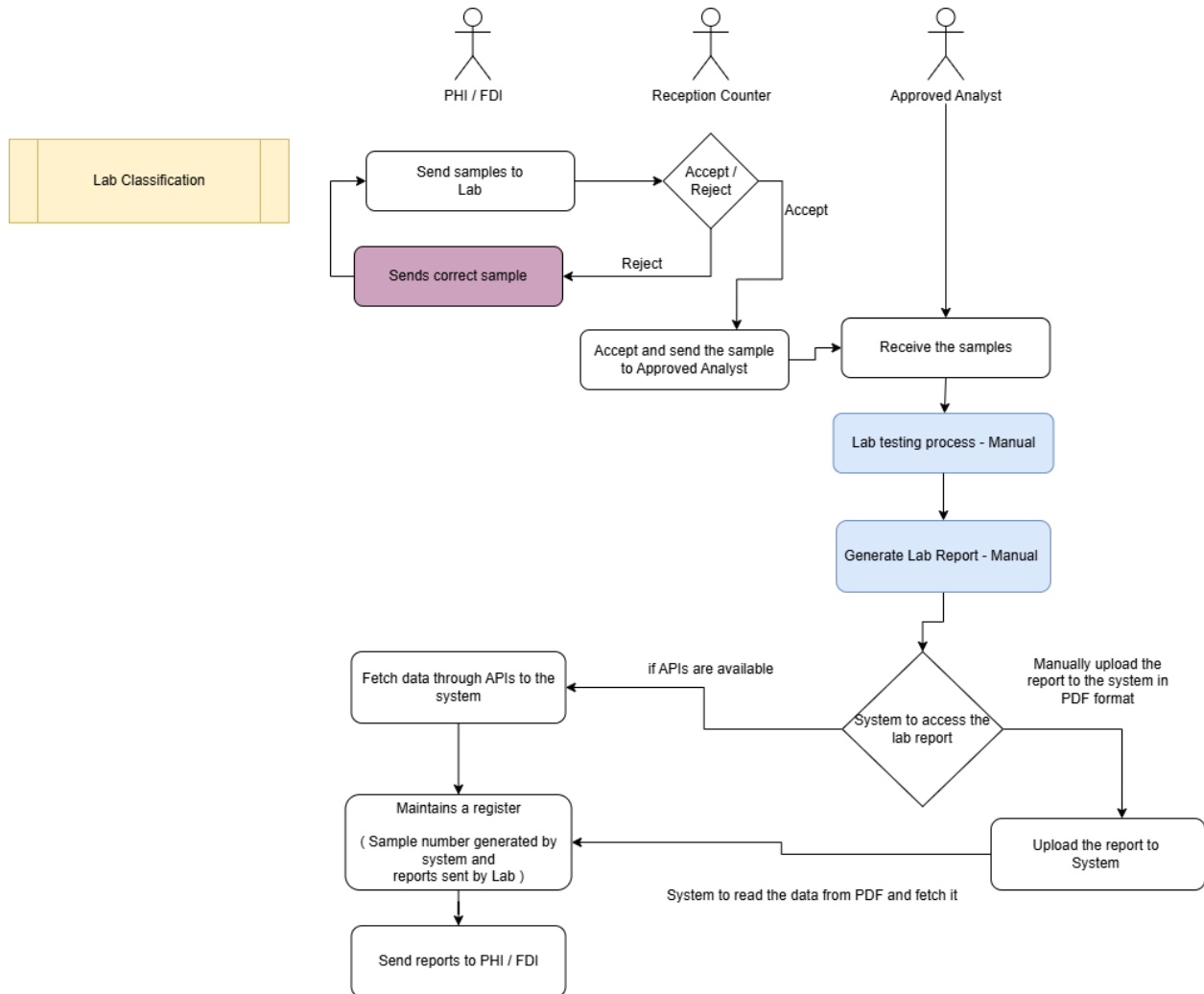


Figure 1.2

1.4 Legal, Prosecution and Enforcement Management Module

1.4.1 Summarized

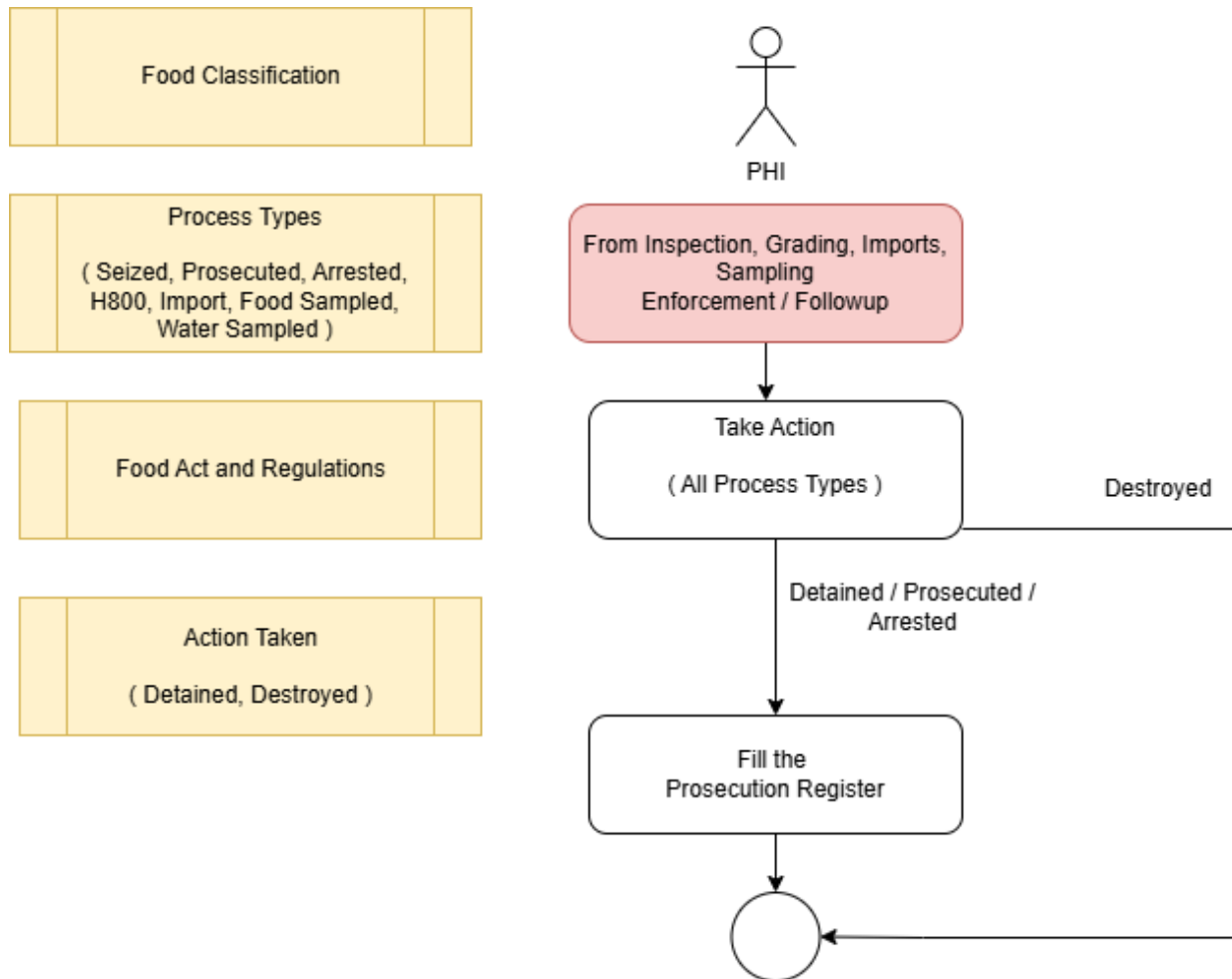


Figure 1.3

1.4.2 Seized

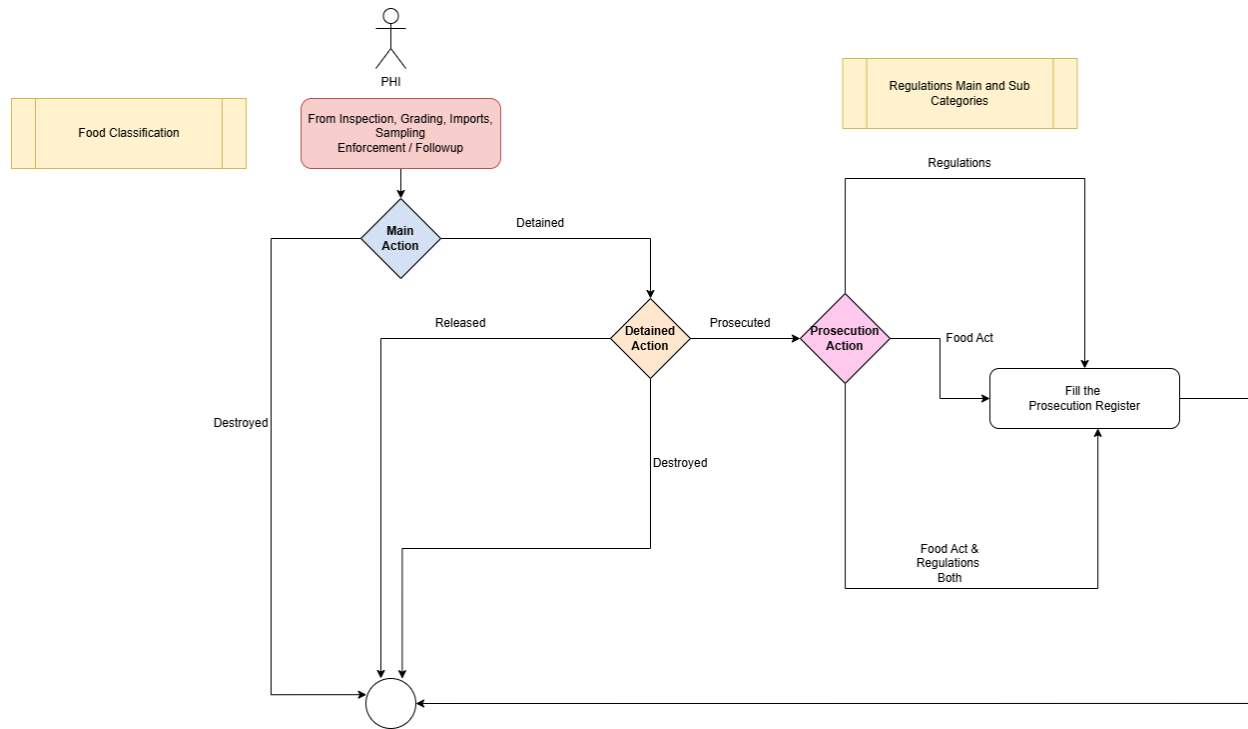


Figure 1.4

1.4.3 Prosecution

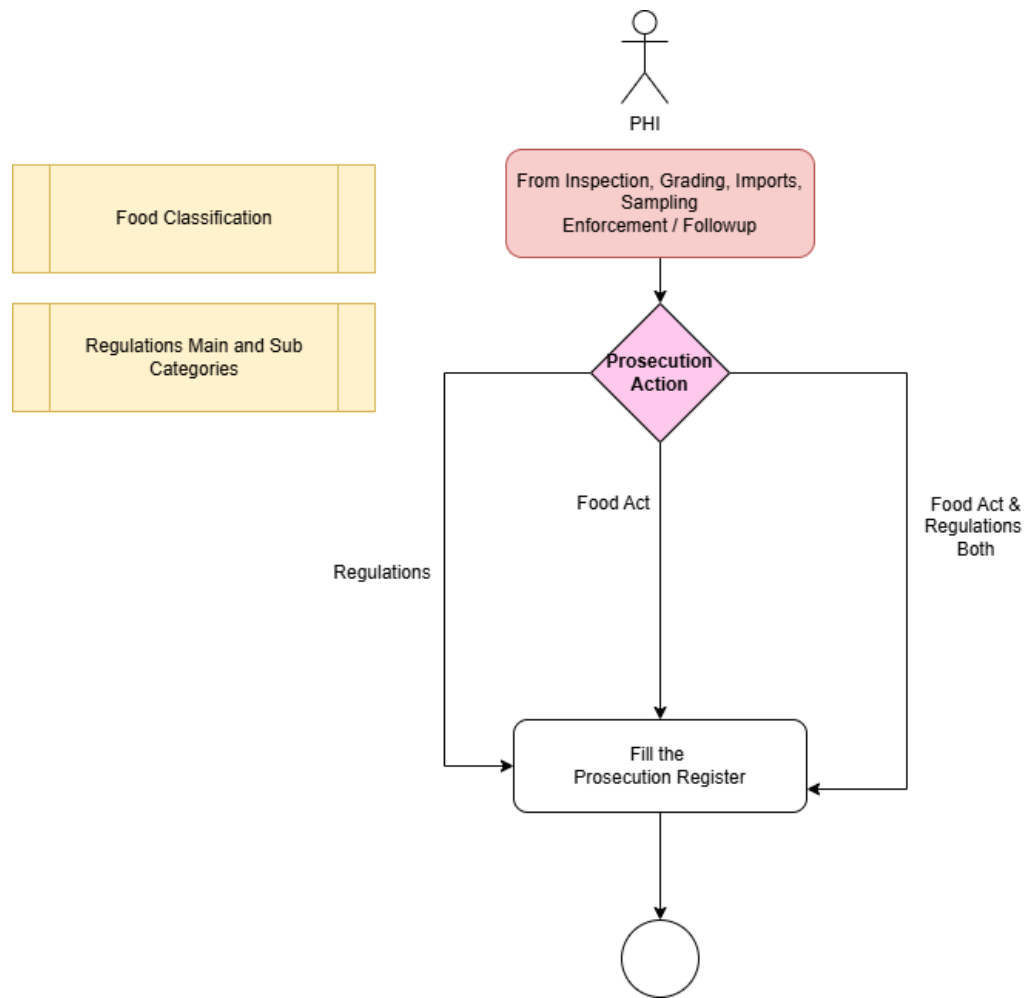


Figure 1.5

1.4.4 Arrested

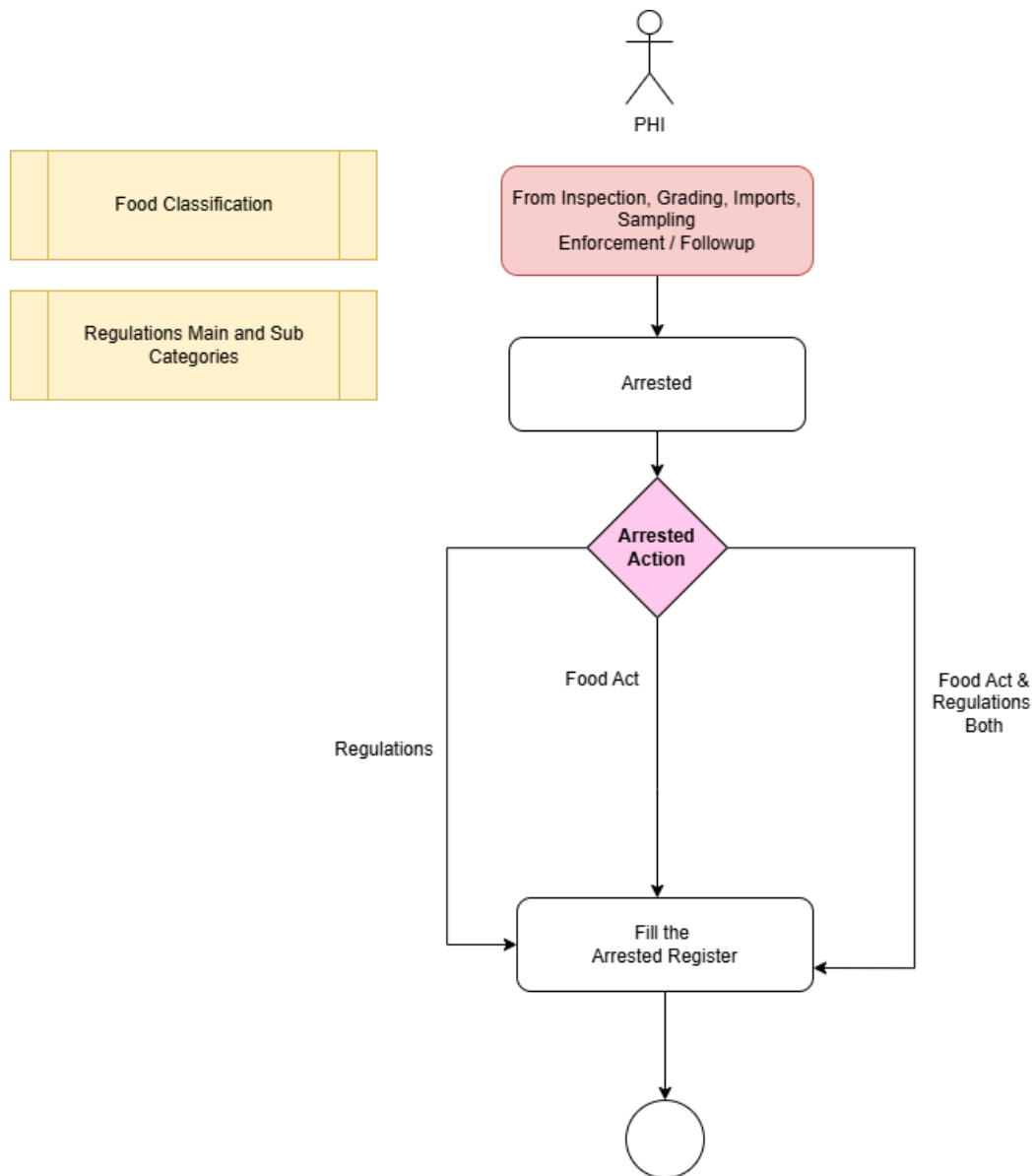


Figure 1.6

1.6 Export Control Module

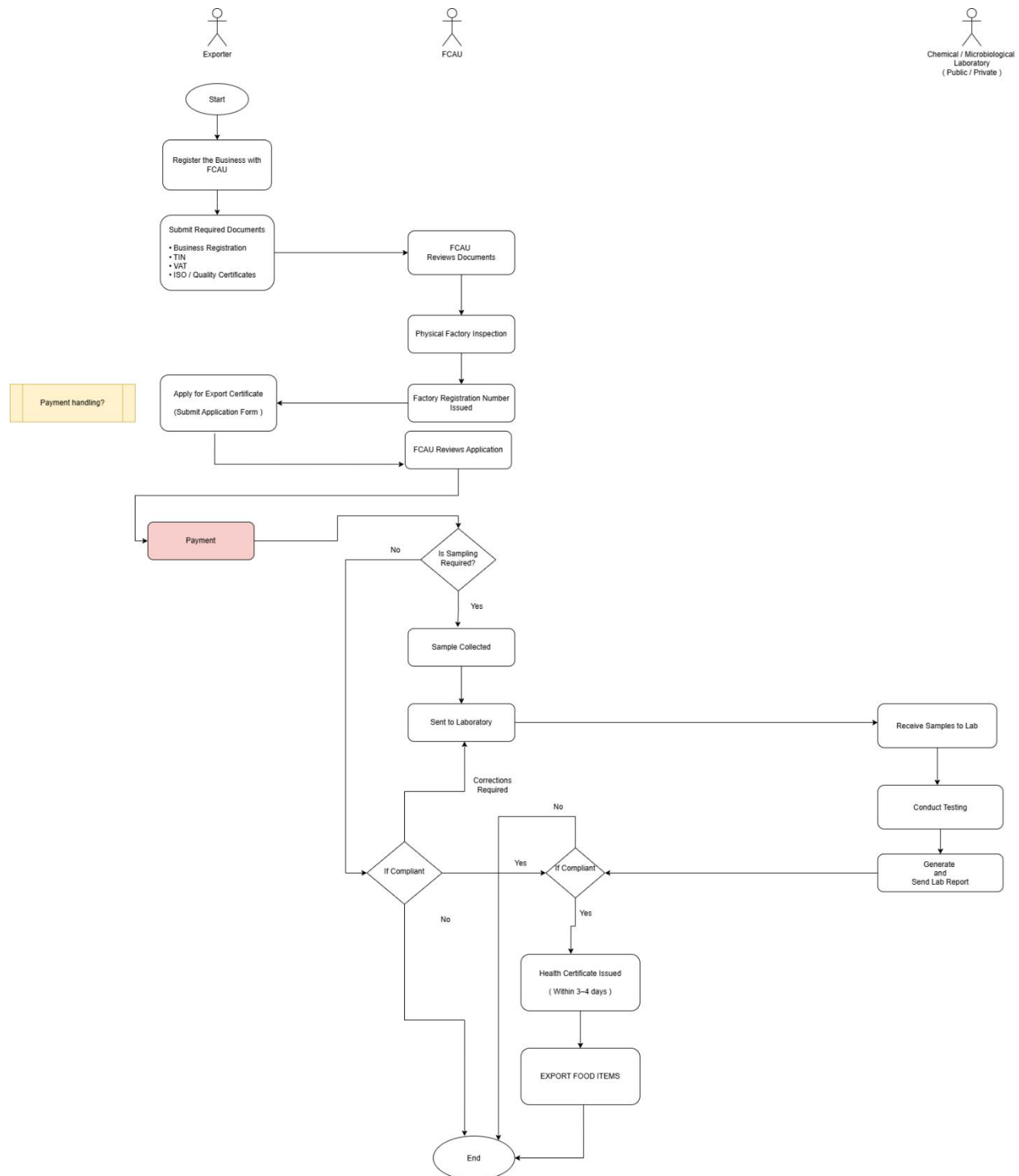


Figure 1.8

1.7 Registration of Common Salt Manufacturing and Transports Management Module

1.7.1 Salt Factory Registration

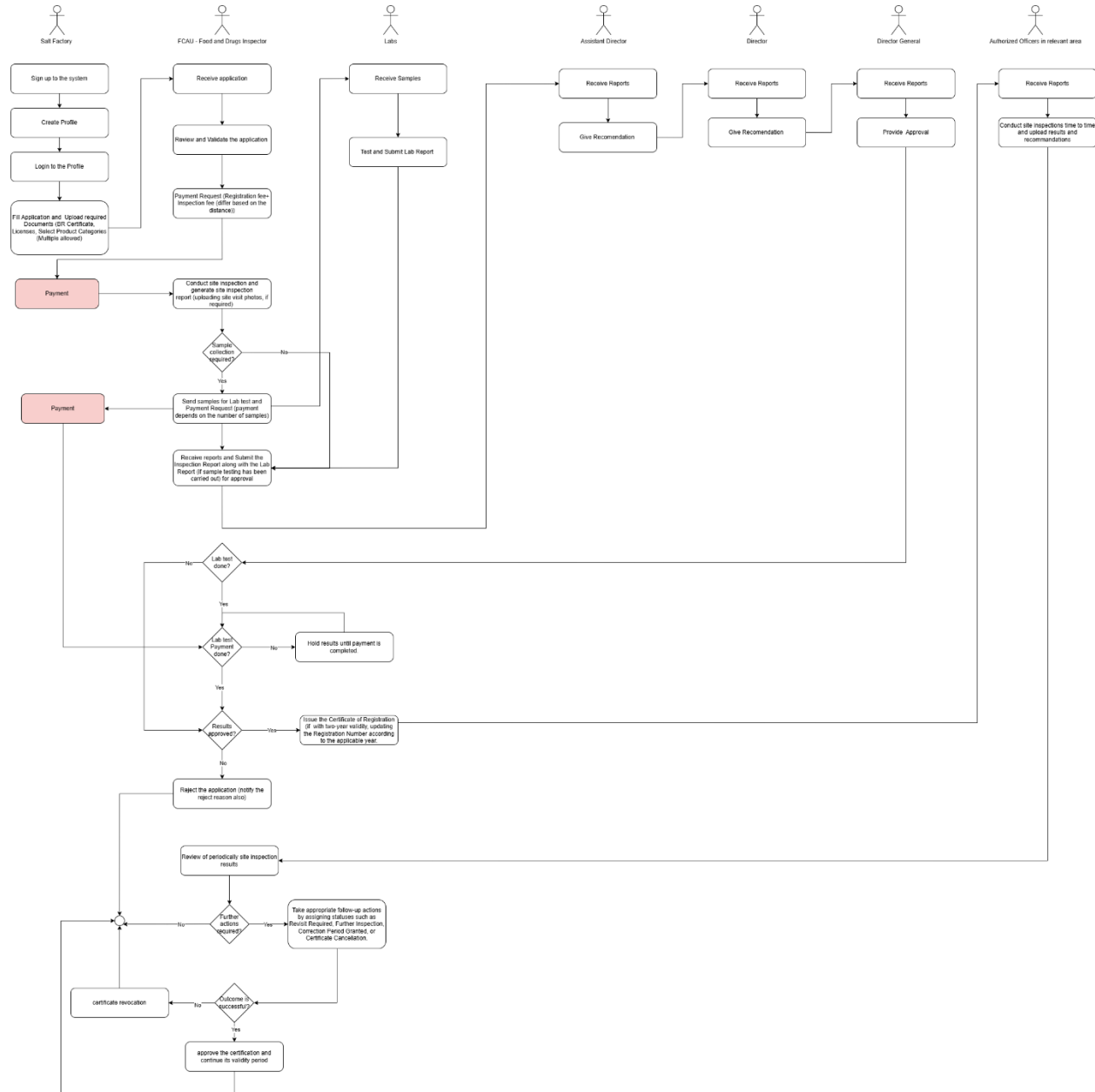


Figure 1.9

1.7.2 Salt Factory Annual Renewal

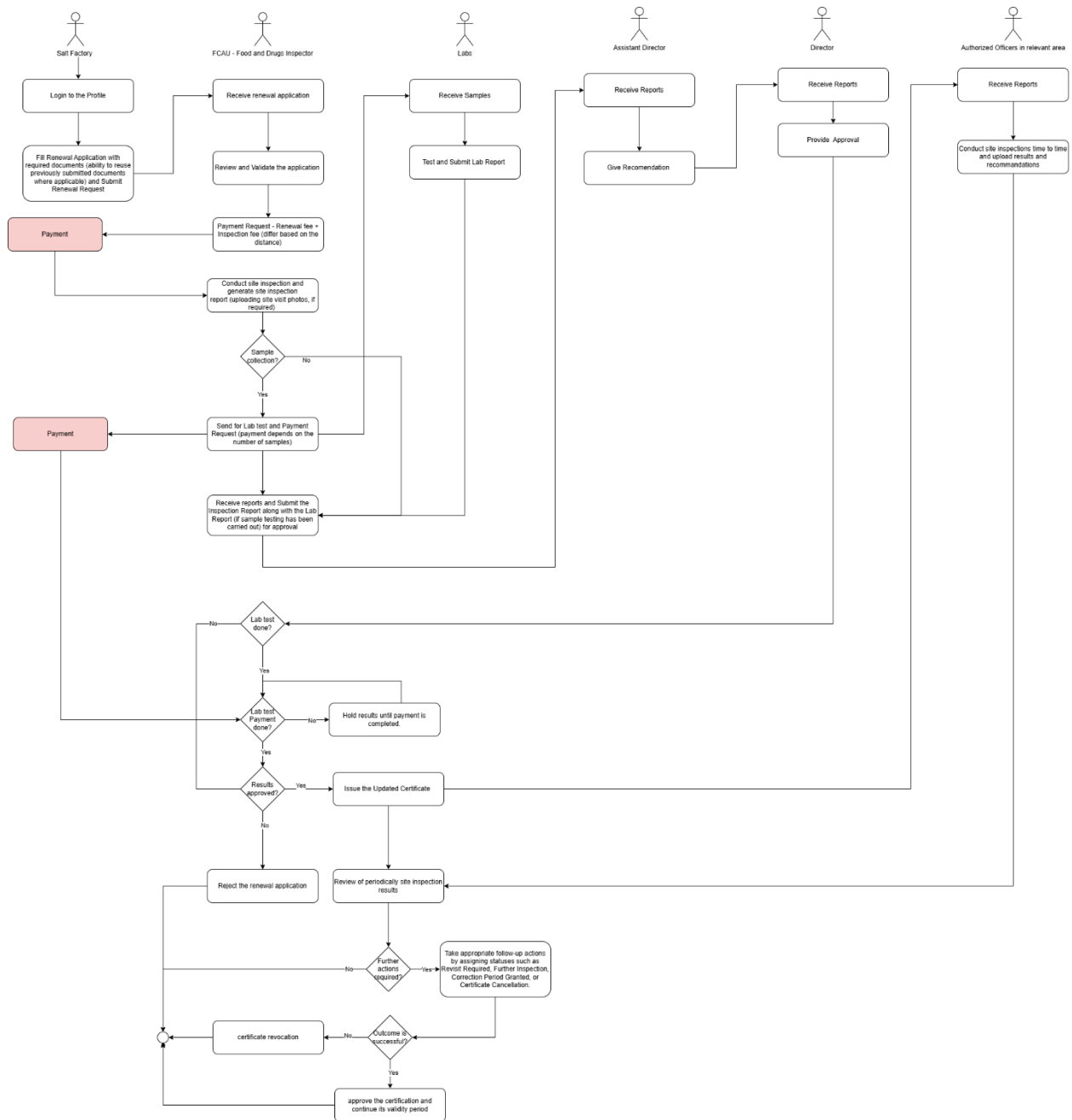


Figure 1.10

1.7.3 Salt Lagoon Registration

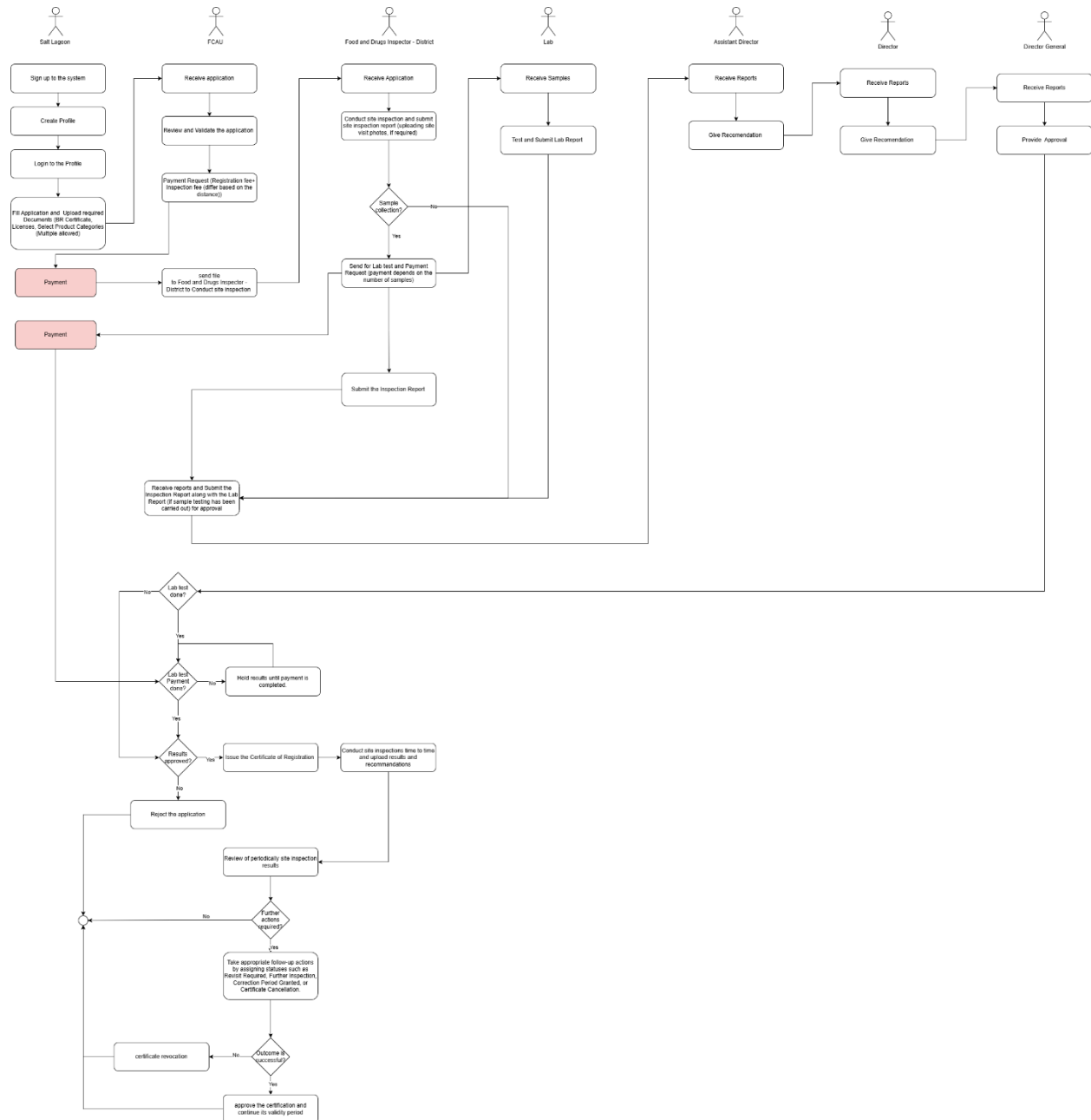


Figure 1.11

1.7.4 Salt Lagoon Annual Renewal

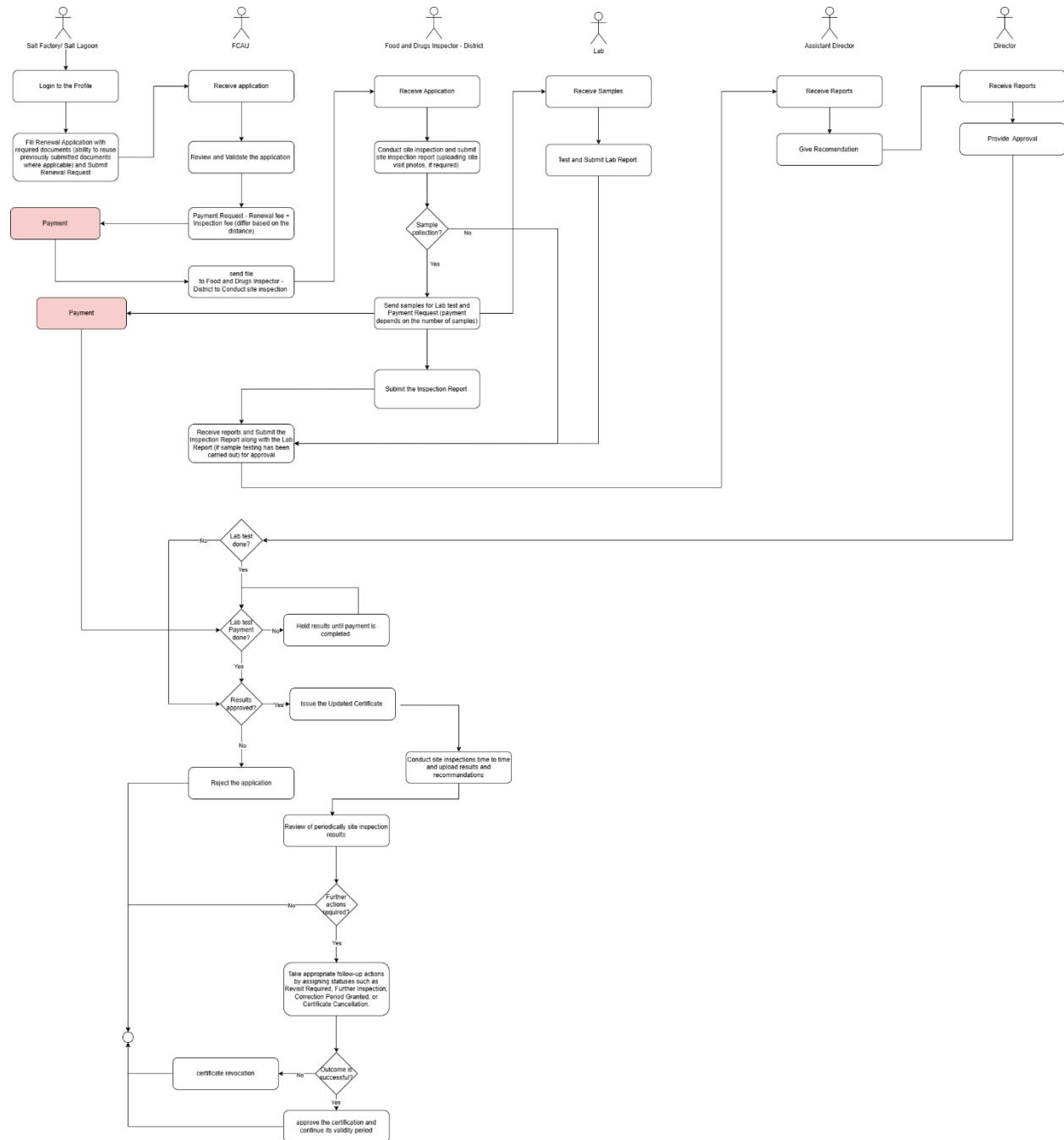


Figure 1.12

1.7.5 Salt Factory/Lagoon Registration Amendment

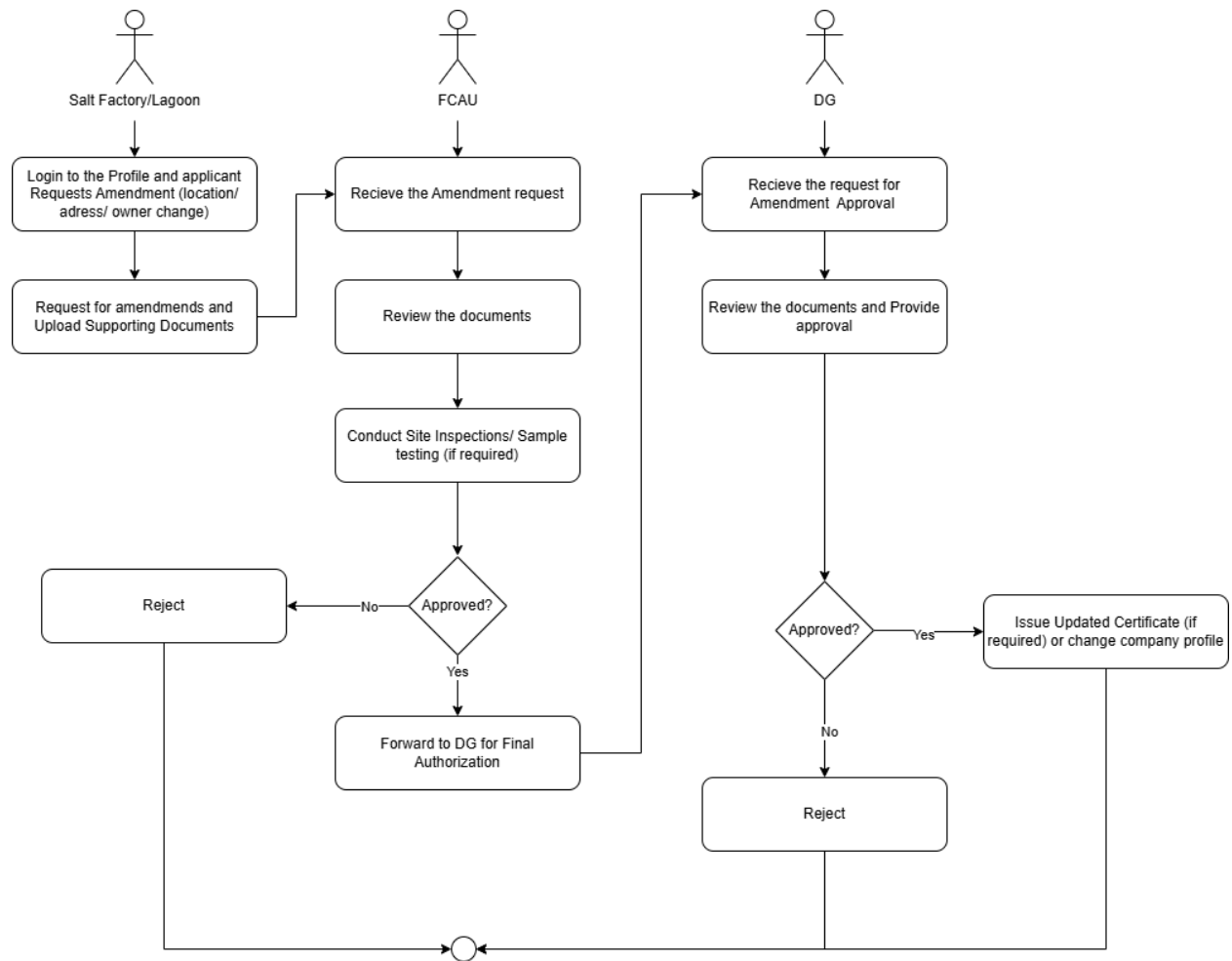


Figure 1.13

1.8 Bottled Water Manufacturing and Registration Management Module

1.8.1 Bottled Water Registration

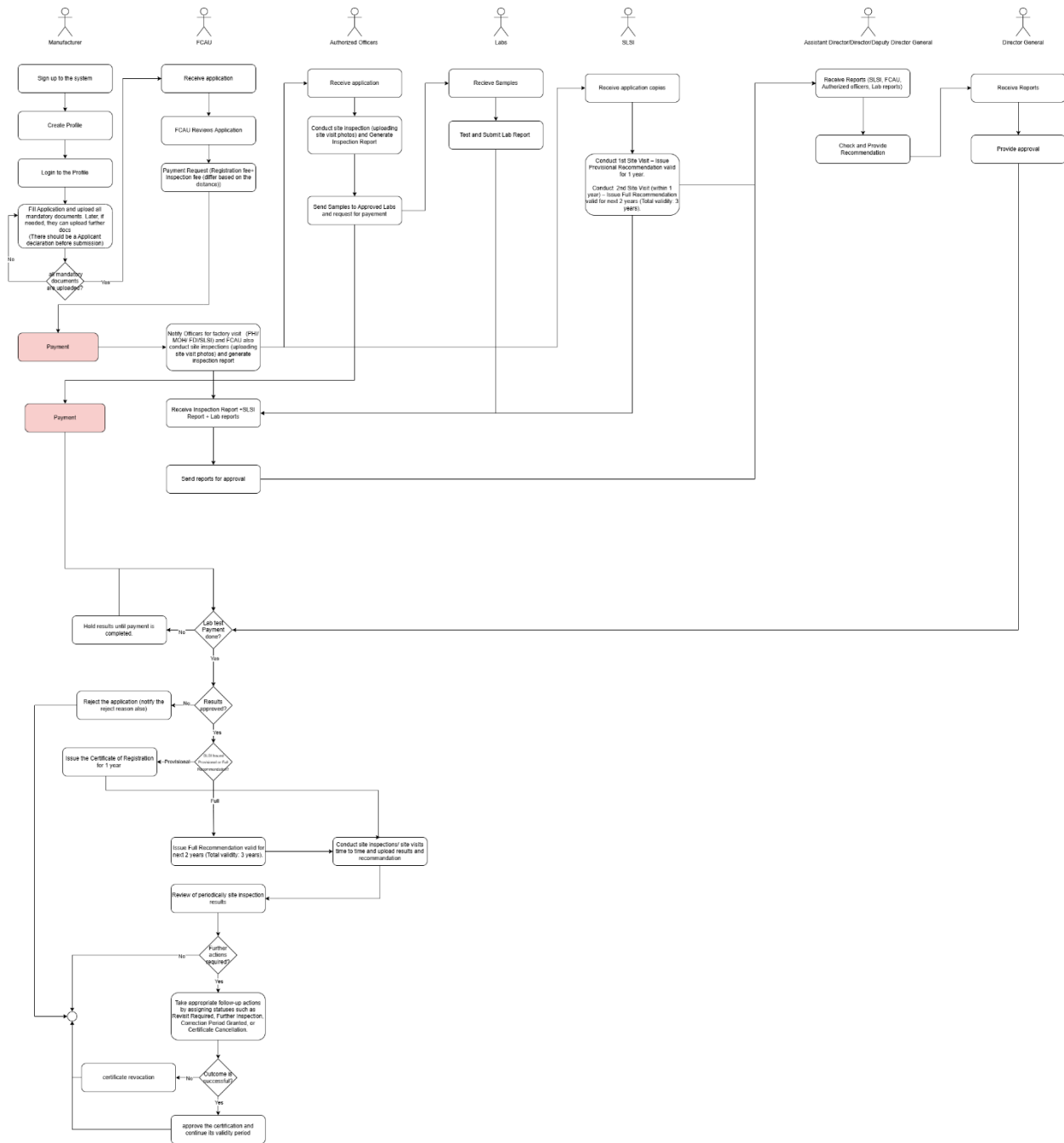


Figure 1.14

1.8.2 Bottled Water Registration Renewal

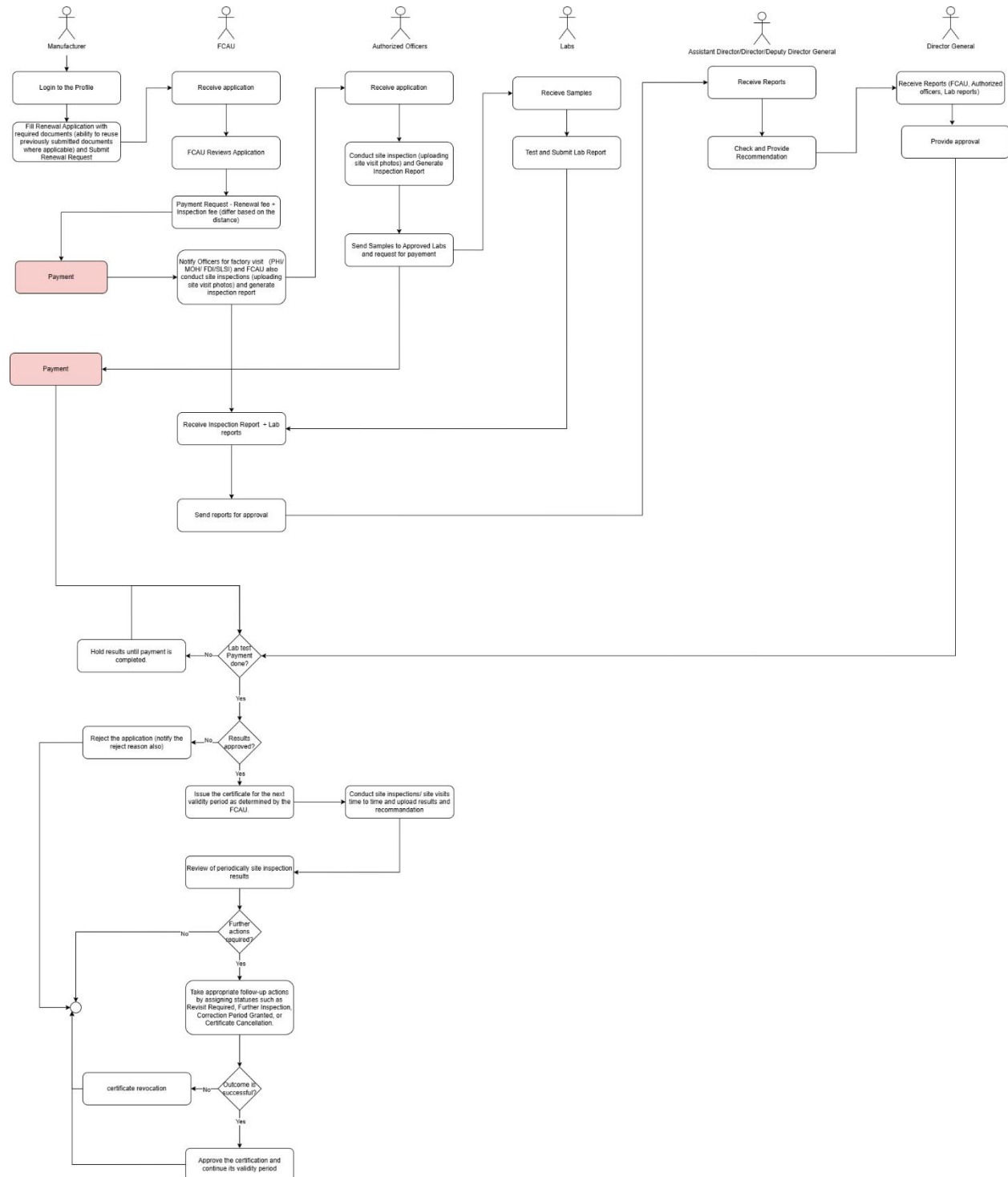


Figure 1.15

1.8.3 Bottled Water Registration Amendment

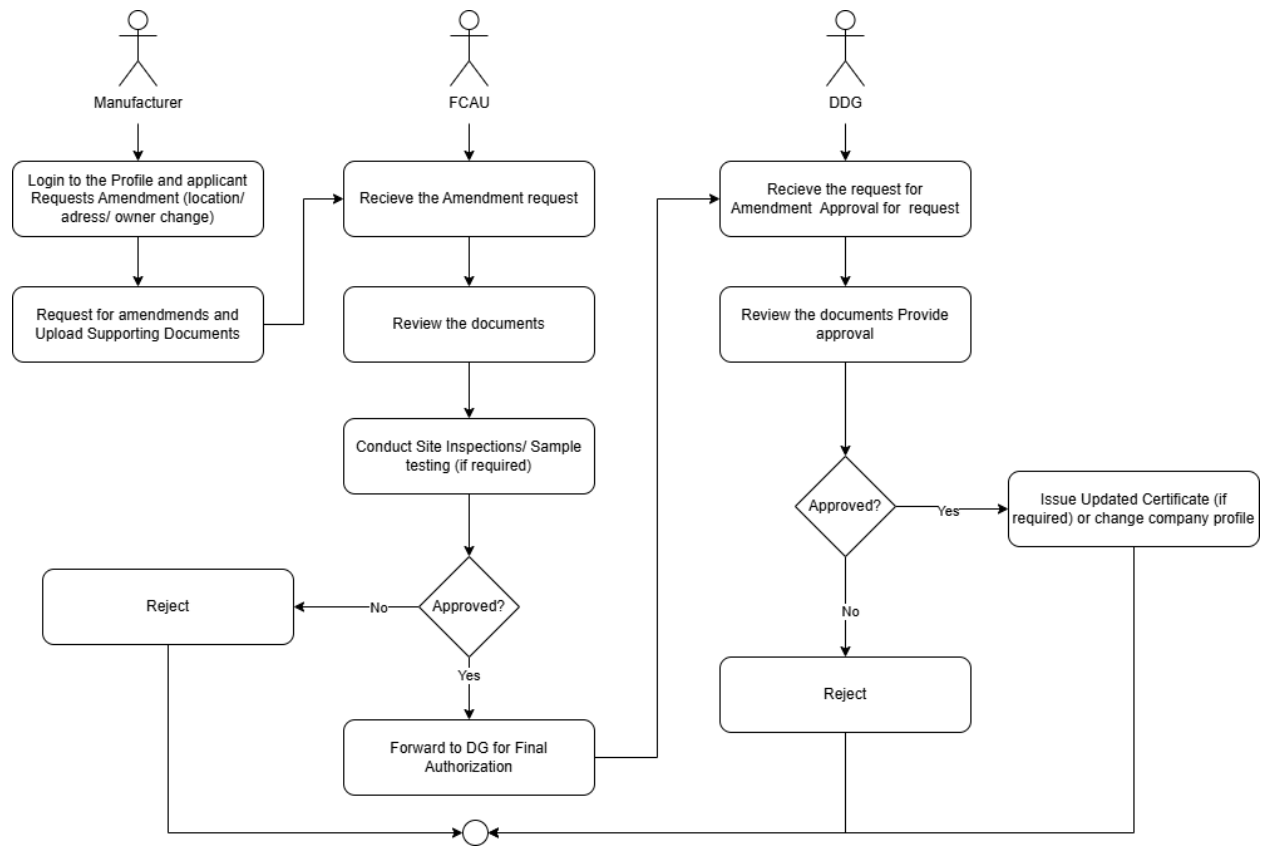


Figure 1.16

1.8.4 Bottled Water Multiple Brands

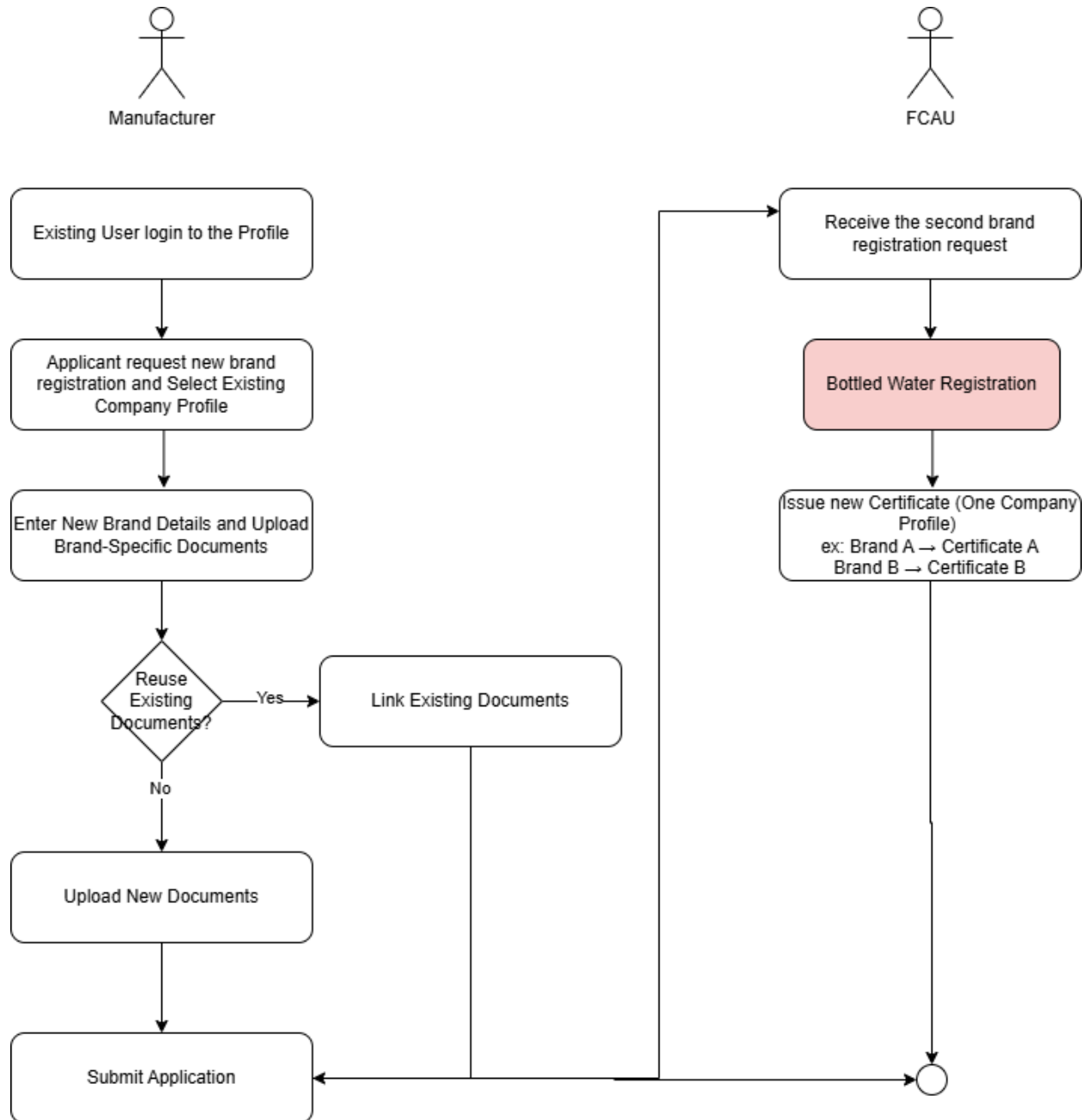


Figure 1.17

2. Use Case Modeling

Use case modeling is a fundamental technique used to capture and represent the functional behavior of a system from the perspective of its users and external actors. Within this Software Design Document (SDD), use case modeling provides a clear and structured way to describe how different stakeholders interact with the system to achieve specific goals.

The primary purpose of use case modeling is to translate high-level functional requirements defined in the Software Requirements Specification (SRS) into user centric scenarios. Each use case outlines a sequence of interactions between an actor (such as a system user, administrator, or external system) and the system, demonstrating how a particular function or service is executed.

Use case models typically consist of:

- **Actors** – entities that interact with the system (e.g., public users, system administrators, regulatory officers)
- **Use Cases** – specific functionalities or services provided by the system (e.g., register premises, submit complaints, generate reports)
- **System Boundary** – defines the scope of the system and its interactions with external entities
- **Relationships** – associations such as include, extend, and generalization that show dependencies and reuse among use cases

In the context of a microservices-based architecture, use case modeling plays a critical role in identifying system functionalities and mapping them to corresponding services. It helps ensure that each microservice is aligned with specific business capabilities while maintaining loose coupling and high cohesion.

Use case diagrams and detailed use case descriptions also support effective communication among stakeholders, including developers, designers, project managers, and domain experts. They provide a shared understanding of system behavior, reduce ambiguity in requirements, and serve as a foundation for further design activities such as sequence diagrams, activity diagrams, and API design.

Overall, use case modeling ensures that the system is designed with a strong focus on user interactions and business processes, enabling the development of a solution that is both functionally complete and aligned with stakeholder expectations.

2.1 Premises Registration and Lifecycle Management Module



Figure 2.0

Use Case Narratives

UC-01: Register Account	
Primary Actor:	Food Business Operator (FBO)
Description:	Allows a new user to create an account in the system.
Preconditions:	User does not already have an account
Postconditions:	- User account is successfully created - User can log in to the system
Main Flow:	- User selects “Register Account” - System displays registration form - User enters required details - System validates input - System creates account
Alternate Flow:	4a. Invalid input → System shows error message

UC-02: Login	
Primary Actor:	FBO
Description:	Allows registered users to access the system.
Preconditions:	User must have a registered account
Postconditions:	User is authenticated and granted access
Main Flow:	- User enters credentials - System validates credentials - System grants access
Alternate Flow:	2a. Invalid credentials → Error message displayed

UC-03: Fill Registration Form	
Primary Actor:	FBO
Description:	User fills details required for premises registration.
Preconditions:	User is logged in
Postconditions:	Form data is saved (draft or ready for submission)
Main Flow:	- User opens registration form - User enters premises details - System validates inputs - User proceeds to next step
Alternate Flow:	N/A

UC-04: Upload Documents	
Primary Actor:	FBO
Description:	Upload supporting documents for registration.
Preconditions:	Registration form initiated
Postconditions:	Documents stored successfully
Main Flow:	- User selects documents - System validates file type/size - Documents are uploaded
Alternate Flow:	2a. Invalid file → Error displayed

UC-05: Save Draft Application	
Primary Actor:	FBO
Description:	Save incomplete application for later completion.
Preconditions:	Registration form partially filled
Postconditions:	Draft stored in system
Main Flow:	User clicks “Save Draft”

	System saves current data
Alternate Flow:	N/A

UC-06: Submit Application	
Primary Actor:	FBO
Description:	Submit completed application for processing.
Preconditions:	- Form completed - Documents uploaded
Postconditions:	- Application submitted - Reference number generated
Main Flow:	- User clicks “Submit” - System validates full application - System performs: - Premises categorization & risk profiling - GN division mapping - Duplicate validation - System generates reference number - Application is forwarded to MOH/PHI
Alternate Flow:	2a. Validation fails → Errors displayed

UC-07: Track Application Status	
Primary Actor:	FBO
Description:	Allows user to track submitted application.
Preconditions:	Application submitted
Postconditions:	Status displayed
Main Flow:	- User selects “Track Status” - System retrieves application details

	Status displayed
Alternate Flow:	N/A

UC-08: Submit Appeal	
Primary Actor:	FBO
Description:	Submit appeal against rejected application.
Preconditions:	Application rejected
Postconditions:	Appeal submitted to system
Main Flow:	- User selects rejected application - User submits appeal with justification - System records appeal - Appeal forwarded to relevant authority
Alternate Flow:	N/A

UC-09: View Assigned Applications	
Primary Actor:	PHI
Description:	View applications assigned for inspection.
Preconditions:	Applications assigned by MOH
Postconditions:	List of applications displayed
Main Flow:	- PHI logs in - System displays assigned applications
Alternate Flow:	N/A

UC-10: Conduct Inspections	
Primary Actor:	PHI
Description:	Perform inspections on premises.
Preconditions:	Applications assigned by MOH

Postconditions:	
Main Flow:	UC-10A: Conduct 1st Inspection • PHI selects application • Conducts inspection • Records observations UC-10B: Conduct 2nd Inspection (if required) • Triggered based on first inspection outcome • PHI performs follow-up inspection
Alternate Flow:	N/A

UC-11: Submit Inspection Report	
Primary Actor:	PHI
Description:	Submit inspection findings.
Preconditions:	• Inspection completed
Postconditions:	• Report submitted to MOH
Main Flow:	• PHI fills report • System validates input • Report submitted
Alternate Flow:	N/A

UC-12: Review PHI Recommendation	
Primary Actor:	MOH
Description:	Review inspection report and recommendation.
Preconditions:	• Inspection report submitted
Postconditions:	• Decision ready for approval/rejection
Main Flow:	• Review inspection

	• System validates input • Submitted for approval
Alternate Flow:	N/A

UC-13: Approve Application	
Primary Actor:	MOH
Description:	Approve premises registration.
Preconditions:	• Review inspection
Postconditions:	• Application marked approved • Notification sent
Main Flow:	• Submit approval/rejection
Alternate Flow:	N/A

2.2 Inspection and Grading Module

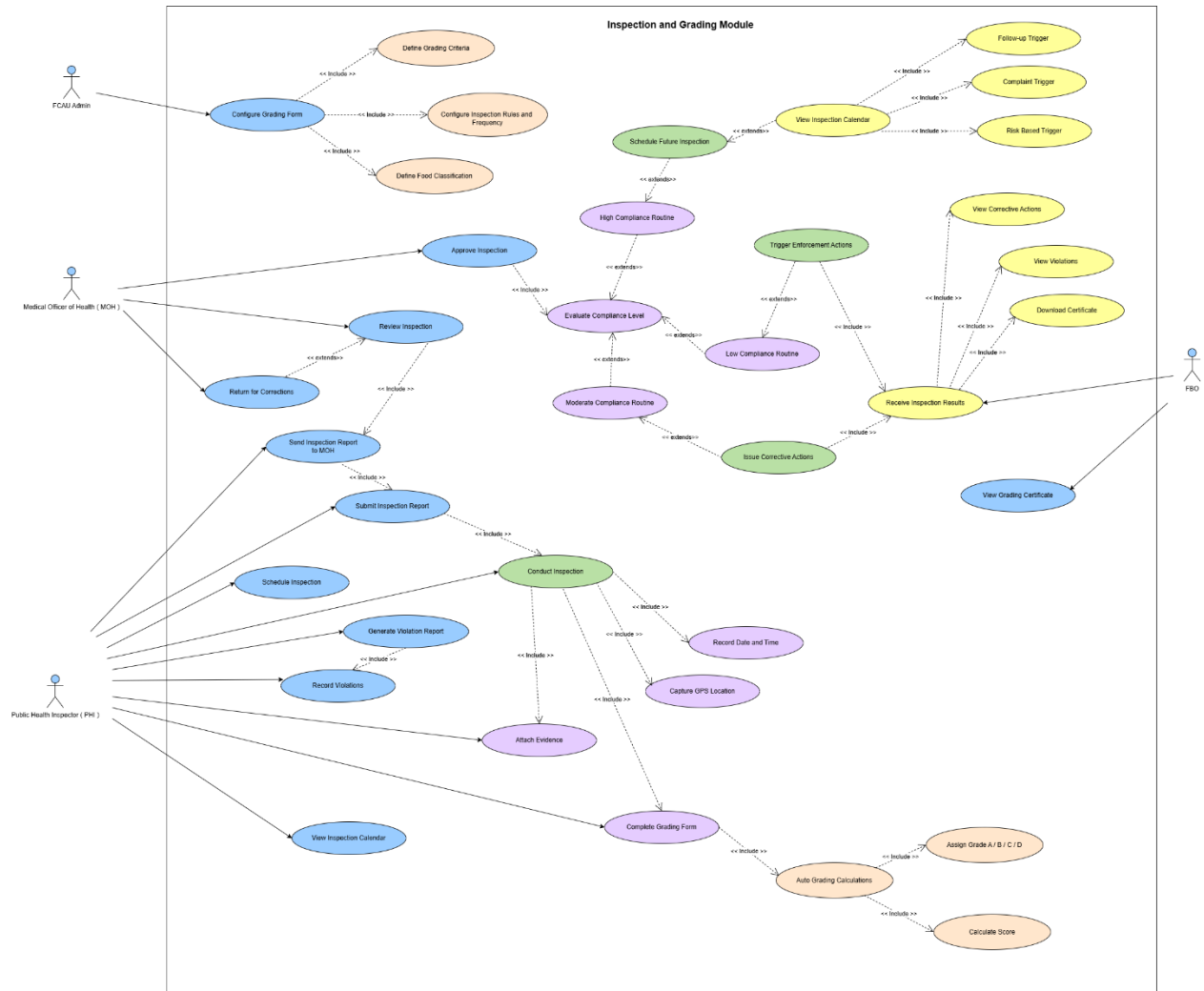


Figure 2.1

Use Case Narratives

UC-01: Configure Grading Framework	
Primary Actor:	FCAU Admin
Description:	Configure grading criteria, inspection rules, and food classifications.
Preconditions:	Admin is authenticated
Postconditions:	Grading framework is configured and available for inspections
Main Flow:	- Admin selects “Configure Grading Form” - Defines grading criteria - Configures inspection rules and frequency - Defines food classification categories - System saves configuration
Alternate Flow:	N/A

UC-02: Schedule Inspection	
Primary Actor:	PHI
Description:	Schedule inspections for registered premises.
Preconditions:	Premises registered and assigned
Postconditions:	Inspection scheduled and visible in calendar
Main Flow:	- PHI selects premises - Chooses inspection date/time - System schedules inspection - Entry added to inspection calendar
Alternate Flow:	N/A

UC-03: View Inspection Calendar	
Primary Actor:	PHI, FBO
Description:	View scheduled inspections and triggers.
Preconditions:	User logged in
Postconditions:	Calendar displayed with relevant events
Main Flow:	- User opens calendar - System displays: - Scheduled inspections - Risk-based triggers - Complaint triggers

	◦ Follow-up triggers
Alternate Flow:	N/A

UC-04: Conduct Inspection	
Primary Actor:	PHI
Description:	Perform on-site inspection and collect data.
Preconditions:	• Inspection scheduled
Postconditions:	• Inspection data recorded
Main Flow:	• PHI starts inspection • Records date and time • Captures GPS location • Completes grading form • Attaches evidence (images/documents)
Alternate Flow:	N/A

UC-05: Record Violations	
Primary Actor:	PHI
Description:	Record any violations observed during inspection.
Preconditions:	• Inspection in progress or completed
Postconditions:	• Violations stored in system
Main Flow:	• PHI identifies violation • Records violation details • System saves record
Alternate Flow:	N/A

UC-06: Generate Violation Report	
Primary Actor:	PHI
Description:	Generate report of violations.
Preconditions:	• Violations recorded
Postconditions:	• Violation report generated
Main Flow:	• PHI selects violations • System compiles report • Report generated
Alternate Flow:	N/A

UC-07: Submit Inspection Report	
Primary Actor:	PHI
Description:	Submit completed inspection report to MOH.
Preconditions:	Inspection completed
Postconditions:	Report sent to MOH
Main Flow:	- PHI reviews inspection data - Submits report - System forwards to MOH
Alternate Flow:	N/A

UC-08: Review Inspection	
Primary Actor:	MOH
Description:	Review submitted inspection report.
Preconditions:	Report submitted by PHI
Postconditions:	Decision taken (approve/return)
Main Flow:	- MOH accesses report - Reviews findings and grading - Proceeds to decision
Alternate Flow:	N/A

UC-09: Approve Inspection	
Primary Actor:	MOH
Description:	Approve inspection results.
Preconditions:	
Postconditions:	- Inspection approved - Results published to FBO
Main Flow:	
Alternate Flow:	N/A

UC-10: Return for Corrections	
Primary Actor:	MOH
Description:	Return report to PHI for corrections.
Preconditions:	
Postconditions:	PHI notified for corrections
Main Flow:	- MOH identifies issues - Sends report back with comments

Alternate Flow:	N/A
-----------------	-----

UC-11: Evaluate Compliance Level	
Primary Actor:	System
Description:	Determine compliance level based on inspection results.
Preconditions:	Inspection data available
Postconditions:	Compliance level assigned
Main Flow:	- System analyzes grading inputs - Categorizes into: - High compliance - Moderate compliance - Low compliance
Alternate Flow:	N/A

UC-12: Compliance-Based Actions	
Primary Actor:	
Description:	
Preconditions:	
Postconditions:	
Main Flow:	UC-12A: High Compliance Routine - Schedule future inspections UC-12B: Moderate Compliance Routine - Issue corrective actions UC-12C: Low Compliance Routine - Trigger enforcement actions
Alternate Flow:	N/A

UC-13: Trigger Enforcement Actions	
Primary Actor:	System / MOH
Description:	Initiate enforcement for low compliance.
Preconditions:	
Postconditions:	Enforcement actions triggered
Main Flow:	
Alternate Flow:	N/A

UC-14: Issue Corrective Actions	
Primary Actor:	MOH / System
Description:	Issue corrective actions to FBO

Preconditions:	
Postconditions:	FBO notified
Main Flow:	
Alternate Flow:	N/A

UC-15: Receive Inspection Results	
Primary Actor:	FBO
Description:	View inspection outcome.
Preconditions:	Inspection approved
Postconditions:	Results visible to FBO
Main Flow:	- FBO logs in - Views inspection results - Accesses: - Violations - Corrective actions - Certification status
Alternate Flow:	N/A

UC-16: View Violations	
Primary Actor:	FBO
Description:	View violations recorded.
Preconditions:	PHI to submit violations
Postconditions:	
Main Flow:	- PHI submit violation record - FBO to view the record
Alternate Flow:	N/A

UC-17: View Corrective Actions	
Primary Actor:	FBO
Description:	View required corrective actions.
Preconditions:	Get corrective actions from PHI
Postconditions:	Update the system after corrective actions implemented
Main Flow:	- Receive actions - Update the system after implemented
Alternate Flow:	N/A

UC-18: Download Certificate	
Primary Actor:	FBO

Description:	Download inspection certificate.
Preconditions:	• Inspection approved
Postconditions:	• Certificate downloaded
Main Flow:	
Alternate Flow:	N/A

UC-19: View Grading Certificate	
Primary Actor:	FBO
Description:	View grading classification (A/B/C/D)
Preconditions:	Final grading should submit
Postconditions:	
Main Flow:	• Authority to submit final grade • View grade
Alternate Flow:	N/A

UC-20: Auto Grading Calculation	
Primary Actor:	System
Description:	Automatically calculate inspection score and grade.
Preconditions:	• Grading form completed
Postconditions:	• Grade assigned
Main Flow:	• System calculates score • Assigns grade (A/B/C/D)
Alternate Flow:	N/A

2.3 Laboratory Coordination Module



Figure 2.2

Use Case Narratives

UC-01: Generate Sample ID and QR Code	
Primary Actor:	PHI / FDI
Description:	Generate a unique identifier and QR code for each collected sample.
Preconditions:	Sample collection initiated
Postconditions:	Unique sample ID and QR code generated
Main Flow:	- User initiates sample registration - System generates unique sample ID - System generates QR code - QR code is linked to sample record
Alternate Flow:	N/A

UC-02: Enter Sample Metadata	
Primary Actor:	PHI / FDI
Description:	Enter details about the collected sample.
Preconditions:	Sample ID generated
Postconditions:	Sample metadata stored
Main Flow:	- User inputs sample details (type, location, date, etc.) - System validates inputs - Data is saved
Alternate Flow:	N/A

UC-03: Select Lab Classification	
Primary Actor:	PHI / FDI
Description:	Select appropriate lab category/type for testing.
Preconditions:	Sample registered
Postconditions:	Lab classification assigned
Main Flow:	- User selects lab classification - System records classification
Alternate Flow:	N/A

UC-04: Send Sample to Lab	
Primary Actor:	PHI / FDI
Description:	Dispatch sample to laboratory.
Preconditions:	Sample ID generated

	• Metadata completed
Postconditions:	• Sample marked as sent
Main Flow:	• User confirms dispatch • System updates sample status • Lab notified
Alternate Flow:	N/A

UC-05: Scan Sample QR Code	
Primary Actor:	Reception Counter
Description:	Scan QR code to retrieve sample information.
Preconditions:	• Sample received at lab
Postconditions:	• Sample details displayed
Main Flow:	• Staff scans QR code • System retrieves sample record • Displays details
Alternate Flow:	N/A

UC-06: Retrieve Digital Requisition	
Primary Actor:	Reception Counter
Description:	Retrieve associated digital requisition form.
Preconditions:	• QR code scanned
Postconditions:	• Requisition displayed
Main Flow:	
Alternate Flow:	N/A

UC-07: Verify Sample Integrity	
Primary Actor:	Reception Counter
Description:	Check condition of received sample.
Preconditions:	• Sample received
Postconditions:	• Sample marked as accepted or rejected
Main Flow:	• Staff inspects sample • Confirms integrity • System records verification
Alternate Flow:	• 2a. Sample compromised → proceed to reject sample

UC-08: Accept Sample	
Primary Actor:	Reception Counter
Description:	Accept valid sample for analysis.
Preconditions:	Sample integrity verified
Postconditions:	Sample accepted and queued for analysis
Main Flow:	
Alternate Flow:	N/A

UC-09: Reject Sample	
Primary Actor:	Reception Counter
Description:	Reject invalid sample.
Preconditions:	
Postconditions:	- Sample marked rejected - Reason recorded
Main Flow:	
Alternate Flow:	N/A

UC-10: Assign to Approved Analyst	
Primary Actor:	Reception Counter
Description:	Assign accepted sample to certified analyst.
Preconditions:	Sample accepted
Postconditions:	Analyst assigned
Main Flow:	- Select analyst - System assigns sample - Analyst notified
Alternate Flow:	N/A

UC-11: Upload Lab Report PDF	
Primary Actor:	Approved Analyst
Description:	Upload finalized lab report.
Preconditions:	Sample analysis completed
Postconditions:	Report stored in system
Main Flow:	- Analyst uploads report - System validates file - Report saved
Alternate Flow:	N/A

UC-12: Generate Manual Report	
Primary Actor:	Approved Analyst
Description:	Manually create lab report if needed.
Preconditions:	Analysis completed
Postconditions:	Report generated and uploaded
Main Flow:	
Alternate Flow:	N/A

UC-13: Fetch Report via API	
Primary Actor:	LIMS (External System)
Description:	Retrieve lab report through integration.
Preconditions:	Report available in LIMS
Postconditions:	Report imported to system
Main Flow:	
Alternate Flow:	N/A

UC-14: Parse PDF Report	
Primary Actor:	System
Description:	Extract structured data from uploaded report.
Preconditions:	Report uploaded
Postconditions:	Data extracted and stored
Main Flow:	- System reads PDF - Extracts relevant data fields - Stores structured data
Alternate Flow:	N/A

UC-15: Maintain Sample Registry	
Primary Actor:	System
Description:	Maintain lifecycle records of samples.
Preconditions:	Sample data available
Postconditions:	Registry updated
Main Flow:	
Alternate Flow:	N/A

UC-16: Evaluate Compliance Thresholds	
Primary Actor:	System

Description:	Evaluate lab results against regulatory thresholds.
Preconditions:	• Parsed report data available
Postconditions:	• Compliance status determined
Main Flow:	• System compares results with thresholds • Determines compliance status
Alternate Flow:	N/A

UC-17: Dispatch Results to PHI	
Primary Actor:	System
Description:	Send analyzed results to PHI.
Preconditions:	• Compliance evaluation completed
Postconditions:	• PHI notified with results
Main Flow:	• System compiles results • Sends notification/report to PHI
Alternate Flow:	N/A

2.4 Legal, Prosecution and Enforcement Management Module

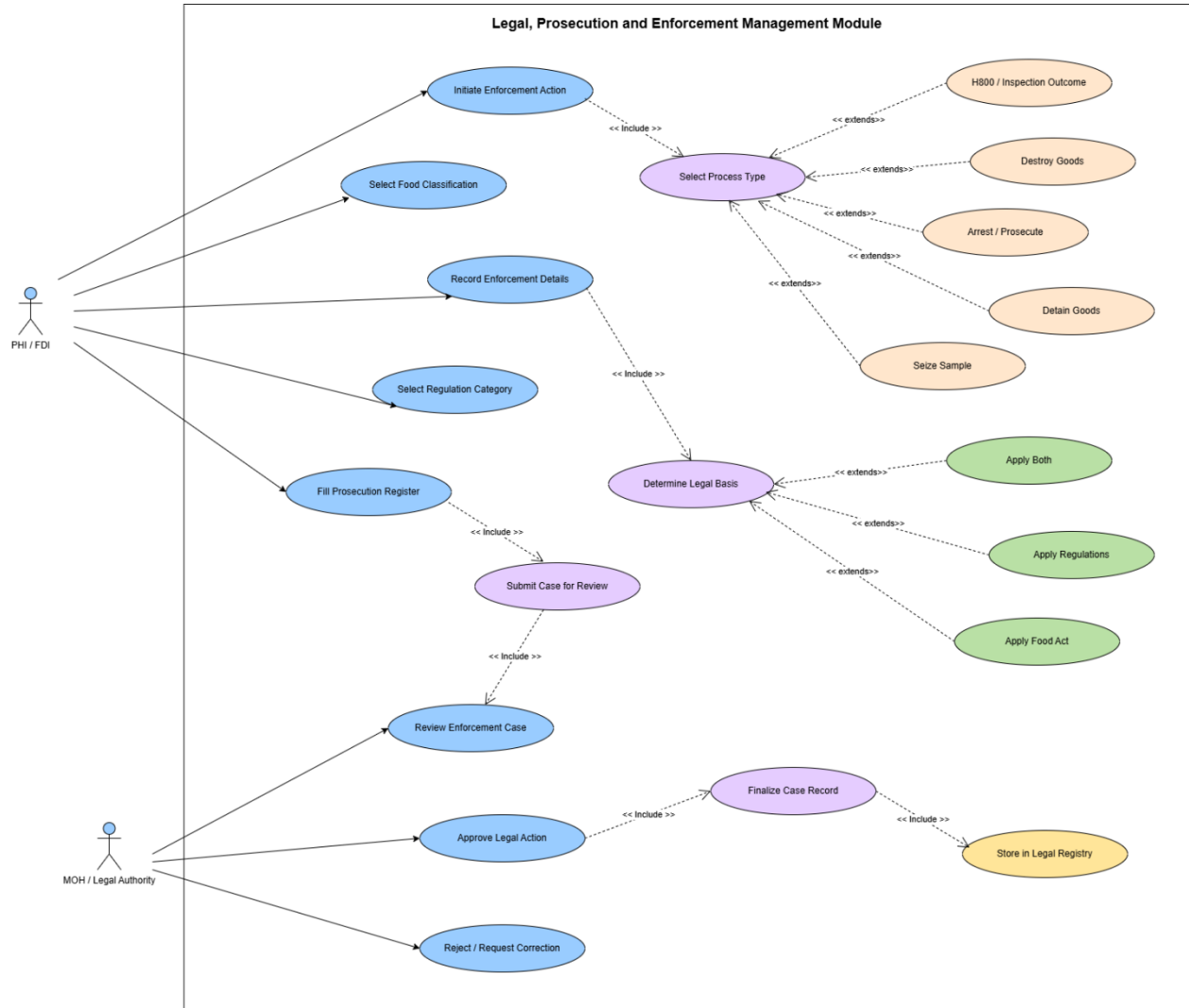


Figure 2.3

Use Case Narratives

UC-01: Initiate Enforcement Action	
Primary Actor:	PHI / FDI
Description:	Initiate enforcement process based on inspection findings or violations.
Preconditions:	Violation identified from inspection or lab results
Postconditions:	Enforcement case initiated
Main Flow:	- Actor selects “Initiate Enforcement Action” - System creates new enforcement case - User proceeds to select process type
Alternate Flow:	N/A

UC-02: Select Process Type	
Primary Actor:	PHI / FDI
Description:	Choose the type of enforcement action to be taken.
Preconditions:	Enforcement action initiated
Postconditions:	Process type selected
Main Flow:	- User selects enforcement type: - H800 / Inspection Outcome - Destroy Goods - Arrest / Prosecute - Detain Goods - Seize Sample - System records selection
Alternate Flow:	N/A

UC-03: Select Food Classification	
Primary Actor:	PHI / FDI
Description:	Define food category relevant to enforcement.
Preconditions:	Enforcement initiated
Postconditions:	Food classification recorded
Main Flow:	
Alternate Flow:	N/A

UC-04: Select Regulation Category	
Primary Actor:	PHI / FDI
Description:	Identify applicable regulatory category.
Preconditions:	
Postconditions:	Regulation category assigned
Main Flow:	
Alternate Flow:	N/A

UC-05: Record Enforcement Details	
Primary Actor:	PHI / FDI
Description:	Capture all enforcement-related details.
Preconditions:	Enforcement process started
Postconditions:	Case details recorded
Main Flow:	- User enters violation details - Adds supporting evidence - System validates and stores data
Alternate Flow:	N/A

UC-06: Determine Legal Basis	
Primary Actor:	System / PHI
Description:	Identify applicable legal framework.
Preconditions:	Enforcement details recorded
Postconditions:	Legal basis determined
Main Flow:	- System evaluates case details - Determines applicable: - Food Act - Regulations - Both
Alternate Flow:	N/A

UC-07: Fill Prosecution Register	
Primary Actor:	PHI / FDI
Description:	Enter prosecution details for legal processing.
Preconditions:	Legal basis determined
Postconditions:	Prosecution record created
Main Flow:	- User fills prosecution form - System validates entries

	Data saved
Alternate Flow:	N/A

UC-08: Submit Case for Review	
Primary Actor:	PHI / FDI
Description:	Submit enforcement case for authority review.
Preconditions:	All required details completed
Postconditions:	Case forwarded to MOH / Legal Authority
Main Flow:	- User submits case - System forwards to authority
Alternate Flow:	N/A

UC-09: Review Enforcement Case	
Primary Actor:	MOH / Legal Authority
Description:	Review submitted enforcement case.
Preconditions:	Case submitted
Postconditions:	Decision prepared
Main Flow:	- Authority reviews case details - Verifies evidence and legal basis
Alternate Flow:	N/A

UC-10: Approve Legal Action	
Primary Actor:	MOH / Legal Authority
Description:	Approve enforcement/legal action.
Preconditions:	
Postconditions:	Case approved for execution
Main Flow:	- Authority approves case - System proceeds to finalization
Alternate Flow:	N/A

UC-11: Reject / Request Correction	
Primary Actor:	MOH / Legal Authority
Description:	Reject case or request corrections.
Preconditions:	
Postconditions:	Case returned to PHI for updates
Main Flow:	- Authority identifies issues - Sends feedback

	Case returned
Alternate Flow:	N/A

UC-12: Finalize Case Record	
Primary Actor:	System
Description:	Finalize approved enforcement case.
Preconditions:	Case approved
Postconditions:	Case closed and finalized
Main Flow:	- System consolidates case data - Marks case as finalized
Alternate Flow:	N/A

UC-13: Store in Legal Registry	
Primary Actor:	System
Description:	Store finalized case in legal registry.
Preconditions:	Case finalized
Postconditions:	Case archived for recordkeeping
Main Flow:	
Alternate Flow:	N/A

2.5 Import Control Module

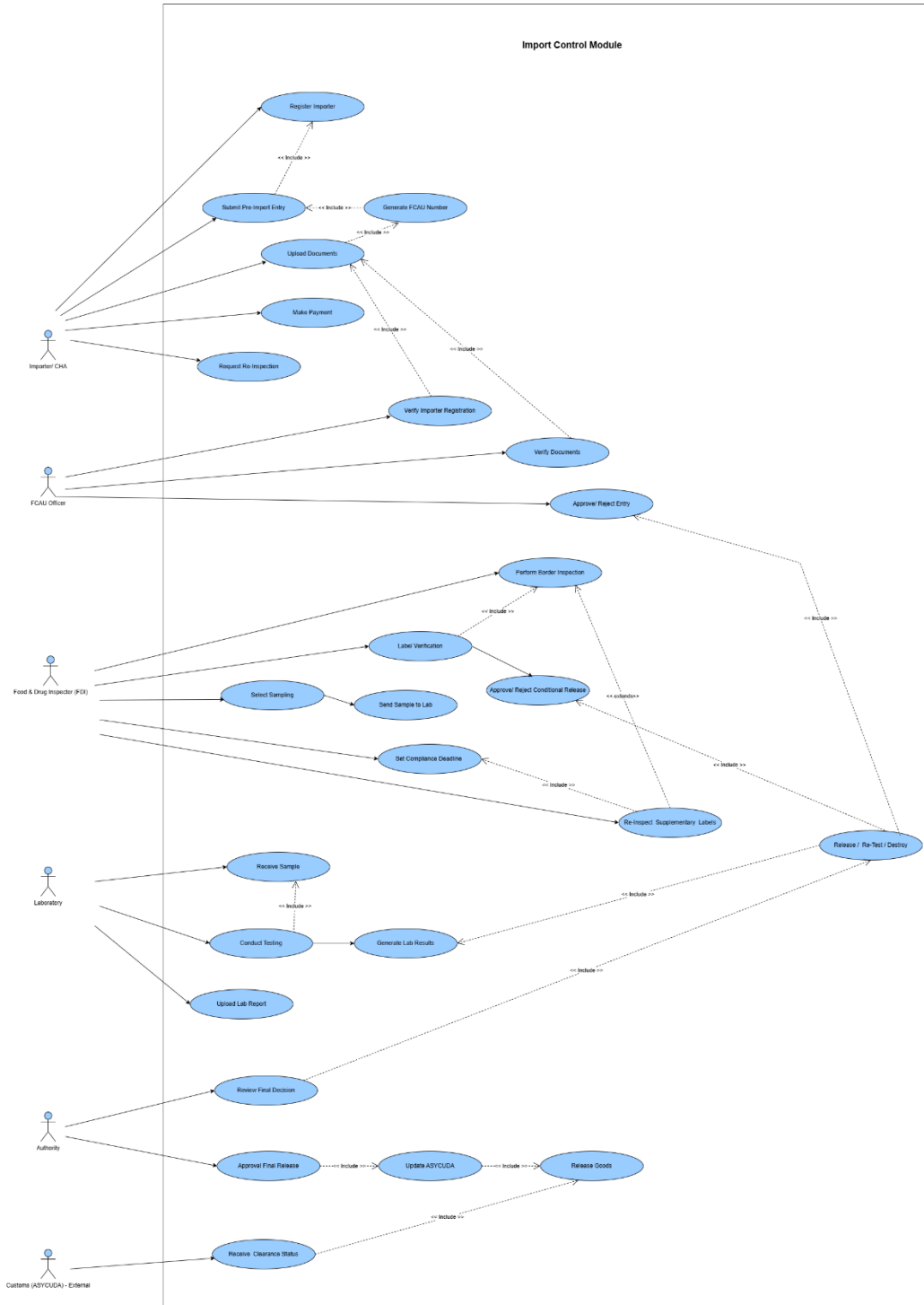


Figure 2.4

Use Case Narratives

UC-01: Register Importer	
Primary Actor:	Importer / CHA
Description:	Register a new importer in the system.
Preconditions:	Importer not previously registered
Postconditions:	Importer account created and verified
Main Flow:	- Importer submits registration details - System validates information - FCAU Officer verifies registration - System activates importer account
Alternate Flow:	N/A

UC-02: Submit Pre-Import Entry	
Primary Actor:	Importer / CHA
Description:	Submit details of goods prior to import.
Preconditions:	Importer registered
Postconditions:	Pre-import entry created with reference number
Main Flow:	- Importer enters shipment details - Uploads required documents - System generates FCAU number - Entry submitted for review
Alternate Flow:	N/A

UC-03: Upload Documents	
Primary Actor:	Importer / CHA
Description:	Upload supporting documents.
Preconditions:	Pre-import entry initiated
Postconditions:	Documents stored
Main Flow:	- User uploads documents - System validates file format - Files stored
Alternate Flow:	N/A

UC-04: Make Payment	
Primary Actor:	Importer / CHA
Description:	Pay applicable inspection or processing fees.

Preconditions:	• Pre-import entry submitted
Postconditions:	• Payment recorded
Main Flow:	
Alternate Flow:	N/A

UC-05: Verify Importer Registration	
Primary Actor:	FCAU Officer
Description:	Verify importer registration details.
Preconditions:	• Importer registered
Postconditions:	• Importer verified or rejected
Main Flow:	
Alternate Flow:	N/A

UC-06: Verify Documents	
Primary Actor:	FCAU Officer
Description:	Validate submitted documents.
Preconditions:	• Documents uploaded
Postconditions:	• Documents approved or flagged
Main Flow:	
Alternate Flow:	N/A

UC-07: Approve / Reject Entry	
Primary Actor:	FCAU Officer
Description:	Approve or reject pre-import entry.
Preconditions:	• Documents verified
Postconditions:	• Entry approved or rejected
Main Flow:	
Alternate Flow:	N/A

UC-08: Perform Border Inspection	
Primary Actor:	Food & Drug Inspector (FDI)
Description:	Conduct inspection at entry point.
Preconditions:	• Entry approved
Postconditions:	• Inspection results recorded
Main Flow:	• Inspector performs inspection • Conducts label verification • Records findings

Alternate Flow:	N/A
------------------------	-----

UC-09: Label Verification	
Primary Actor:	FDI
Description:	Verify compliance of product labeling.
Preconditions:	• Inspection initiated
Postconditions:	• Label compliance recorded
Main Flow:	
Alternate Flow:	N/A

UC-10: Select Sampling	
Primary Actor:	FDI
Description:	Select samples for lab testing.
Preconditions:	• Inspection ongoing
Postconditions:	• Samples selected and recorded
Main Flow:	
Alternate Flow:	N/A

UC-11: Send Sample to Lab	
Primary Actor:	FDI
Description:	Dispatch selected samples for testing.
Preconditions:	
Postconditions:	• Sample sent to laboratory
Main Flow:	
Alternate Flow:	N/A

UC-12: Set Compliance Deadline	
Primary Actor:	FDI
Description:	Define deadline for corrective actions.
Preconditions:	
Postconditions:	• Deadline recorded
Main Flow:	
Alternate Flow:	N/A

UC-13: Approve / Reject Conditional Release	
Primary Actor:	FDI
Description:	Decide conditional release of goods.

Preconditions:	• Inspection completed
Postconditions:	• Conditional release decision recorded
Main Flow:	
Alternate Flow:	N/A

UC-14: Re-Inspect / Supplementary Labels	
Primary Actor:	FDI
Description:	Perform follow-up inspection for compliance.
Preconditions:	• Conditional release granted
Postconditions:	• Compliance verified
Main Flow:	
Alternate Flow:	N/A

UC-15: Laboratory Processing	
Primary Actor:	Laboratory
Description:	Lab processing
Preconditions:	Samples to receive
Postconditions:	Upload the lab report
Main Flow:	UC-15A: Receive Sample • Sample received and logged UC-15B: Conduct Testing • Perform analysis UC-15C: Generate Lab Results • Results generated UC-15D: Upload Lab Report • Report uploaded to system
Alternate Flow:	N/A

UC-16: Release / Re-Test / Destroy Decision	
Primary Actor:	System / Authority
Description:	Determine final disposition based on results.
Preconditions:	• Inspection and/or lab results available
Postconditions:	• Decision recorded: ◦ Release ◦ Re-test ◦ Destroy
Main Flow:	
Alternate Flow:	N/A

UC-17: Review Final Decision	
Primary Actor:	Authority
Description:	Review final outcome before release.
Preconditions:	Decision prepared
Postconditions:	Decision confirmed
Main Flow:	
Alternate Flow:	N/A

UC-18: Approve Final Release	
Primary Actor:	Authority
Description:	Approve goods for release.
Preconditions:	
Postconditions:	Goods approved for clearance
Main Flow:	
Alternate Flow:	N/A

UC-19: Update ASYCUDA System	
Primary Actor:	System
Description:	Update customs system with clearance status.
Preconditions:	Final approval granted
Postconditions:	External system updated
Main Flow:	
Alternate Flow:	N/A

UC-20: Release Goods	
Primary Actor:	System
Description:	Complete goods release process.
Preconditions:	Approval completed
Postconditions:	Goods released
Main Flow:	
Alternate Flow:	N/A

UC-21: Receive Clearance Status	
Primary Actor:	Customs (ASYCUDA – External)
Description:	Receive updated clearance status.
Preconditions:	

Postconditions:	• Clearance status reflected in customs system
Main Flow:	
Alternate Flow:	N/A

UC-22: Request Re-Inspection	
Primary Actor:	Importer / CHA
Description:	Request re-inspection if required.
Preconditions:	• Initial inspection completed
Postconditions:	• Re-inspection scheduled
Main Flow:	
Alternate Flow:	N/A

2.6 Export Control Module

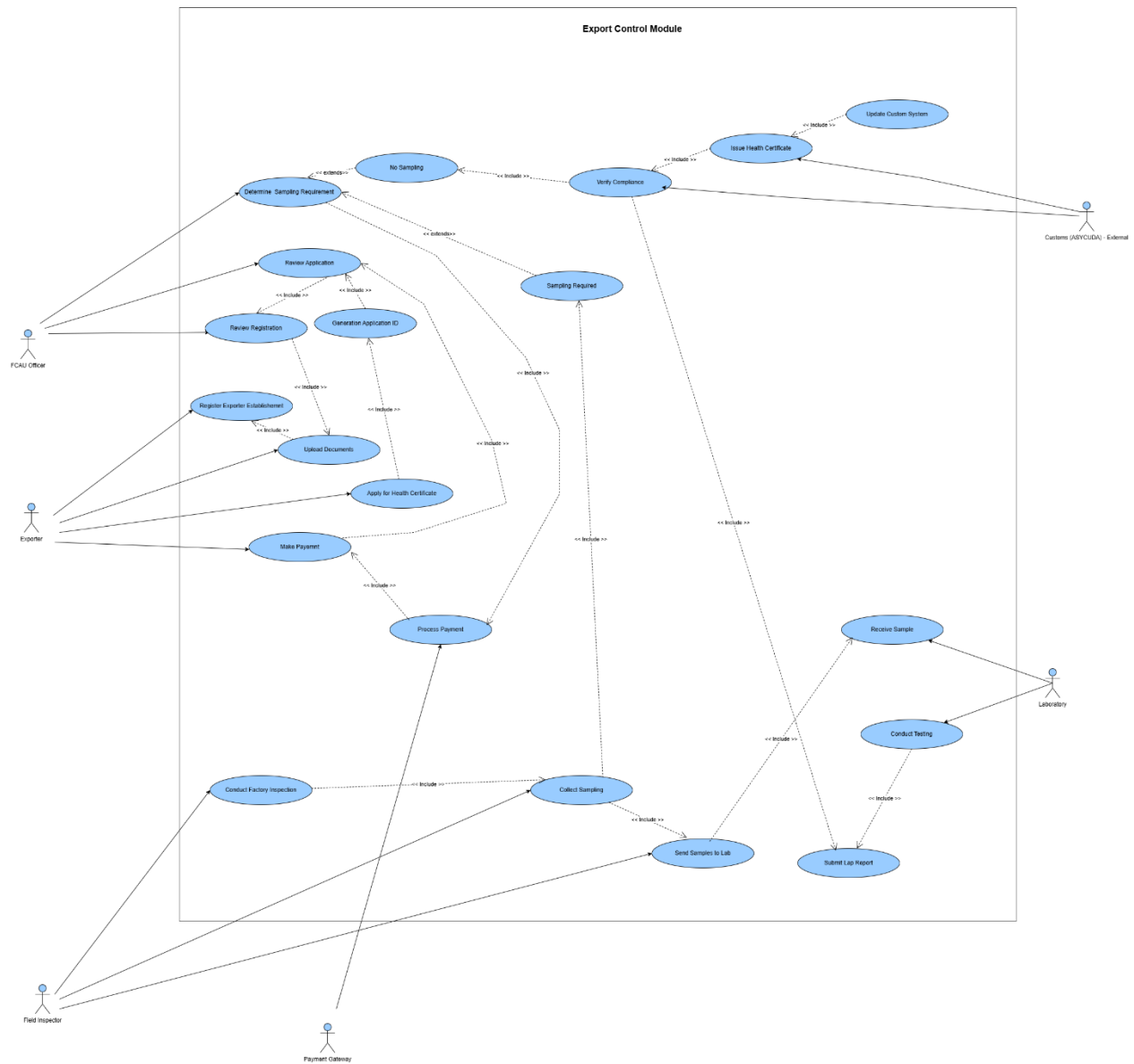


Figure 2.5

Use Case Narratives

UC-01: Register Exporter Establishment	
Primary Actor:	Exporter
Description:	Register an exporter and their establishment in the system.
Preconditions:	• Exporter is not registered
Postconditions:	• Exporter establishment registered and available for review
Main Flow:	• Exporter submits registration details • Uploads required documents • System stores registration data • FCAU Officer reviews registration
Alternate Flow:	N/A

UC-02: Upload Documents	
Primary Actor:	Exporter
Description:	Upload required documents for registration or certification.
Preconditions:	• Registration or application initiated
Postconditions:	• Documents stored and linked to application
Main Flow:	
Alternate Flow:	N/A

UC-03: Review Registration	
Primary Actor:	FCAU Officer
Description:	Review exporter registration details.
Preconditions:	• Registration submitted
Postconditions:	• Registration approved or rejected
Main Flow:	
Alternate Flow:	N/A

UC-04: Apply for Health Certificate	
Primary Actor:	Exporter
Description:	Apply for export health certification.
Preconditions:	• Exporter registered
Postconditions:	• Application submitted
Main Flow:	• Exporter submits application • System generates application ID

	Application sent for review
Alternate Flow:	N/A

UC-05: Generate Application ID	
Primary Actor:	System
Description:	Generate a unique identifier for the application.
Preconditions:	
Postconditions:	Unique application ID assigned
Main Flow:	
Alternate Flow:	N/A

UC-06: Make Payment	
Primary Actor:	Exporter
Description:	Pay certification or inspection fees.
Preconditions:	Application submitted
Postconditions:	Payment initiated
Main Flow:	
Alternate Flow:	N/A

UC-07: Process Payment	
Primary Actor:	Payment Gateway (External)
Description:	Process payment transaction.
Preconditions:	
Postconditions:	Payment confirmed or failed
Main Flow:	
Alternate Flow:	N/A

UC-08: Review Application	
Primary Actor:	FCAU Officer
Description:	Review health certificate application.
Preconditions:	Application submitted and payment completed
Postconditions:	Application proceeds to sampling decision
Main Flow:	
Alternate Flow:	N/A

UC-09: Determine Sampling Requirement	
Primary Actor:	FCAU Officer / System

Description:	Decide whether sampling is required.
Preconditions:	• Application reviewed
Postconditions:	• Sampling decision recorded
Main Flow:	• System evaluates risk criteria • Determines: ◦ No sampling required ◦ Sampling required
Alternate Flow:	N/A

UC-10: No Sampling	
Primary Actor:	System
Description:	Skip sampling and proceed to compliance verification.
Preconditions:	
Postconditions:	• Application moves directly to compliance verification
Main Flow:	
Alternate Flow:	N/A

UC-11: Sampling Required	
Primary Actor:	FCAU Officer / Field Inspector
Description:	Initiate sampling process.
Preconditions:	
Postconditions:	• Sampling scheduled
Main Flow:	
Alternate Flow:	N/A

UC-12: Conduct Factory Inspection	
Primary Actor:	Field Inspector
Description:	Inspect exporter's facility.
Preconditions:	• Sampling required
Postconditions:	• Inspection findings recorded
Main Flow:	
Alternate Flow:	N/A

UC-13: Collect Sampling	
Primary Actor:	Field Inspector
Description:	Collect product samples.
Preconditions:	

Postconditions:	Samples collected and logged
Main Flow:	
Alternate Flow:	N/A

UC-14: Send Samples to Lab	
Primary Actor:	Field Inspector
Description:	Dispatch samples for laboratory testing.
Preconditions:	
Postconditions:	Samples sent to laboratory
Main Flow:	
Alternate Flow:	N/A

UC-15: Laboratory Processing	
Primary Actor:	Laboratory
Description:	Laboratory Processing
Preconditions:	Sample received
Postconditions:	Submit Lab report
Main Flow:	UC-15A: Receive Sample - Sample received and recorded UC-15B: Conduct Testing - Perform analysis UC-15C: Submit Lab Report - Upload test results
Alternate Flow:	N/A

UC-16: Verify Compliance	
Primary Actor:	FCAU Officer / System
Description:	Verify compliance based on inspection and lab results.
Preconditions:	All required checks completed
Postconditions:	Compliance status determined
Main Flow:	
Alternate Flow:	N/A

UC-17: Issue Health Certificate	
Primary Actor:	FCAU Officer
Description:	Issue certificate for compliant exports.
Preconditions:	Compliance verified
Postconditions:	Health certificate issued

Main Flow:	
Alternate Flow:	N/A

UC-18: Update Customs System	
Primary Actor:	System
Description:	Update customs (ASYCUDA) with export clearance.
Preconditions:	• Certificate issued
Postconditions:	• External system updated
Main Flow:	
Alternate Flow:	N/A

2.7 Registration of Common Salt Manufacturing and Transports Management Module

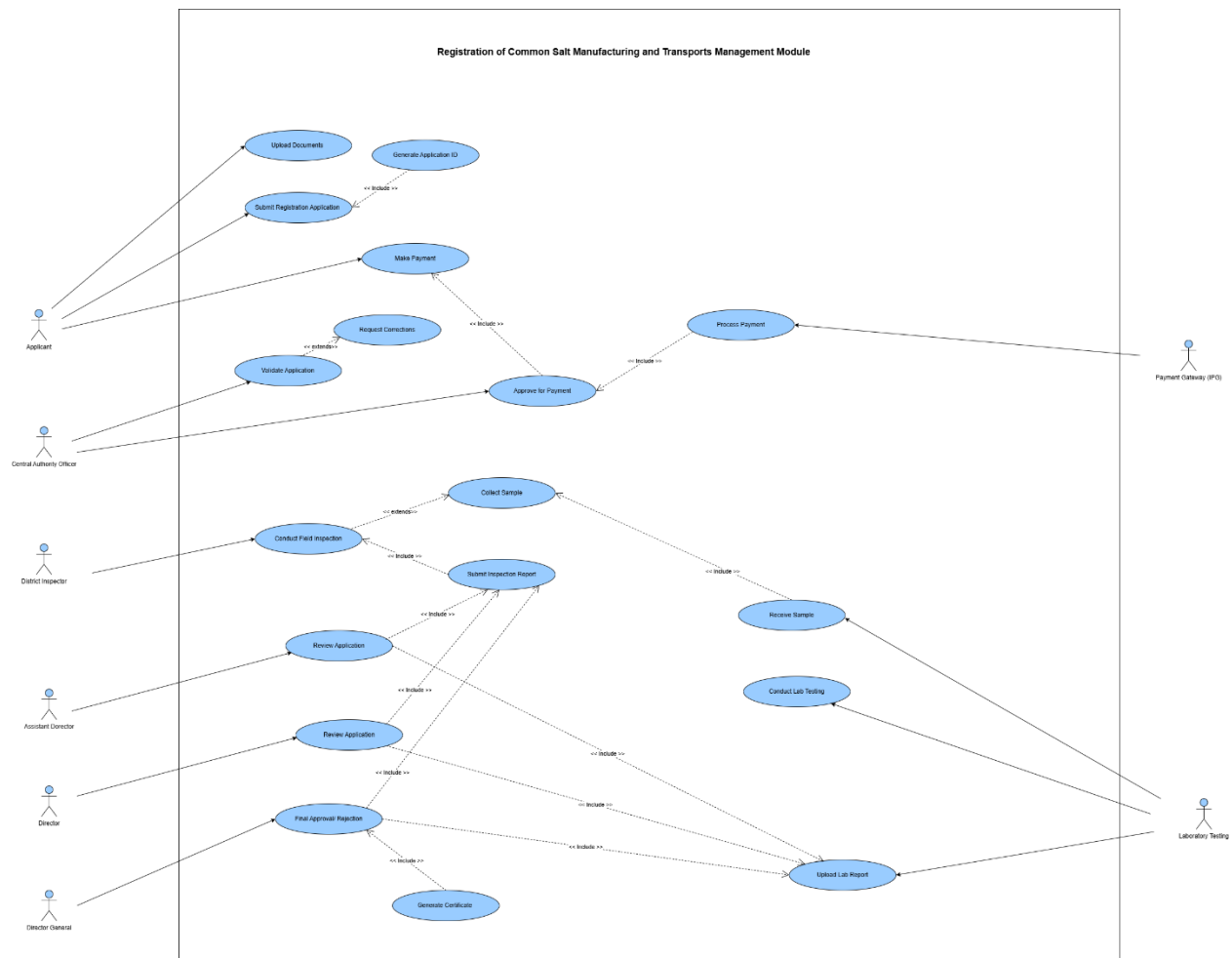


Figure 2.6

Use Case Narratives

UC-01: Submit Registration Application	
Primary Actor:	Applicant
Description:	Submit application for registration of salt manufacturing or transport operations.
Preconditions:	Applicant has access to the system
Postconditions:	Application submitted with unique ID
Main Flow:	- Applicant fills application form - Uploads required documents - System generates application ID - Application submitted
Alternate Flow:	N/A

UC-02: Upload Documents	
Primary Actor:	Applicant
Description:	Upload supporting documents for registration.
Preconditions:	Application initiated
Postconditions:	Documents stored successfully
Main Flow:	
Alternate Flow:	N/A

UC-03: Generate Application ID	
Primary Actor:	System
Description:	Generate unique identifier for application.
Preconditions:	
Postconditions:	Unique ID assigned to application
Main Flow:	
Alternate Flow:	N/A

UC-04: Validate Application	
Primary Actor:	Central Authority Officer
Description:	Validate submitted application details.
Preconditions:	Application submitted
Postconditions:	Application validated or flagged for corrections
Main Flow:	- Officer reviews application - Checks completeness and accuracy

	Approves or requests corrections
Alternate Flow:	N/A

UC-05: Request Corrections	
Primary Actor:	Central Authority Officer
Description:	Request applicant to correct application.
Preconditions:	
Postconditions:	Applicant notified to update application
Main Flow:	
Alternate Flow:	N/A

UC-06: Approve for Payment	
Primary Actor:	Central Authority Officer
Description:	Approve application for payment processing.
Preconditions:	Application validated
Postconditions:	Payment enabled
Main Flow:	
Alternate Flow:	N/A

UC-07: Make Payment	
Primary Actor:	Applicant
Description:	Pay registration fees.
Preconditions:	Application approved for payment
Postconditions:	Payment initiated
Main Flow:	
Alternate Flow:	N/A

UC-08: Process Payment	
Primary Actor:	Payment Gateway (External)
Description:	Process payment transaction.
Preconditions:	
Postconditions:	Payment confirmed or rejected
Main Flow:	
Alternate Flow:	N/A

UC-09: Conduct Field Inspection	
Primary Actor:	District Inspector

Description:	Conduct on-site inspection of facility.
Preconditions:	• Payment completed
Postconditions:	• Inspection conducted
Main Flow:	
Alternate Flow:	N/A

UC-10: Collect Sample	
Primary Actor:	District Inspector
Description:	Collect salt samples during inspection.
Preconditions:	• Inspection in progress
Postconditions:	• Samples collected
Main Flow:	
Alternate Flow:	N/A

UC-11: Submit Inspection Report	
Primary Actor:	District Inspector
Description:	Submit findings from field inspection.
Preconditions:	• Inspection completed
Postconditions:	• Report submitted for review
Main Flow:	
Alternate Flow:	N/A

UC-12: Laboratory Processing	
Primary Actor:	Laboratory Testing
Description:	
Preconditions:	Sample received
Postconditions:	Lab report submitted
Main Flow:	UC-12A: Receive Sample • Sample received and logged UC-12B: Conduct Lab Testing • Perform chemical/quality analysis UC-12C: Upload Lab Report • Upload test results
Alternate Flow:	N/A

UC-13: Review Application (Assistant Director)	
Primary Actor:	Assistant Director
Description:	Review application and inspection report.

Preconditions:	• Inspection report available
Postconditions:	• Recommendation provided
Main Flow:	
Alternate Flow:	N/A

UC-14: Review Application (Director)	
Primary Actor:	Director
Description:	Review application and make decision.
Preconditions:	• Assistant Director review completed
Postconditions:	• Application approved or rejected
Main Flow:	
Alternate Flow:	N/A

UC-15: Final Approval / Rejection	
Primary Actor:	Director General
Description:	Provide final approval or rejection.
Preconditions:	• Director decision available
Postconditions:	• Final decision recorded
Main Flow:	
Alternate Flow:	N/A

UC-16: Generate Certificate	
Primary Actor:	System
Description:	Generate registration certificate.
Preconditions:	• Application approved
Postconditions:	• Certificate issued to applicant
Main Flow:	
Alternate Flow:	N/A

2.8 Bottled Water Manufacturing and Registration Management Module

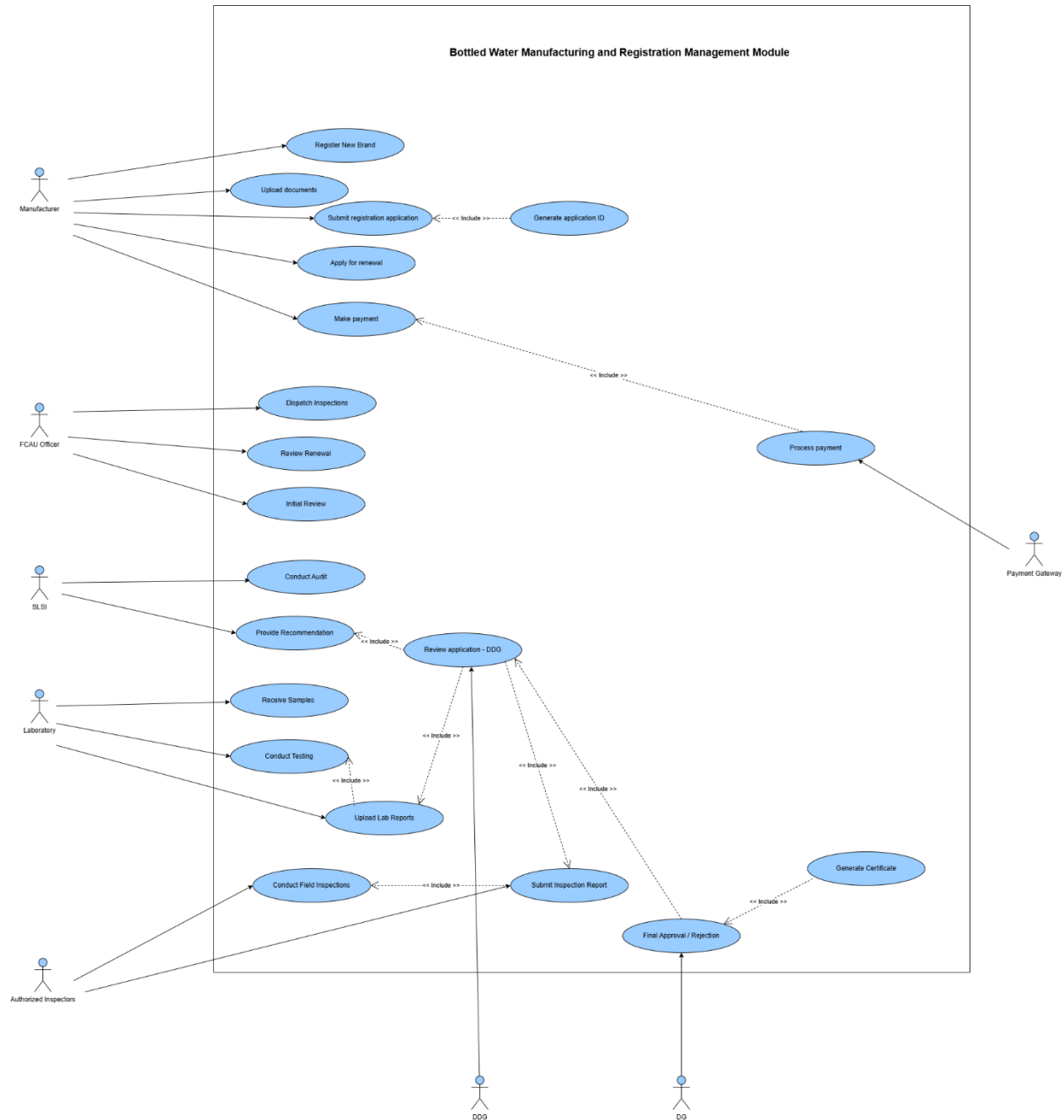


Figure 2.7

Use Case Narratives

UC-01: Register New Brand	
Primary Actor:	Manufacturer
Description:	Register a new bottled water brand in the system.
Preconditions:	Manufacturer is registered in the system
Postconditions:	Brand registration initiated
Main Flow:	- Manufacturer selects “Register New Brand” - Enters brand details - System stores brand information
Alternate Flow:	N/A

UC-02: Upload Documents	
Primary Actor:	Manufacturer
Description:	Upload required documents for registration or renewal.
Preconditions:	Application initiated
Postconditions:	Documents uploaded and stored
Main Flow:	
Alternate Flow:	N/A

UC-03: Submit Registration Application	
Primary Actor:	Manufacturer
Description:	Submit application for bottled water registration.
Preconditions:	Documents uploaded
Postconditions:	Application submitted with ID
Main Flow:	- Manufacturer fills application form - System generates application ID - Application submitted for review
Alternate Flow:	N/A

UC-04: Generate Application ID	
Primary Actor:	System
Description:	Generate unique identifier for application.
Preconditions:	
Postconditions:	Application ID created

Main Flow:	
Alternate Flow:	N/A

UC-05: Apply for Renewal	
Primary Actor:	Manufacturer
Description:	Apply to renew existing registration.
Preconditions:	Existing valid registration
Postconditions:	Renewal application submitted
Main Flow:	
Alternate Flow:	N/A

UC-06: Make Payment	
Primary Actor:	Manufacturer
Description:	Pay registration or renewal fees.
Preconditions:	Application submitted
Postconditions:	Payment initiated
Main Flow:	
Alternate Flow:	N/A

UC-07: Process Payment	
Primary Actor:	Payment Gateway (External)
Description:	Process payment transaction.
Preconditions:	
Postconditions:	Payment confirmed or failed
Main Flow:	
Alternate Flow:	N/A

UC-08: Initial Review	
Primary Actor:	FCAU Officer
Description:	Conduct preliminary review of application.
Preconditions:	Application submitted
Postconditions:	Application moves to inspection stage or requires clarification
Main Flow:	
Alternate Flow:	N/A

UC-09: Review Renewal	
-----------------------	--

Primary Actor:	FCAU Officer
Description:	Review renewal applications.
Preconditions:	Renewal request submitted
Postconditions:	Renewal proceeds for further processing
Main Flow:	
Alternate Flow:	N/A

UC-10: Dispatch Inspections	
Primary Actor:	FCAU Officer
Description:	Assign inspections to authorized inspectors.
Preconditions:	Application under review
Postconditions:	Inspection scheduled
Main Flow:	
Alternate Flow:	N/A

UC-11: Conduct Field Inspections	
Primary Actor:	Authorized Inspectors
Description:	Perform on-site inspection of manufacturing facility.
Preconditions:	Inspection assigned
Postconditions:	Inspection findings recorded
Main Flow:	
Alternate Flow:	N/A

UC-12: Submit Inspection Report	
Primary Actor:	Authorized Inspectors
Description:	Submit inspection findings.
Preconditions:	Inspection completed
Postconditions:	Report submitted to system
Main Flow:	
Alternate Flow:	N/A

UC-13: Laboratory Processing	
Primary Actor:	Laboratory
Description:	Lab processing
Preconditions:	Receive sample
Postconditions:	Submit Report
Main Flow:	UC-13A: Receive Samples Samples received

	UC-13B: Conduct Testing - Perform quality testing UC-13C: Upload Lab Reports - Upload test results
Alternate Flow:	N/A

UC-14: Conduct Audit	
Primary Actor:	SLSI
Description:	Conduct audit on manufacturing processes.
Preconditions:	Application under review
Postconditions:	Audit completed
Main Flow:	
Alternate Flow:	N/A

UC-15: Provide Recommendation	
Primary Actor:	SLSI
Description:	Provide recommendation based on audit.
Preconditions:	Audit completed
Postconditions:	Recommendation submitted
Main Flow:	
Alternate Flow:	N/A

UC-16: Review Application (DDG)	
Primary Actor:	DDG
Description:	Review application, inspection, and lab reports.
Preconditions:	All reports and recommendations available
Postconditions:	Application forwarded for final decision
Main Flow:	
Alternate Flow:	N/A

UC-17: Final Approval / Rejection	
Primary Actor:	DG
Description:	Make final decision on application.
Preconditions:	DDG review completed
Postconditions:	Application approved or rejected
Main Flow:	
Alternate Flow:	N/A

UC-18: Generate Certificate	
Primary Actor:	System
Description:	Generate registration certificate.
Preconditions:	• Application approved
Postconditions:	• Certificate issued
Main Flow:	
Alternate Flow:	N/A

3. Sequence Modeling

Sequence modeling is a dynamic design technique used to represent how system components interact over time to execute a specific functionality. Within the context of this Software Design Document (SDD), sequence modeling provides a time-ordered view of interactions between actors, system components, and microservices, illustrating how individual use cases are realized step by step.

Unlike static models, sequence models focus on behavioral flow, showing how messages are exchanged between different entities in a chronological sequence. These entities may include users (actors), application services, APIs, databases, and external systems. By capturing these interactions, sequence modeling helps in understanding not just *what* the system does, but *how* it performs operations internally.

Sequence diagrams, which are the primary artifact of sequence modeling, typically include:

- **Actors and Lifelines** – Represent users and system components involved in the interaction
- **Messages** – Show communication between components (synchronous or asynchronous)
- **Activation Bars** – Indicate processing time within a component
- **Control Structures** – Such as loops, conditions, and parallel processing

In a microservices-based architecture, sequence modeling plays a critical role in visualizing service orchestration and communication patterns. It helps define:

- API interactions between services
- Event-driven communication (e.g., message queues, publish–subscribe patterns)
- Data flow across distributed services
- Error handling and fallback mechanisms

Sequence models are especially valuable for complex workflows such as application submission, inspection processes, lab coordination, enforcement actions, and import/export handling. They ensure clarity in system behavior, reduce ambiguity in implementation, and support developers in designing efficient and scalable interactions.

Overall, sequence modeling acts as a bridge between use case narratives and technical implementation, enabling a precise and shared understanding of runtime system behavior among architects, developers, and stakeholders.

3.1 Premises Registration and Lifecycle Management Module

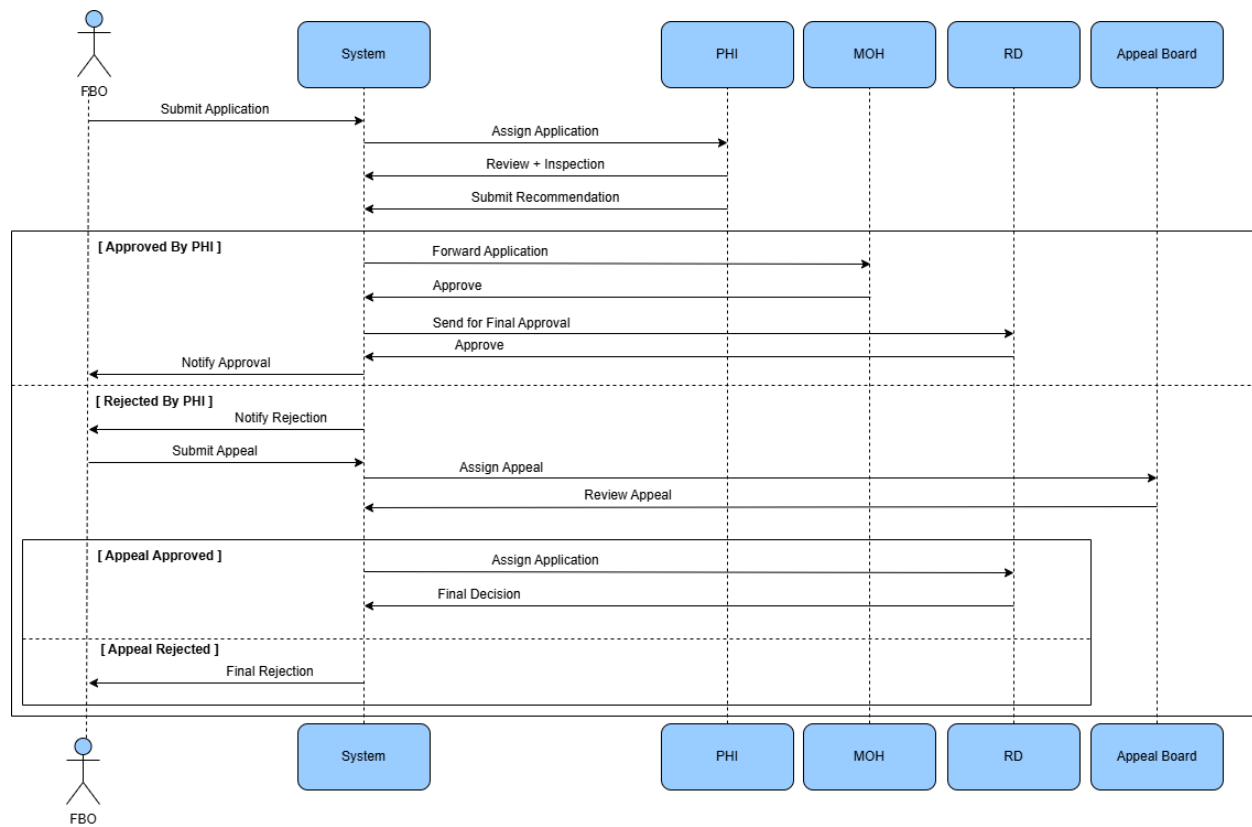


Figure 3.0

3.2 Inspection and Grading Module

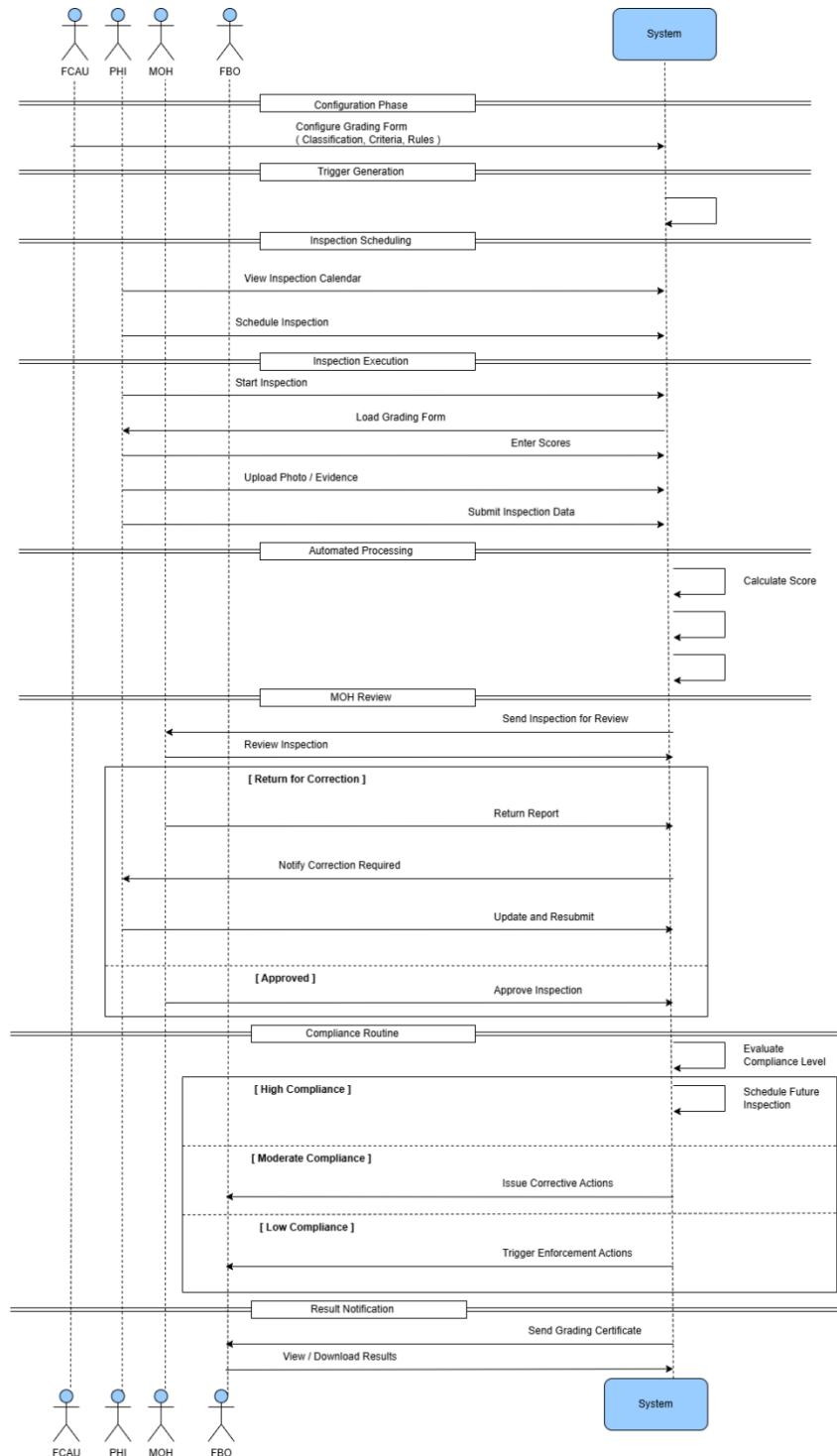


Figure 3.1

3.3 Laboratory Coordination Module

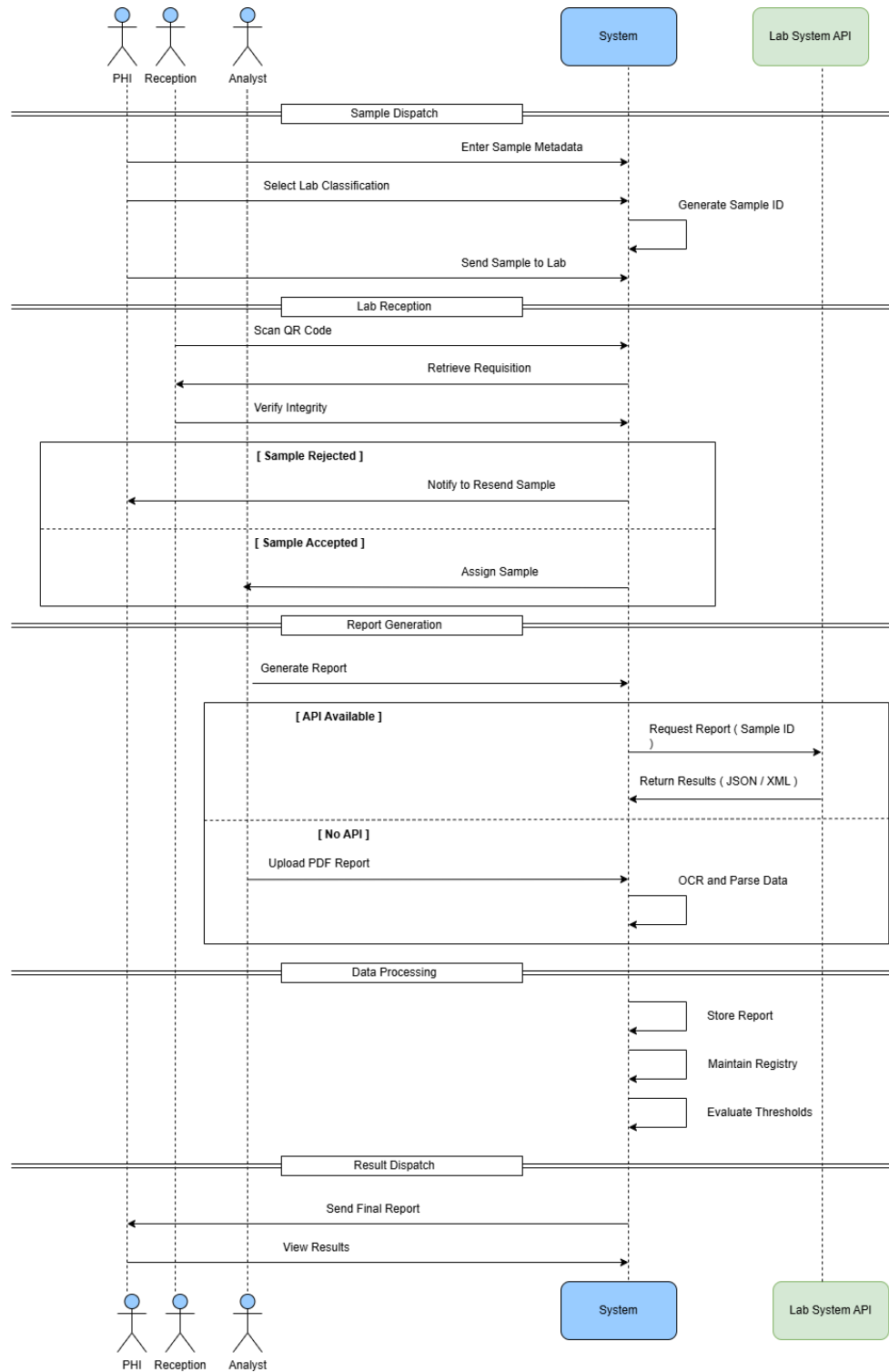


Figure 3.2

3.4 Legal, Prosecution and Enforcement Management Module

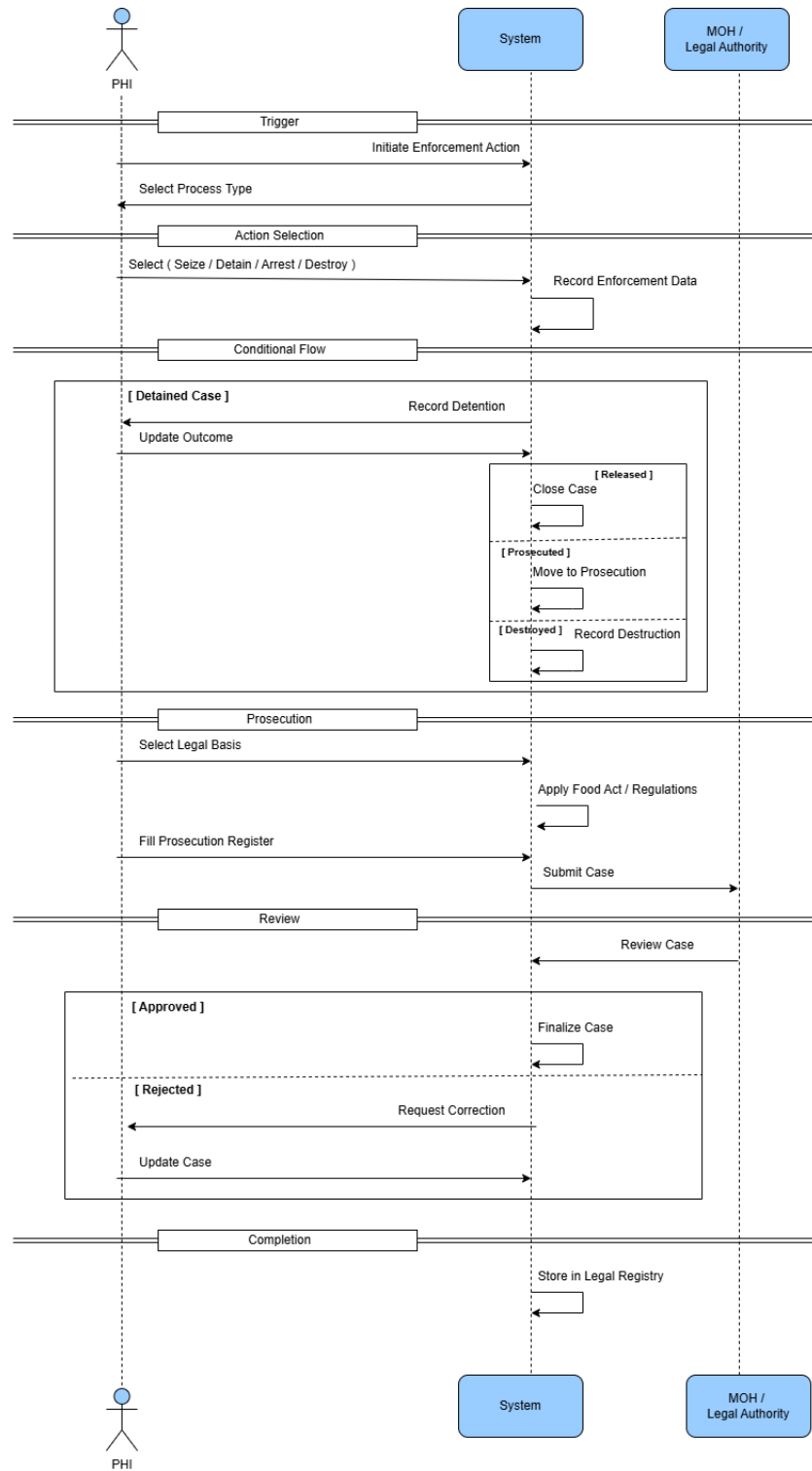


Figure 3.3

3.5 Import Control Module

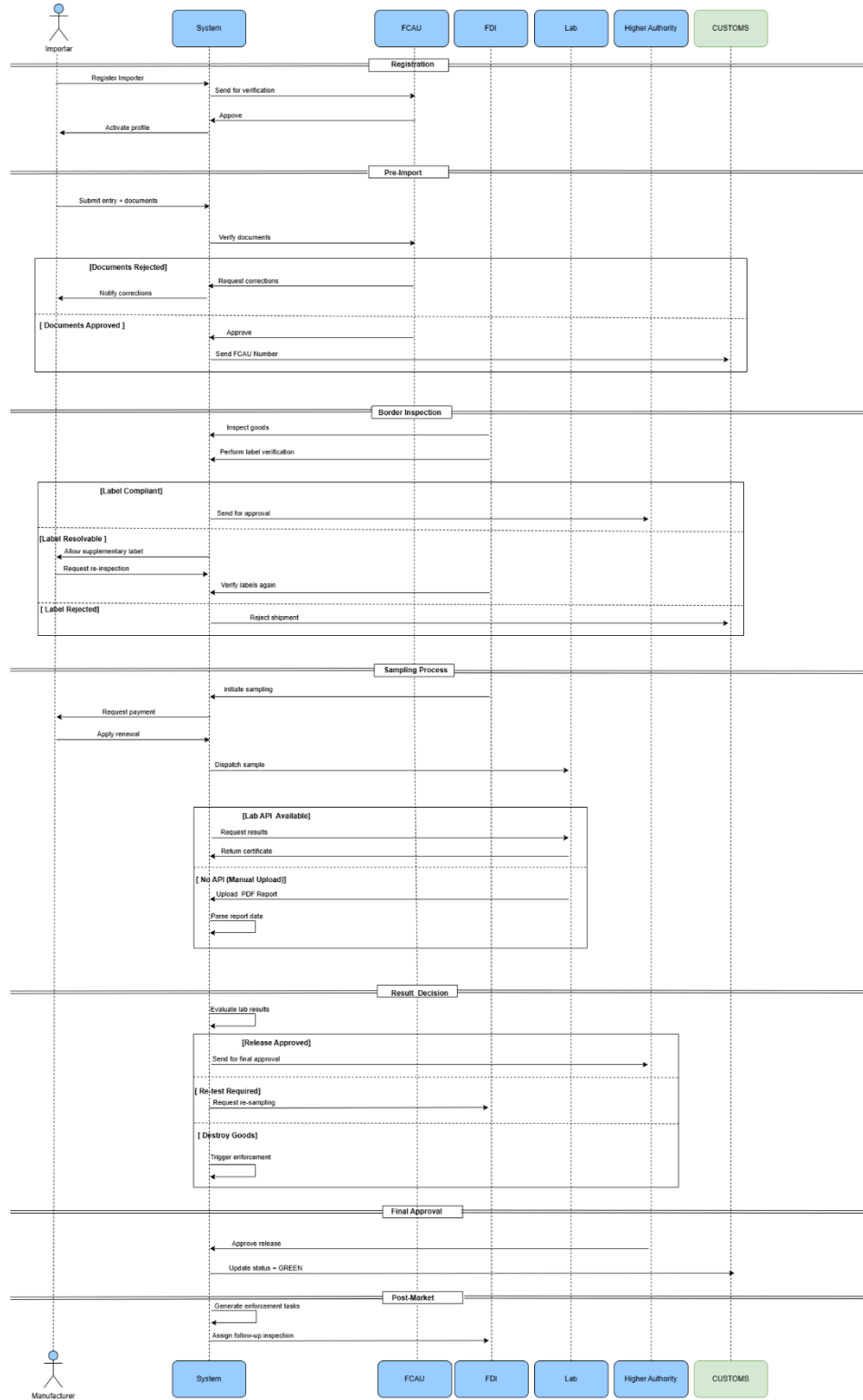


Figure 3.4

3.6 Export Control Module

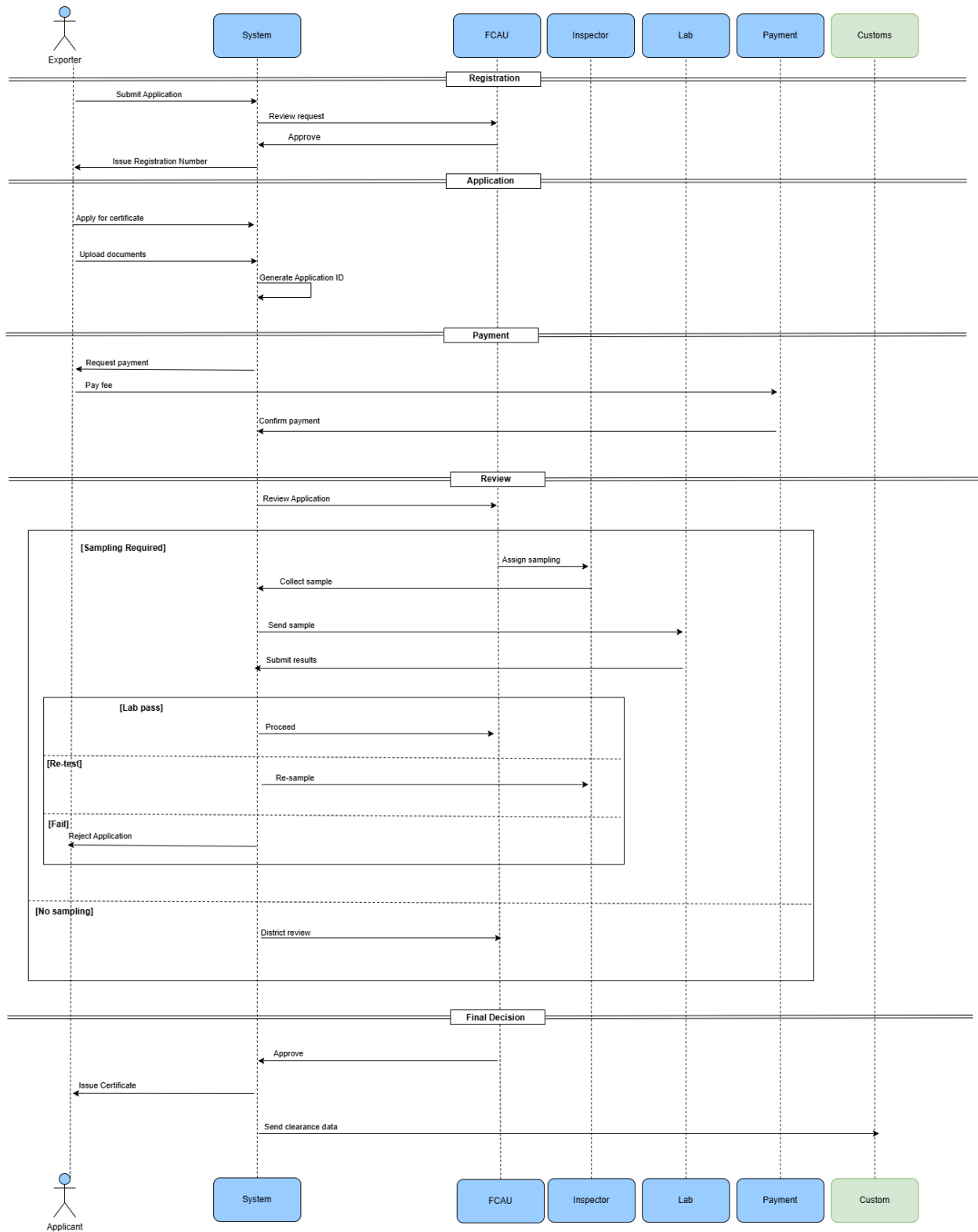


Figure 3.5

3.7 Registration of Common Salt Manufacturing and Transports Management Module

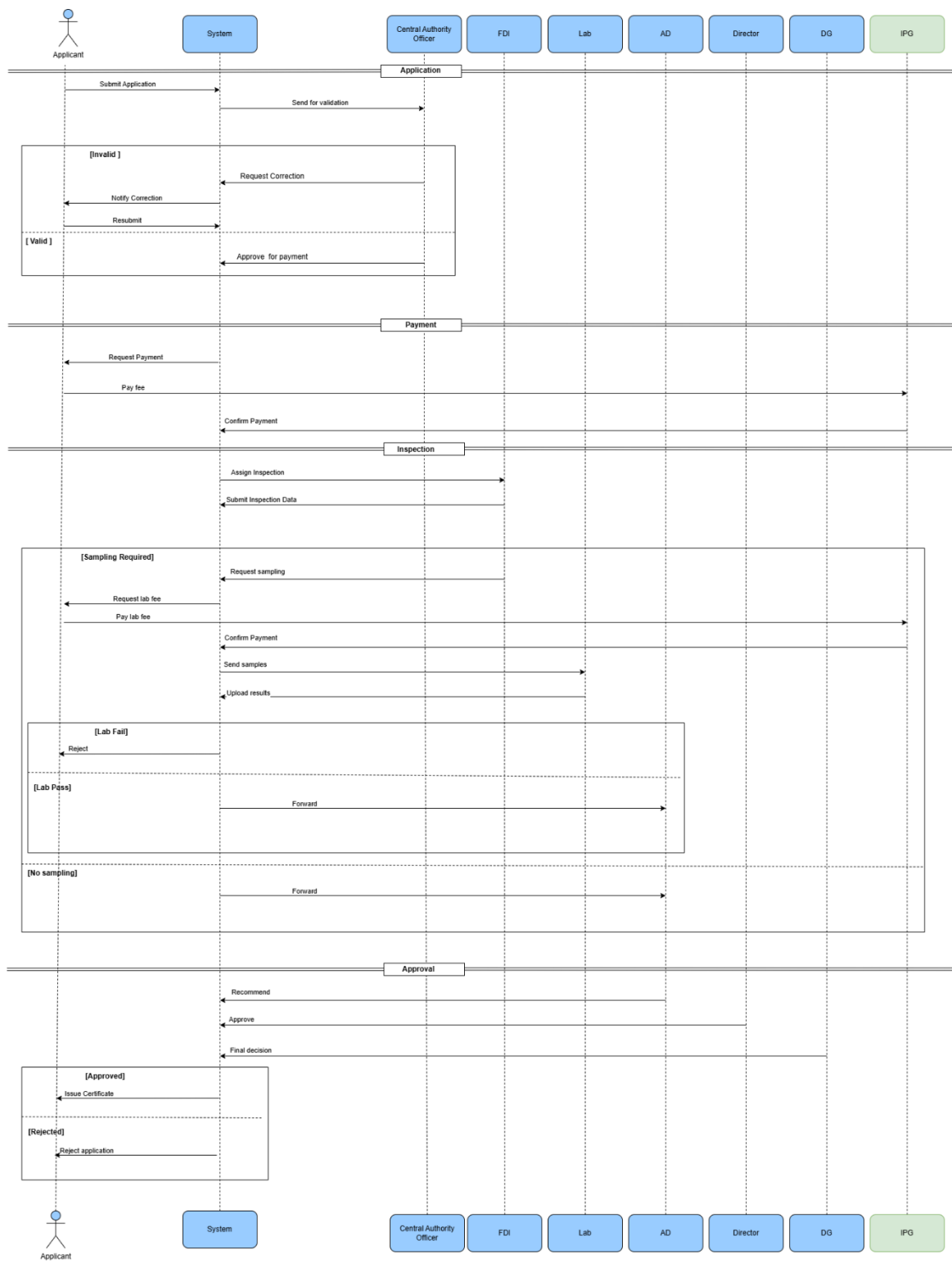


Figure 3.6

3.8 Bottled Water Manufacturing and Registration Management Module

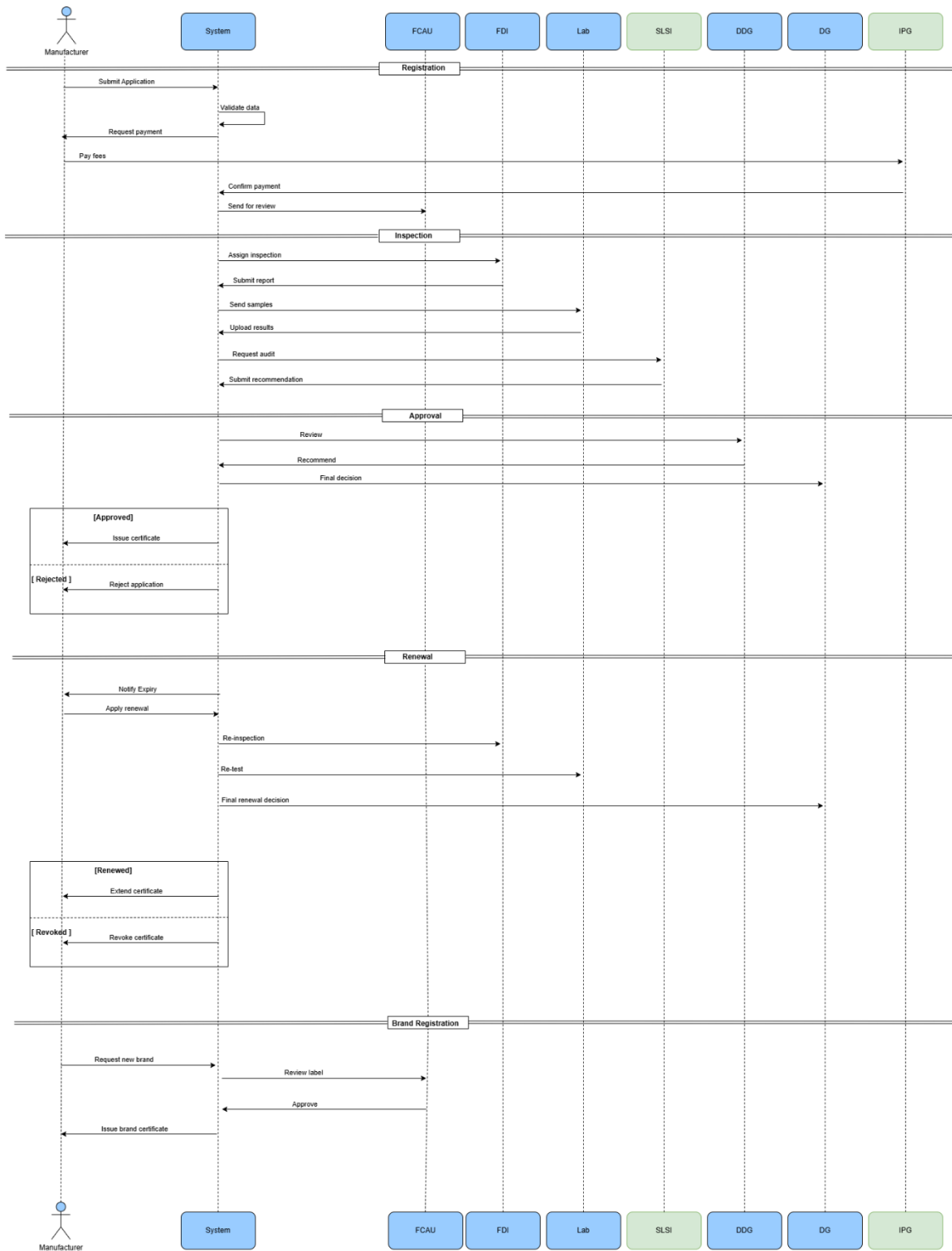


Figure 3.7

4. Activity Modeling

Activity modeling is a behavioral modeling technique used to represent the flow of control and activities within a system, capturing how processes are executed from start to end. In this Software Design Document (SDD), activity modeling provides a clear visualization of business processes, system workflows, and decision logic across different modules.

Unlike sequence modeling, which focuses on interactions over time, activity modeling emphasizes the overall process flow, including parallel activities, decision points, and transitions between different states. It is particularly useful for illustrating complex workflows that involve multiple actors, conditional paths, and iterative steps.

Activity diagrams, the primary artifact of activity modeling typically include:

- **Initial and Final Nodes** – Indicating the start and end of a process
- **Activities/Actions** – Representing individual steps or tasks
- **Control Flows** – Showing the sequence of execution
- **Decision Nodes** – Capturing branching logic based on conditions
- **Merge Nodes** – Combining alternative flows
- **Fork and Join Nodes** – Representing parallel processing

In the context of a microservices based architecture, activity modeling helps to:

- Visualize end-to-end business workflows across multiple services
- Identify service boundaries and responsibilities
- Highlight parallel processing opportunities for performance optimization
- Define decision logic and exception handling paths
- Support orchestration and choreography of distributed services

Activity models are especially valuable for processes such as registration workflows, inspection and grading procedures, lab coordination, enforcement handling, and import/export operations.

Overall, activity modeling serves as a bridge between business process understanding and system design, ensuring that workflows are logically structured, efficient, and aligned with system requirements. It enhances communication among stakeholders and provides a solid foundation for developing robust and scalable system implementations.

4.1 Premises Registration and Lifecycle Management Module

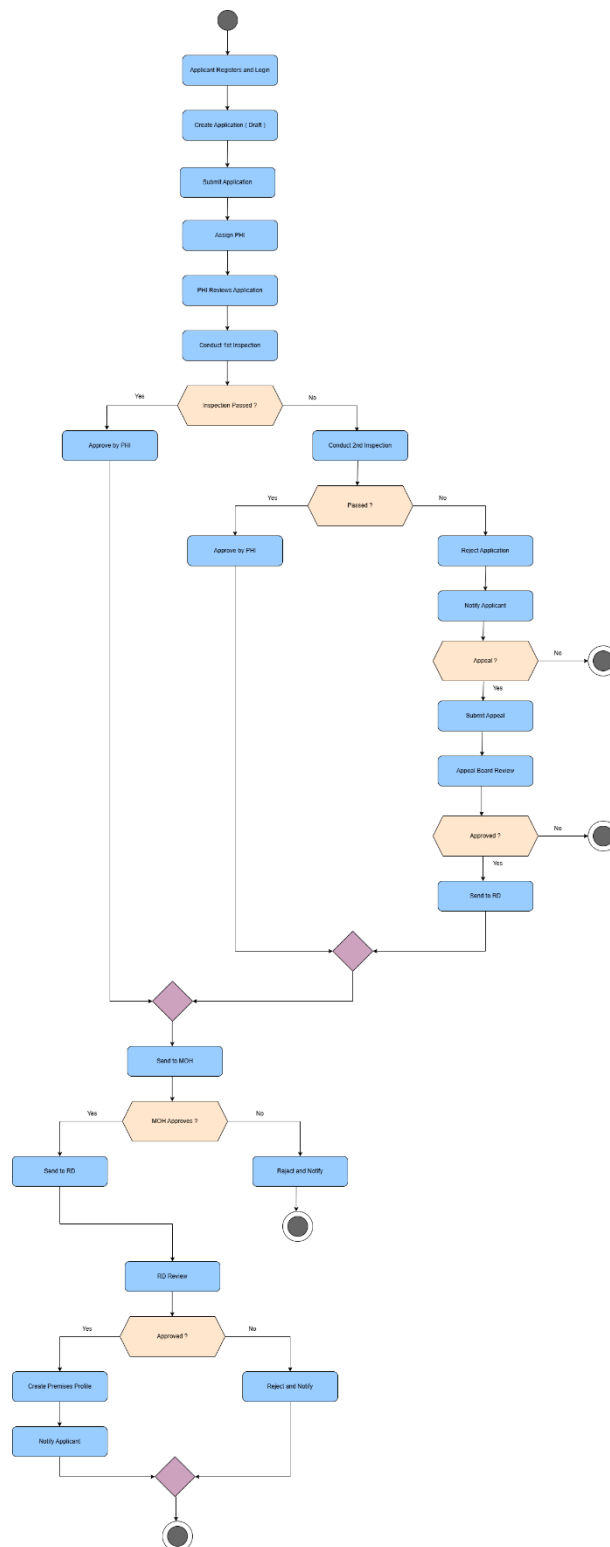


Figure 4.0

4.2 Inspection and Grading Module

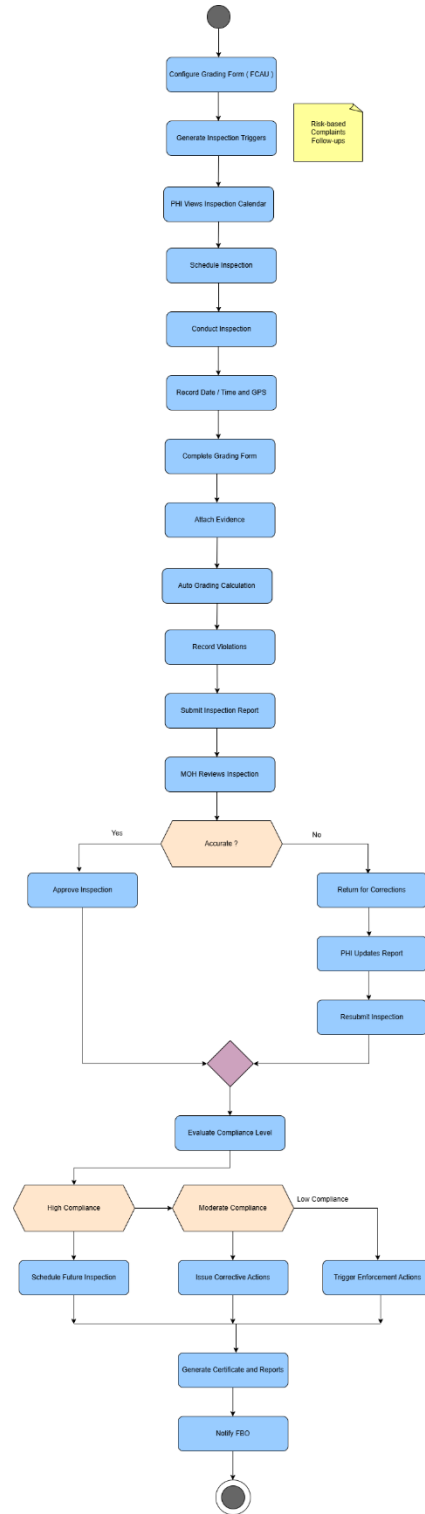


Figure 4.1

4.3 Laboratory Coordination Module

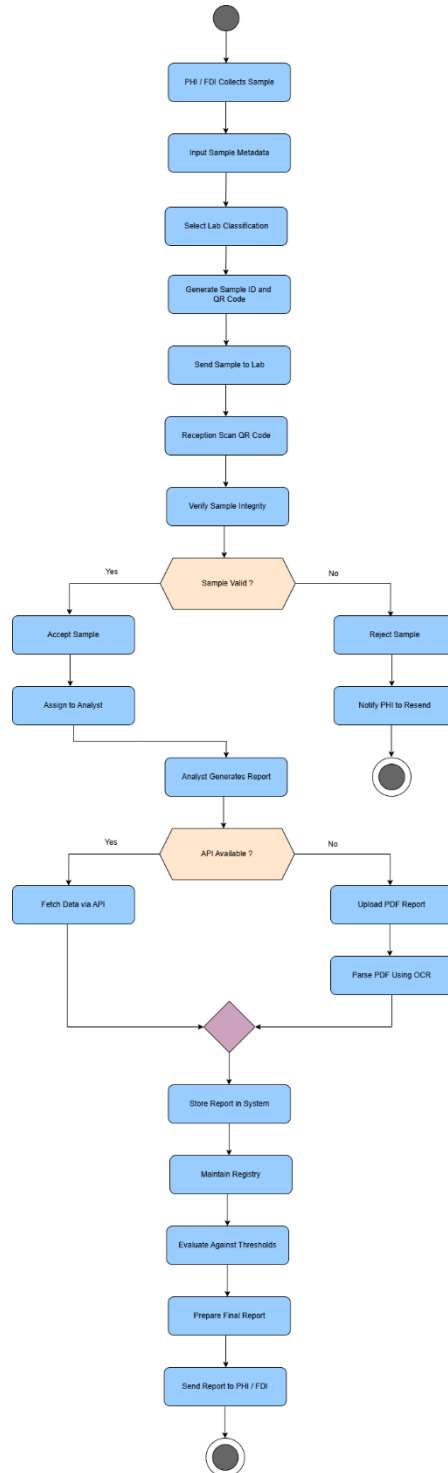


Figure 4.2

4.4 Legal, Prosecution and Enforcement Management Module

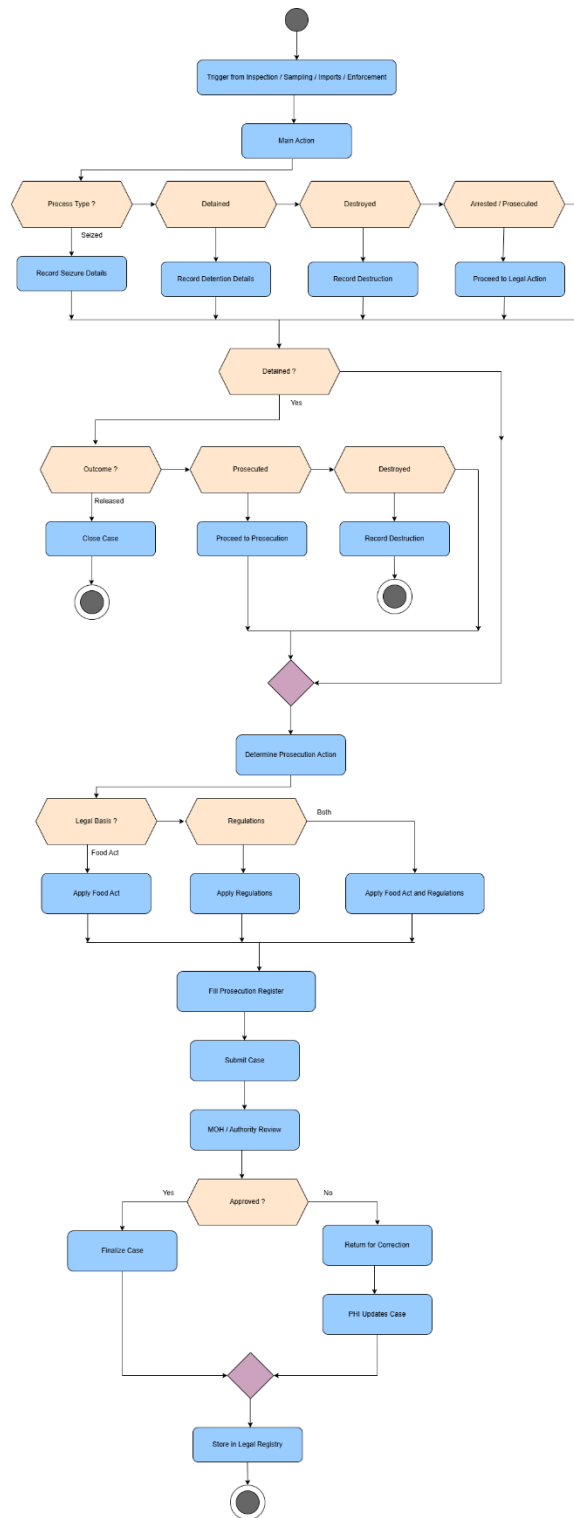


Figure 4.3

4.5 Import Control Module

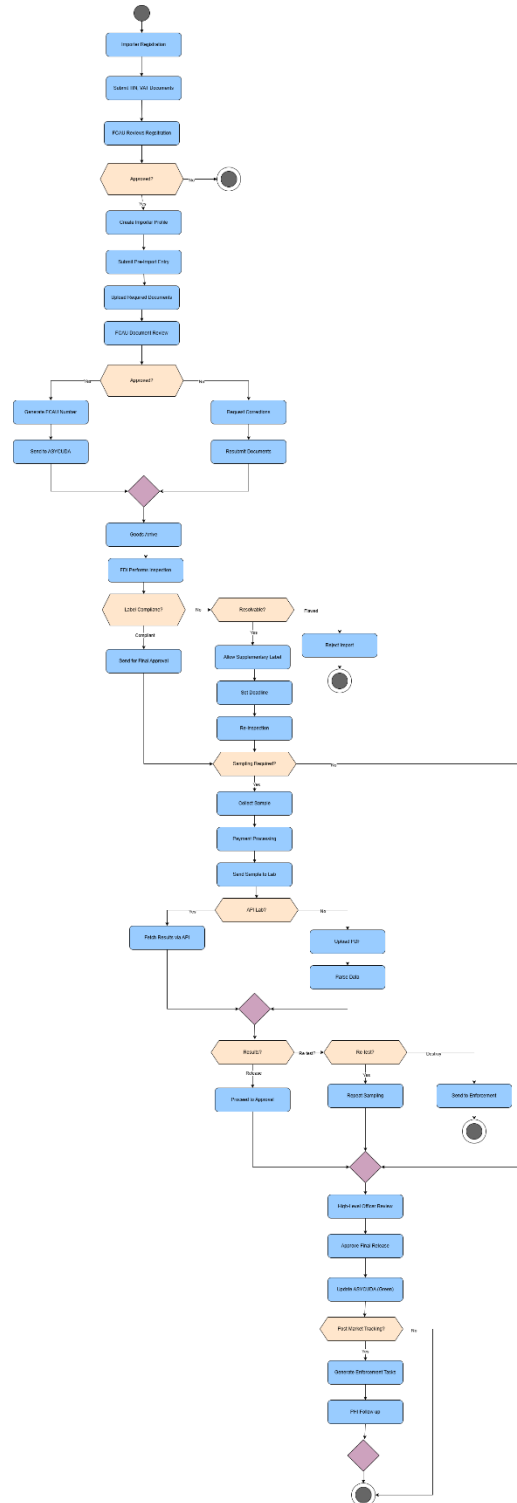


Figure 4.4

4.6 Export Control Module

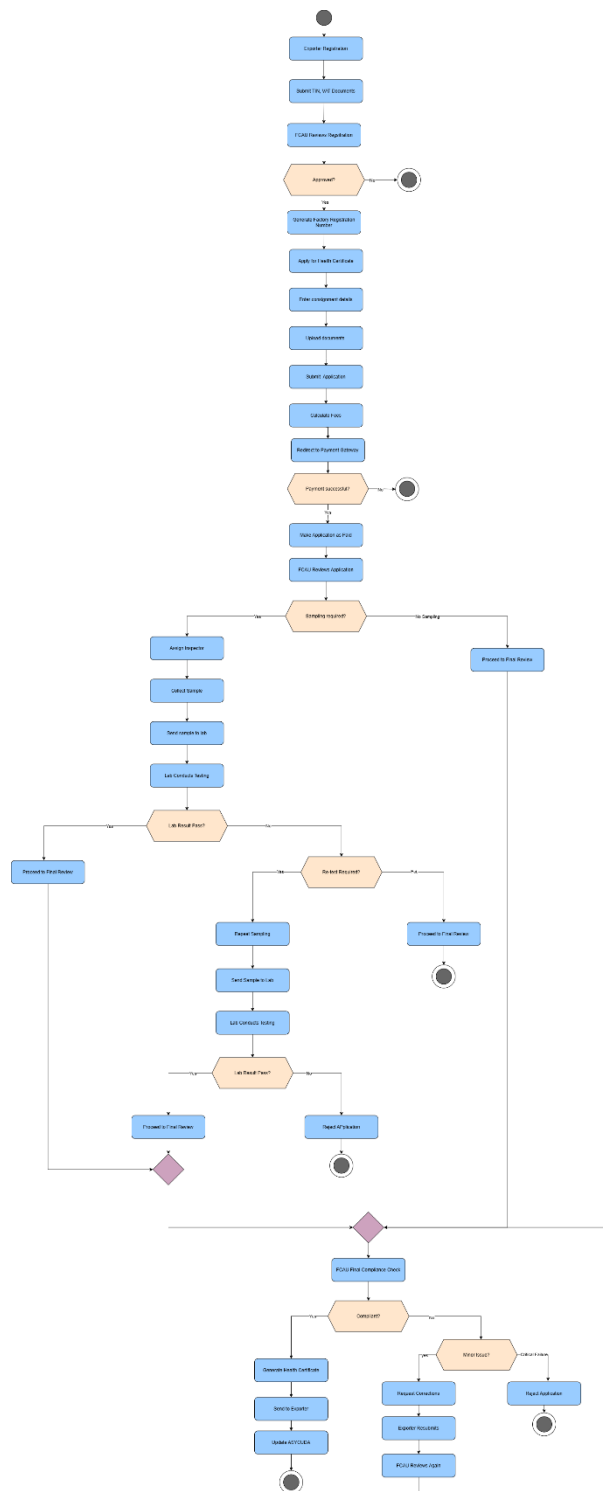


Figure 4.5

4.7 Registration of Common Salt Manufacturing and Transports Management Module

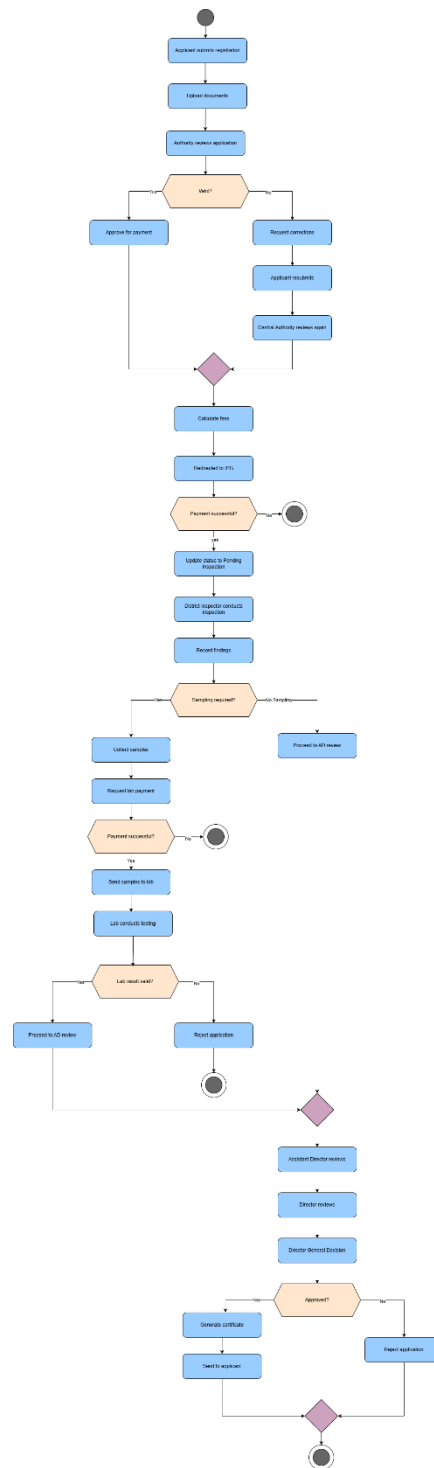


Figure 4.6

4.8 Bottled Water Manufacturing and Registration Management Module

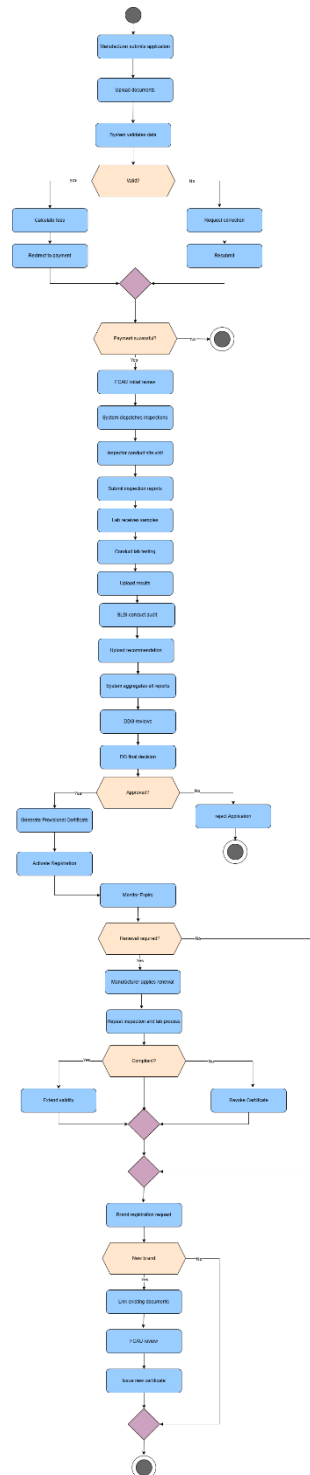


Figure 4.7

5. System Architecture

The proposed system architecture for the Interoperable Risk-Based Food Safety Inspection Information Management System (IIMS) is designed as a scalable, resilient, and service-oriented architecture that supports nationwide food safety regulatory operations. It adopts a microservices-based approach combined with containerization, API-driven communication, and event-based processing, enabling the system to efficiently manage complex workflows such as registration, inspection, laboratory coordination, enforcement, and reporting.

This architecture is aligned with the SRS objective of creating a centralized yet interoperable platform capable of integrating multiple stakeholders, external systems, and regulatory processes into a unified digital ecosystem. The design ensures high availability, fault tolerance, and extensibility, while supporting real-time operations and asynchronous processing where required.

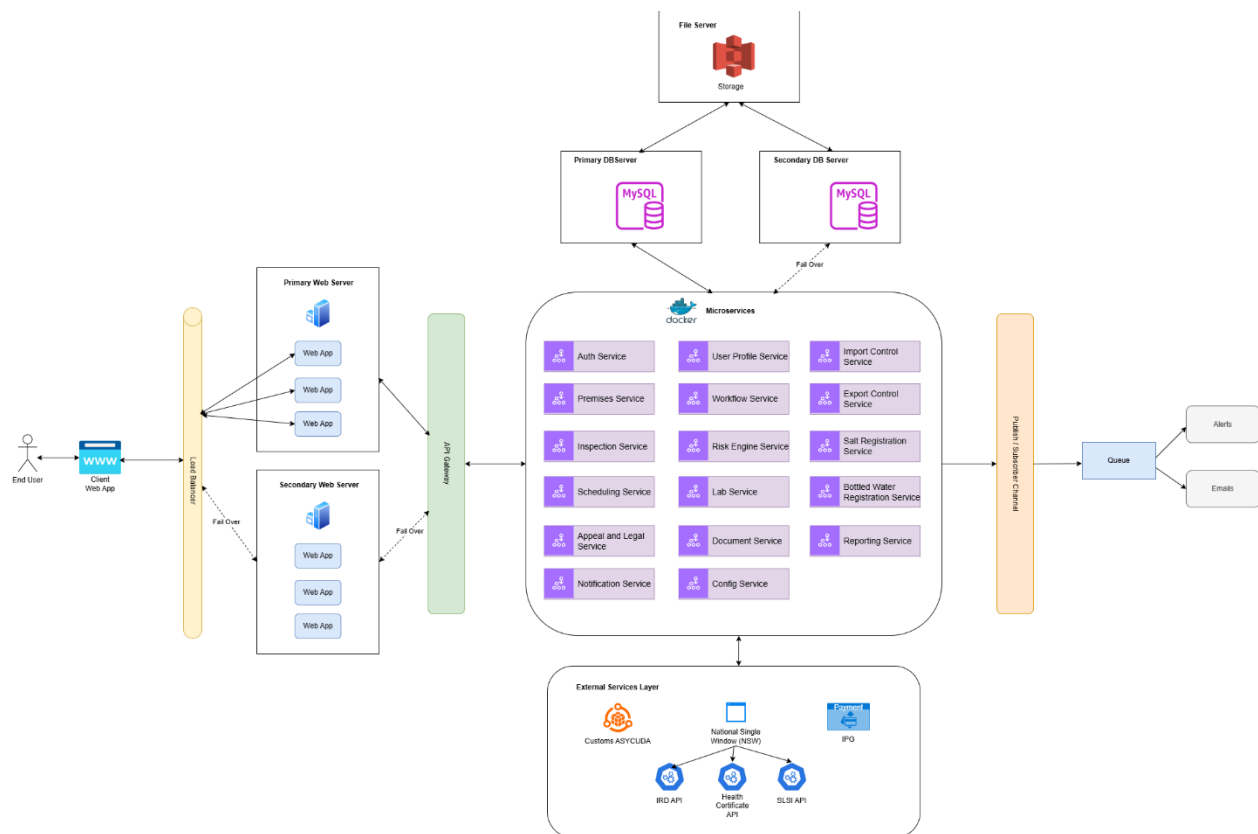


Figure 5.0

5.1 Comprehensive Explanation of the Proposed Architecture

The architecture is composed of multiple layers and components, each serving a specific responsibility. It follows a layered and distributed design, ensuring separation of concerns and operational efficiency.

5.1.1 Client and Access Layer

At the entry point of the system:

- End Users (FBOs, PHIs, MOHs, FCAU Admins, Public users etc)
- Access the system through a Web Application (Client Layer)

Key Characteristics:

- Browser-based access (no frontend framework, server-rendered UI)
- Supports multiple user roles
- Mobile-responsive for field inspections

5.1.2 Load Balancer Layer

- Acts as the first control point for incoming traffic
- Distributes requests across multiple web servers

Key Functions:

- Load distribution
- High availability
- Failover handling (redirects traffic if a server is down)

Ensures the system remains accessible under high traffic conditions.

5.1.3 Web Server Layer

Two main components:

- Primary Web Server
- Secondary Web Server (Failover)

Responsibilities:

- Hosts Laravel-based web applications
- Handles:
 - User interface rendering
 - Session handling
 - Initial request processing

Key Benefits:

- Redundancy (failover support)
- Scalability (horizontal scaling possible)

5.1.4 API Gateway Layer

- Serves as the single-entry point to backend microservices

Responsibilities:

- Routes requests to appropriate services
- Handles:
 - Authentication validation
 - Rate limiting
 - Request aggregation
 - Logging

Benefits:

- Decouples frontend from backend services
- Simplifies client interactions

- Enhances security and governance

5.1.5 Microservices Layer (Core System)

This is the heart of the architecture, deployed using Docker containers.

Key Services (as shown in diagram):

Core Services

- Auth Service
- User Profile Service
- Premises Service
- Workflow Service
- Inspection Service
- Risk Engine Service
- Scheduling Service
- Lab Service

Regulatory Services

- Import Control Service
- Export Control Service
- Salt Registration Service
- Bottled Water Registration Service

Support Services

- Appeal & Legal Service
- Document Service
- Notification Service
- Reporting Service
- Config Service

Key Characteristics:

- Each service is independently deployable
- Domain-driven design (aligned with SRS modules)
- Communicates via:
 - REST APIs
 - Internal service calls

Benefits:

- High modularity
- Independent scaling
- Easier maintenance and upgrades

5.1.6 Database Layer

- Primary MySQL Database Server
- Secondary MySQL Database Server (Failover/Replication)

Features:

- Centralized relational database
- Data replication between primary and secondary
- Automatic failover capability

Ensures data consistency, durability, and availability

5.1.7 File Storage Layer

- Dedicated File Server (Object Storage)

Stores:

- Uploaded documents
- Inspection images
- Lab reports

- Certificates

Benefits:

- Scalable storage
- Decouples file handling from database
- Supports secure file access

5.1.8 Messaging & Event Processing Layer

Components:

- Publish/Subscribe Channel
- Queue System

Functions:

- Handles asynchronous processing for:
 - Notifications
 - Email delivery
 - Alerts
 - Background jobs

Outputs:

- Alerts
- Emails

Benefits:

- Non-blocking operations
- Improved performance
- Reliable event handling

5.1.9 External Services Layer

The system integrates with multiple external systems:

- Customs ASYCUDA → import/export validation
- National Single Window (NSW) → cross-agency coordination
- IRD API → tax/registration validation
- Health Certificate API
- SLSI API → standards compliance
- IPG (Payment Gateway) → payments

Role:

- Enables interoperability with national systems
- Supports regulatory compliance and verification

The proposed architecture provides a robust, scalable, and future-ready foundation for the IIMS. By combining microservices, API gateway, event-driven processing, and high-availability infrastructure, the system is well-equipped to handle complex regulatory workflows, large-scale data processing, and multi-agency integration, ensuring efficient and reliable food safety governance across Sri Lanka.

6. CI/CD Pipeline Design

The CI/CD (Continuous Integration and Continuous Deployment) pipeline for the IIMS is designed to ensure rapid, reliable, and repeatable delivery of software across all microservices. Given the architecture (Laravel-based microservices, Docker containers, API Gateway, and shared database), the pipeline emphasizes:

- Automation of build, test, and deployment processes
- Consistency across all services
- Environment isolation (Dev → QA → UAT → Production)
- High availability with zero/minimal downtime deployments

This pipeline supports independent service deployments while maintaining overall system integrity.

6.1 High-Level CI/CD Flow

The CI/CD pipeline for the system follows a structured and automated sequence to ensure reliable and efficient software delivery. The process begins with a code commit, where developers push changes to the source repository, triggering the pipeline. In the build stage, the system compiles the application and installs all required dependencies to prepare it for execution. This is followed by the testing phase, where automated unit, integration, and API tests are executed to validate functionality and ensure that new changes do not introduce defects. Next, a security scan is performed to identify vulnerabilities in dependencies, code, and configurations, ensuring compliance with security standards. Once validated, the application is packaged during the Docker build stage, creating a containerized image that guarantees consistency across environments. This image is then uploaded to a container registry, making it available for deployment. In the deployment stage, the latest stable image is released to the target environment (such as development, QA, or production) using automated deployment strategies like rolling updates. Finally, the pipeline concludes with a verification stage, where health checks, smoke tests, and monitoring mechanisms confirm that the application is functioning correctly in the deployed environment. This end-to-end automated workflow ensures faster delivery, improved quality, and reduced deployment risks.

Code Commit → Build → Test → Security Scan → Docker Build → Push Registry → Deploy → Verify

6.2 Pipeline Architecture Overview

Core Components

- Source Control: Git (GitHub / GitLab / Bitbucket)
- CI/CD Tool: GitHub Actions / GitLab CI / Jenkins
- Containerization: Docker
- Container Registry: Docker Hub / GitLab Registry
- Deployment: Kubernetes / VM-based
- Infrastructure: Cloud (As provided by the Client)

6.3 Pipeline Stages

6.3.1 Source Code Management (SCM)

Strategy:

- Branching model:
 - main → production
 - develop → integration
 - feature/* → new features
 - hotfix/* → urgent fixes

Trigger:

- Push / Pull Request → triggers pipeline

6.3.2 Build Stage

Activities:

- Install dependencies: `composer install`
- Environment validation
- Code compilation (if needed)

6.3.3 Code Quality & Static Analysis

Tools:

- Laravel

Purpose:

- Detect bugs, vulnerabilities, and code smells early

6.3.4 Testing Stage

Types:

- Unit Tests (PHPUnit)
- Feature Tests
- API Tests

Requirement:

- Minimum coverage threshold (70%)

6.3.5 Security Scanning

Checks:

- Dependency vulnerabilities (e.g., Composer audit)
- Secrets detection
- Container scanning

6.3.6 Docker Build Stage and Push to Container Registry

Each microservice builds its own image:

```
docker build -t fcav/premises-service:latest .
```

```
docker push fcav/premises-service:latest
```

Best Practices:

- Use multi-stage builds
- Keep images lightweight
- Use .env only at runtime

6.3.7 Deployment Stage

Strategy:

- Environment-based deployment:

Environment	Purpose
Dev	Development testing
QA	Functional testing
UAT	Stakeholder validation
Prod	Live system

6.3.8 Post-Deployment Verification

- Health checks (/health endpoint)
- Smoke tests
- API validation

6.3.9 Rollback Mechanism

- Revert to previous Docker image
- DB rollback (if safe)

7. OpenAPI 3.0 Design

7.1 AUTH SERVICE

openapi: 3.0.3

info:

title: FCAU Auth Service API

version: 1.0.0

servers:

- url: https://api.fcau.gov.lk/auth

components:

securitySchemes:

bearerAuth:

type: http

scheme: bearer

bearerFormat: JWT

paths:

/register:

post:

summary: Register user

tags: [Auth]

requestBody:

required: true

content:

application/json:

schema:

type: object

required: [full_name, email, password, role]

properties:

```
        full_name: { type: string }
        email: { type: string }
        password: { type: string }
        role: { type: string }
    responses:
        "200": { description: Registered }

/login:
    post:
        summary: Login
        tags: [Auth]
        requestBody:
            content:
                application/json:
                    schema:
                        type: object
                        properties:
                            email: { type: string }
                            password: { type: string }
        responses:
            "200": { description: JWT issued }

/verify:
    post:
        summary: Verify MFA
        tags: [Auth]
        responses:
            "200": { description: Verified }

/logout:
    post:
        summary: Logout
        security: [{ bearerAuth: [] }]
```

```
tags: [Auth]

responses:
  "200": { description: Logged out }
```

7.2 USER PROFILE SERVICE

openapi: 3.0.3

info:

title: User Profile Service

version: 1.0.0

servers:

- url: https://api.fcau.gov.lk/users

paths:

/me:

get:

summary: Get current user

security: [{ bearerAuth: [] }]

responses:

"200": { description: User profile }

put:

summary: Update profile

security: [{ bearerAuth: [] }]

responses:

"200": { description: Updated }

/assign-role:

post:

summary: Assign role

security: [{ bearerAuth: [] }]

```
responses:
  "200": { description: Role assigned }
```

7.3 PREMISES SERVICE

```
openapi: 3.0.3
```

```
info:
```

```
  title: Premises Registration Service
  version: 1.0.0
```

```
servers:
```

```
  - url: https://api.fcau.gov.lk/premises
```

```
paths:
```

```
  /:
```

```
    post:
```

```
      summary: Create application
      security: [{ bearerAuth: [] }]
      responses:
        "200": { description: Created }
```

```
    get:
```

```
      summary: List applications
      security: [{ bearerAuth: [] }]
      responses:
        "200": { description: List }
```

```
  /{id}:
```

```
    get:
```

```
      summary: Get application
      security: [{ bearerAuth: [] }]
      parameters:
        - in: path
```

```
      name: id
      schema: { type: integer }
    responses:
      "200": { description: Retrieved }

/{id}/submit:
  post:
    summary: Submit application
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Submitted }

/{id}/documents:
  post:
    summary: Upload documents
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Uploaded }
```

7.4 WORKFLOW SERVICE

```
openapi: 3.0.3
info:
  title: Workflow Service
  version: 1.0.0

servers:
  - url: https://api.fcau.gov.lk/workflow

paths:

  /assign-phi:
    post:
```



```
summary: Assign PHI
security: [{ bearerAuth: [] }]
responses:
  "200": { description: Assigned }
```

/approve:

```
post:
  summary: Approve application
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Approved }
```

/reject:

```
post:
  summary: Reject application
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Rejected }
```

/application_id/history:

```
get:
  summary: Workflow history
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: History }
```

7.5 INSPECTION SERVICE

openapi: 3.0.3

info:

```
title: Inspection & Grading Service
version: 1.0.0
```

servers:

- url: <https://api.fcau.gov.lk/inspections>

paths:

/:

post:

summary: Create inspection

security: [{ bearerAuth: [] }]

responses:

"200": { description: Created }

/ {id} /submit:

post:

summary: Submit H800 grading

security: [{ bearerAuth: [] }]

responses:

"200": { description: Submitted }

/ {id} :

get:

summary: Get inspection

security: [{ bearerAuth: [] }]

responses:

"200": { description: Inspection data }

/ {id} /evidence:

post:

summary: Upload evidence

security: [{ bearerAuth: [] }]

responses:

"200": { description: Uploaded }

7.6 RISK ENGINE SERVICE

```
openapi: 3.0.3

info:
  title: Risk Analysis Service
  version: 1.0.0

servers:
  - url: https://api.fcau.gov.lk/risk

paths:

  /calculate:
    post:
      summary: Calculate risk
      security: [{ bearerAuth: [] }]
      responses:
        "200": { description: Calculated }

  /{premises_id}:
    get:
      summary: Get risk profile
      security: [{ bearerAuth: [] }]
      responses:
        "200": { description: Risk profile }
```

7.7 SCHEDULING SERVICE

```
openapi: 3.0.3

info:
  title: Scheduling Service
  version: 1.0.0

servers:
```

- url: <https://api.fcau.gov.lk/schedule>

paths:

/calendar:

get:

summary: PHI calendar

security: [{ bearerAuth: [] }]

responses:

"200": { description: Calendar }

/trigger:

post:

summary: Create trigger

security: [{ bearerAuth: [] }]

responses:

"200": { description: Created }

7.8 LAB SERVICE

openapi: 3.0.3

info:

title: Laboratory Service

version: 1.0.0

servers:

- url: <https://api.fcau.gov.lk/lab>

paths:

/results:

post:

summary: Submit lab result (external)

```
    responses:
      "200": { description: Accepted }

/upload-report:
  post:
    summary: Upload PDF report
    responses:
      "200": { description: Uploaded }

/samples/{id}:
  get:
    summary: Get sample status
    responses:
      "200": { description: Sample }
```

7.9 APPEALS SERVICE

```
openapi: 3.0.3
info:
  title: Appeals & Legal Service
  version: 1.0.0

servers:
  - url: https://api.fcau.gov.lk/appeals

paths:
  /:
    post:
      summary: Submit appeal
      security: [{ bearerAuth: [] }]
      responses:
        "200": { description: Submitted }
```

```
/{id}/board-decision:
  post:
    summary: Board decision
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Recorded }
```

```
/{id}/rd-decision:
  post:
    summary: RD decision
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Finalized }
```

7.10 DOCUMENT SERVICE

```
openapi: 3.0.3
info:
  title: Document Management Service
  version: 1.0.0

servers:
  - url: https://api.fcau.gov.lk/documents
```

```
paths:

  /upload:
    post:
      summary: Upload file
      security: [{ bearerAuth: [] }]
      requestBody:
        content:
          multipart/form-data:
            schema:
```

```
      type: object
      properties:
        file:
          type: string
          format: binary
      responses:
        "200": { description: Uploaded }
```

7.11 NOTIFICATION SERVICE

openapi: 3.0.3

info:

title: Notification Service

version: 1.0.0

servers:

- url: https://api.fcau.gov.lk/notifications

paths:

/send:

post:

summary: Send notification

responses:

"200": { description: Sent }

/me:

get:

summary: Get notifications

security: [{ bearerAuth: [] }]

responses:

"200": { description: List }

7.12 CONFIG SERVICE

```
openapi: 3.0.3

info:
  title: Configuration Service
  version: 1.0.0

servers:
  - url: https://api.fcau.gov.lk/config

paths:

  /grading:
    get:
      summary: Get grading rules
      responses:
        "200": { description: Rules }

    put:
      summary: Update grading rules
      security: [{ bearerAuth: [] }]
      responses:
        "200": { description: Updated }
```

7.13 REPORTING SERVICE

```
openapi: 3.0.3

info:
  title: Reporting Service
  version: 1.0.0

servers:
  - url: https://api.fcau.gov.lk/reports

paths:
```



```
/compliance:
  get:
    summary: Compliance dashboard
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Data }
```

```
/export:
  get:
    summary: Export reports
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: File }
```

7.14 IMPORT CONTROL SERVICE

```
openapi: 3.0.3
info:
  title: Import Control Service API
  version: 1.0.0

servers:
  - url: https://api.fcau.gov.lk/imports

components:
  securitySchemes:
    bearerAuth:
      type: http
      scheme: bearer
      bearerFormat: JWT

paths:
```

```
/importers:
  post:
    summary: Register importer
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Importer registered }

/consignments:
  post:
    summary: Create pre-import entry
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Consignment created }

  get:
    summary: List consignments
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: List }

/consignments/{id}:
  get:
    summary: Get consignment details
    security: [{ bearerAuth: [] }]
    parameters:
      - in: path
        name: id
        schema: { type: integer }
    responses:
      "200": { description: Details }

/consignments/{id}/verify-documents:
  post:
```

```
summary: Verify documents
security: [{ bearerAuth: [] }]
responses:
  "200": { description: Verified }
```

/inspections:

```
post:
  summary: Record border inspection
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Inspection recorded }
```

/consignments/{id}/decision:

```
post:
  summary: Final decision (Release / Sample / Destroy)
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Decision recorded }
```

/consignments/{id}/lab-sample:

```
post:
  summary: Send sample to lab
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Sample sent }
```

7.15 EXPORT CONTROL SERVICE

openapi: 3.0.3

info:

```
title: Export Control Service API
version: 1.0.0
```

servers:

- url: <https://api.fcau.gov.lk/exports>

components:

securitySchemes:

bearerAuth:

type: http

scheme: bearer

bearerFormat: JWT

paths:

/exporters:

post:

summary: Register exporter

security: [{ bearerAuth: [] }]

responses:

"200": { description: Exporter registered }

/applications:

post:

summary: Apply for export certificate

security: [{ bearerAuth: [] }]

responses:

"200": { description: Application created }

get:

summary: List applications

security: [{ bearerAuth: [] }]

responses:

"200": { description: List }

/applications/{id}:

get:

```
summary: Get application details
security: [{ bearerAuth: [] }]
parameters:
  - name: id
    in: path
    schema: { type: integer }
responses:
  "200": { description: Details }
```

/applications/{id}/review:

```
post:
  summary: FCAU review application
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Reviewed }
```

/applications/{id}/lab-sample:

```
post:
  summary: Send sample to lab
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Sample sent }
```

/applications/{id}/issue-certificate:

```
post:
  summary: Issue health certificate
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Certificate issued }
```

7.16 SALT REG SERVICE

openapi: 3.0.3

info:

title: Salt Manufacturing Registration Service API

version: 1.0.0

servers:

- url: https://api.fcau.gov.lk/salt

components:

securitySchemes:

bearerAuth:

type: http

scheme: bearer

bearerFormat: JWT

paths:

/applications:

post:

summary: Submit salt manufacturing registration

security: [{ bearerAuth: [] }]

responses:

"200": { description: Application submitted }

get:

summary: List applications

security: [{ bearerAuth: [] }]

responses:

"200": { description: List }

/applications/{id}:

get:

```
summary: Get application details
security: [{ bearerAuth: [] }]
parameters:
  - name: id
    in: path
    schema: { type: integer }
responses:
  "200": { description: Details }
```

/applications/{id}/inspection:

```
post:
  summary: Record inspection and lab testing
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Inspection recorded }
```

/applications/{id}/approve:

```
post:
  summary: Approve and issue certificate
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Approved }
```

/applications/{id}/renew:

```
post:
  summary: Renew registration
  security: [{ bearerAuth: [] }]
  responses:
    "200": { description: Renewed }
```

/applications/{id}/amend:

```
post:
  summary: Amend registration
```

```
security: [{ bearerAuth: [] }]
responses:
  "200": { description: Updated }
```

7.18 BOTTLED WATER REG SERVICE

openapi: 3.0.3

info:

```
title: Bottled Water Registration Service API
version: 1.0.0
```

servers:

```
- url: https://api.fcau.gov.lk/water
```

components:

securitySchemes:

bearerAuth:

type: http

scheme: bearer

bearerFormat: JWT

paths:

/applications:

post:

summary: Submit bottled water registration

security: [{ bearerAuth: [] }]

responses:

```
"200": { description: Application submitted }
```

get:

summary: List applications

security: [{ bearerAuth: [] }]

responses:


```
"200": { description: List }

/applications/{id}:
  get:
    summary: Get application
    security: [{ bearerAuth: [] }]
    parameters:
      - name: id
        in: path
        schema: { type: integer }
    responses:
      "200": { description: Details }

/applications/{id}/brands:
  post:
    summary: Register multiple brands
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Brand added }

/applications/{id}/lab-test:
  post:
    summary: Record lab compliance
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Lab recorded }

/applications/{id}/approve:
  post:
    summary: Approve and issue certificate
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Approved }
```

```
/applications/{id}/renew:
  post:
    summary: Renew registration
    security: [{ bearerAuth: [] }]
    responses:
      "200": { description: Renewed }
```

*Following API designs can be changed with actual implementation where more parameters, functions might add depending on implementation decisions.

8. Tech Stack

8.1 Development

Layer	Technology
Frontend	Blade, Bootstrap
Backend	Laravel (PHP)
API	REST
Database	MySQL
Queue	Laravel Library
Storage	File Server - Blob Storage
Container	Docker
CI/CD	Bitbucket
Notifications	SMTP

8.2 Deployment

Deployment Scope	Application Layer	Database Layer	Storage	Availability Level
Pilot / Limited Institutional Rollout	1 Server 8 vCPU 16GB RAM 500GB NVMe	1 Server 12 vCPU 32GB RAM 1TB NVMe	500GB Backup	Single server per layer
Full National Deployment	2 Servers 8 vCPU each 16GB RAM each 500GB NVMe each	1 Primary DB 16 vCPU 32-64GB RAM 1TB NVMe Optional: Fail over DB server	1-2TB Backup	Load Balanced Layer

9. UI Prototype Samples

Following are few core UI samples where completed front end UI prototype will submit under the Deliverable 03.

9.1 New System User Creation

System Identity Creation

[Home](#) • [User Management](#) • [Add User](#)

Create New Internal User Identity

Full Name *

e.g. Kamal Perera

NIC Number *

e.g. 198XXXXXXXXV

Email Address *

e.g. k.perera@fcau.gov.lk

Mobile Number

e.g. 077 XXX XXXX

Assigned Role *

Select Role... ▾

Administrative Territory / Facility *

Select Territory... ▾

Official Designation / ID

e.g. PHI/COL/O07

☒

An invitation email with a temporary password and MFA setup instructions will be sent to the user upon creation.

Cancel

Create Identity

9.2 FBO Registration

FBO Registration

Home • Registration

1 Identity

2 Business Info

3 Location

4 Documents

5 Review

Applicant Identity Verification

Full Name *

Enter full name as per NIC

NIC / Passport Number *

e.g., 200012345678

Email Address *

verified@email.com

Contact Number *

+94 7X XXX XXXX

Business Registration Number

Optional for sole traders

Applicant Type *

Individual

< Previous

Save Draft

Next >

FBO Registration

Home • Registration

1 Identity

2 Business Info

3 Location

4 Documents

5 Review

Business & Premises Information

Trading Name *

Business trading name

Food Category *

Select category...

Floor Area (sq. ft.) *

e.g., 1200

Number of Employees *

e.g., 15

Seating Capacity

If applicable

Water Source *

NWS&DB Pipeline

Certification (if any)

None

< Previous

Save Draft

Next >

FBO Registration

Home • Registration

1 Identity

2 Business Info

3 Location

4 Documents

5 Review

Location & GN Division Mapping

Province *

Western

District *

Colombo

MOH Area *

Colombo MC

GN Division *

Select GN division...

PHI Area *

Auto-assigned based on GN

Full Address *

Enter complete premises address

Latitude

e.g., 6.9271

Longitude

e.g., 79.8612

GIS Map Integration - Click to pin location

← Previous

Save Draft

Next →

FBO Registration

Home • Registration

1 Identity

2 Business Info

3 Location

4 Documents

5 Review

Statutory Document Upload

Upload scanned copies (PDF/JPG, max 5MB each). Prohibited file types: .exe, .zip, .rar

Trade License *

Choose File

No file chosen

Business Registration Certificate

Choose File

No file chosen

NIC Copy (Owner) *

Choose File

No file chosen

Medical Certificates (Schedule VI) *

Choose Files

No file chosen

Floor Plan / Layout

Choose File

No file chosen

Certification Documents (HACCP/ISO)

Choose File

No file chosen

← Previous

Save Draft

Next →





FBO Registration

Home • Registration

Identity — Business Info — Location — Documents — Review

Application Review & Submit

⚠ Please review all details before submission. Once submitted, the application enters the FCAU workflow for review.

 **Applicant Details**

Name: [From Step 1]

NIC: [From Step 1]

Email: [From Step 1]

Contact: [From Step 1]

 **Business Details**

Trading Name: [From Step 2]

Category: [From Step 2]

Location: [From Step 3]

GN Division: [From Step 3]

 **Uploaded Documents**

Trade License.pdf
NIC_Copy.jpg
Medical_Cert.pdf

☐ I hereby declare that all information provided is true and accurate. I understand that providing false information is a punishable offence under the Food Act No. 26 of 1980.

← Previous
Save Draft
Submit Application

9.3 Premises List

[illegible]

9.4 Grading Form Configuration




H-800 Form Configuration

Home • Inspection Config

Manage Scoring Cages & Weighting

[+ Add New Cage](#)

Configure the 10 compliance cages and their respective line items for the H-800 digital form.

#	Cage Name	Description	Max Score	Line Items	Action
1	General Environmental Conditions	Surrounding hygiene and conditior	10	5 Items	Edit Items
2	Building Structure & Design	Structural soundness, drainage	10	8 Items	Edit Items
Total Maximum Score:			100		

Grading Thresholds

Grade A (Excellent)

75%

Grade B (Good)

50%

Grade C (Fair)

25%

Grade D (Poor)

0%

[Save Configuration](#)
[Reset to Default](#)

9.5 Inspection Calendar




Inspection Calendar

Home • Calendar

Overdue
Risk-Based
Follow-Up
Completed
Complaint

All PHI Areas
+ New Inspection

<
May 2026
>

Mon	Tue	Wed	Thu	Fri	Sat	Sun
28	29	30	1 Fresh Mart - Risk	2 City Bakery - Ov...	3	4
5 Golden Rice - RL... Harbor Fish - Fol...	6 Lanka Spice - D...	7 Complaint #1842	8 Star Hotel - Risk	9	10	11
12 Rasa Bojun - Risk	13	14 City Bakery - 2nd	15 ABC Foods - Risk	16	17	18
19	20 Fish Hub - Over...	21	22 Green Leaf - Risk	23	24	25
26 Fresh Mart - Fol...	27	28	29	30 2 inspections	31	1

Today's Itinerary (May 5, 2026)

Time	Establishment	Type	Priority	Action
09:00 AM	Golden Rice Restaurant	Risk-Based	Medium	Start Inspection
11:30 AM	Harbor Fish Market	Follow-Up	Critical	Start Inspection
-	Fresh Mart (Completed)	Review	Normal	Audit Result

9.6 Inspection

H-800 Compliance Assessment

Home • Premises • Inspection Form

Establishment

Visit Type

Category

Inspection Date

Inspector (PHI)

Golden Rice Restaurant (FE)

1st Visit

A2 - Restaurant

05/05/2026

K. Perera - PHI/COL/007

1. General Environmental Conditions

Overall premises hygiene and surrounding conditions

0 / 10

2. Building Structure & Design

Structural soundness, drainage efficacy, pest exclusion

0 / 10

3. Ventilation & Lighting

Adequate ventilation, lighting conditions, temperature

0 / 10

4. Equipment & Utensils

Cleanliness, maintenance, food-grade materials

0 / 10

5. Food Handling & Processing

Operational flow, cross-contamination barriers, thermal logs

0 / 10

6. Water Supply Standards

Potable water supply, storage facility sanitation

0 / 10

7. Waste Management & Toilet Facilities

Solid waste collection, closed receptacles, toilet condition

0 / 10

8. Food Storage & Transportation

Cold chain integrity, raw/cooked segregation, FIFO rotation

0 / 10

9. Personnel Hygiene & Health

Medical examinations (Schedule VI), personal hygiene, protective clothing

0 / 10

10. Training & Certification

GHP/GMP training documentation, food safety awareness

0 / 10

Violations & Advisory Notes

Record violations and non-compliance findings...

Photo Evidence

Choose Files No file chosen

Upload timestamped photographic evidence of violations

Score Summary

0%

Grade: -

Total Score: 0 / 100

Grade A: 75-100%

Grade B: 50-74%

Grade C: 25-49%

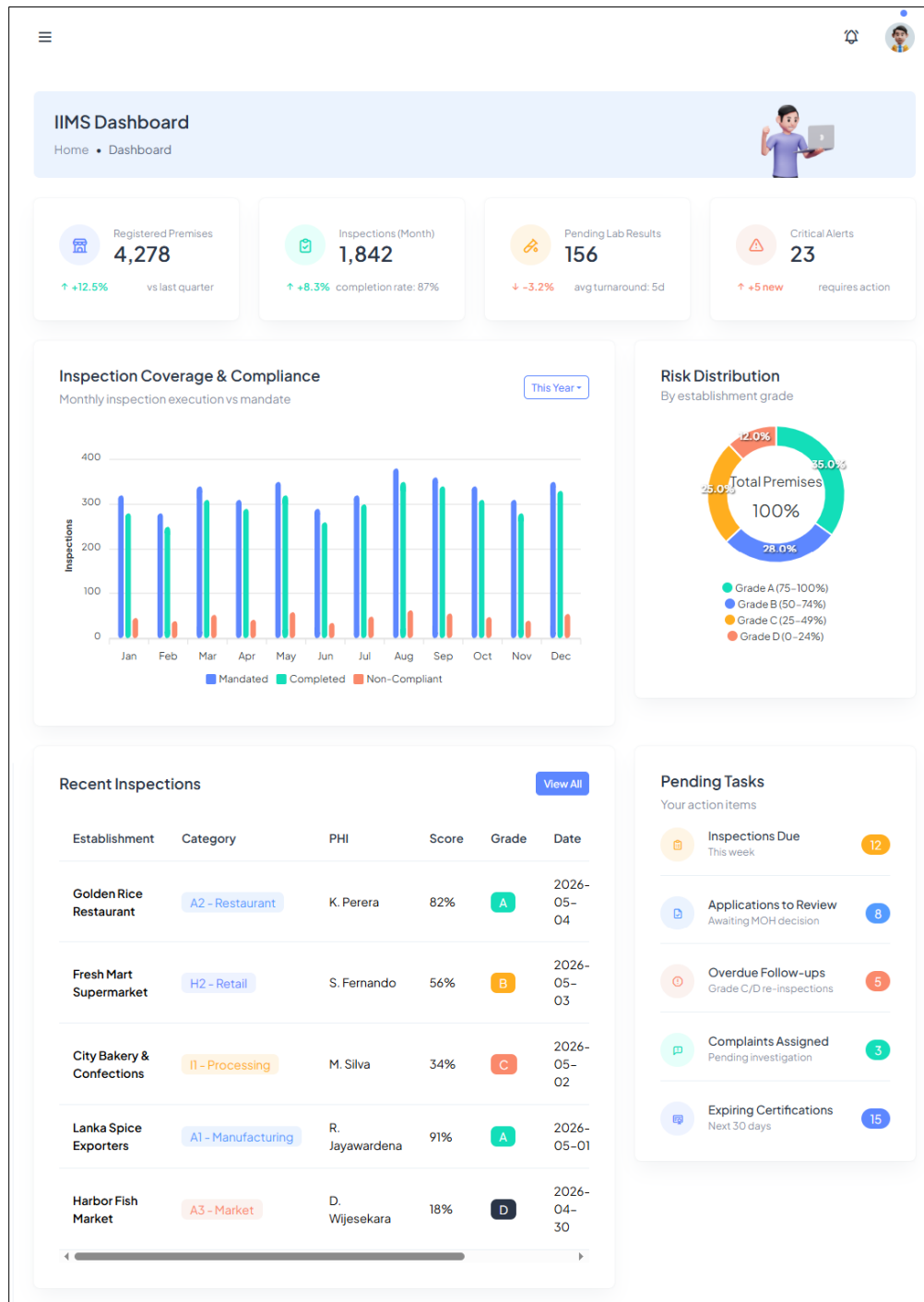
Grade D: 0-24%

Next Inspection: -

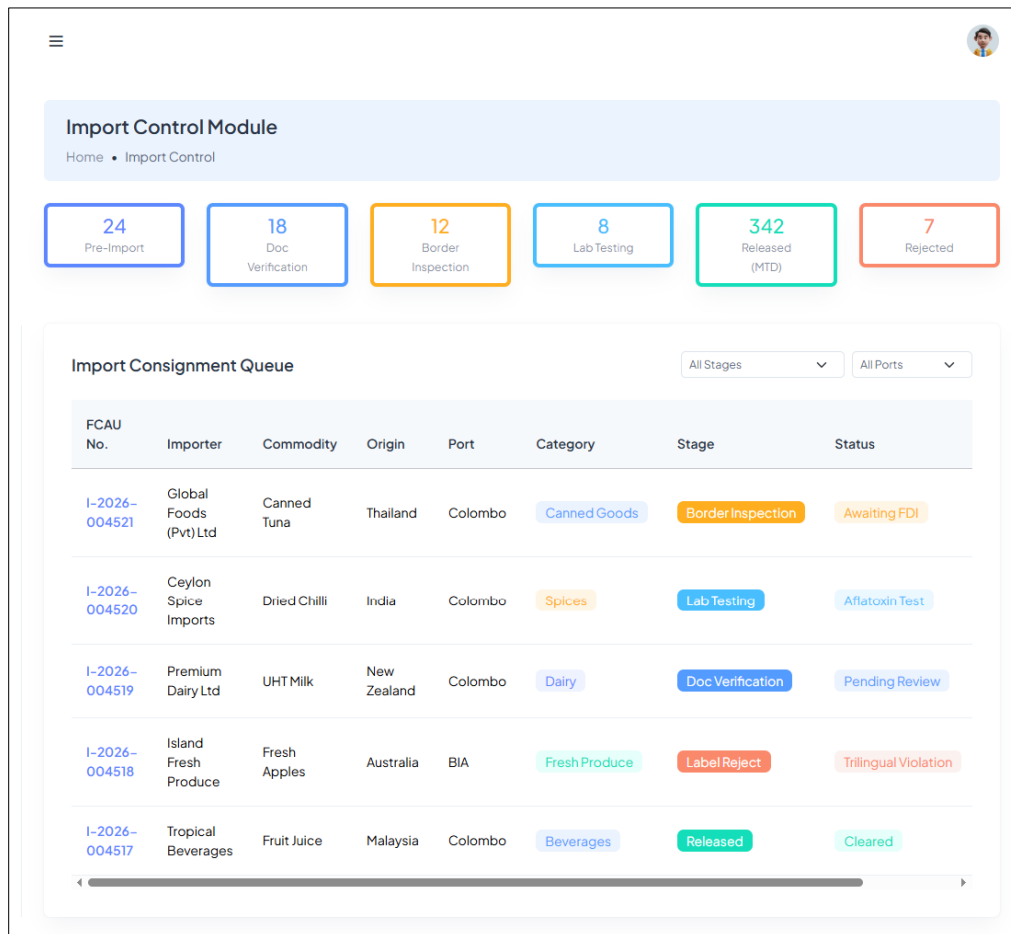
Submit for Review

Save Draft

9.7 FCAU User Dashboard



9.8 Import Control



9.9 Sample Management




Sample Custody Register

Home • Sample Management

156
Pending Lab Results

42
In Transit

1,284
Compliant (YTD)

38
Non-Compliant

Register New Sample

Sample Reference No.

Establishment

Commodity

Batch/Lot No.

2026/PHI/COL/048

Golden Rice Restaurant

e.g., UHT Pasteurized Milk

Manufacturer's batch ID

Sampling Objective

Target Laboratory

Collection Date

Generate QR & Register

Routine Monitoring

Medical Research Institute

05/05/2026

Sample Registry

All Status  Export

Ref No.	Establishment	Commodity	Batch	Objective	Laboratory	Status	Result
2026/PHI/COL/047	Golden Rice	Cooking Oil	LOT-A892	Routine	MRI	Completed	Compliant
2026/PHI/COL/046	Harbor Fish Market	Frozen Prawns	FP-2026-331	Enforcement	ITI	Completed	Salmonella De
2026/PHI/COL/045	Fresh Mart	Packaged Rice	RIC-8844	Surveillance	SGS Lanka	Testing	Awaiting
2026/PHI/COL/044	City Bakery	Wheat Flour	WF-2026-112	Routine	MRI	In Transit	-

10. Error Handling and Logging Strategy

As per the microservices-based architecture of the IIMS comprising multiple services, asynchronous processing, external integrations, and a shared database an effective Error Handling and Logging Strategy is essential to ensure system reliability, traceability, and maintainability. The strategy must provide consistent error management across all services, enable rapid fault diagnosis, and support operational monitoring and audit requirements.

This approach ensures that errors are handled gracefully without disrupting critical regulatory workflows, while maintaining visibility into system behavior for both technical teams and administrative stakeholders.

10.1 Error Handling Strategy

10.1.1 Standardized Error Handling Approach

All services will follow a uniform error handling framework:

- Use consistent HTTP status codes:
 - 200 – Success
 - 400 – Validation errors
 - 401/403 – Authentication/authorization failures
 - 404 – Resource not found
 - 500 – Internal server errors
- Standard error response format:

```
{
  "success": false,
  "message": "Error description",
  "errors": [
    {
      "field": "email",
      "message": "Invalid format"
    }
  ],
  "trace_id": "abc123"
}
```

10.1.2 Error Categorization

Errors should be classified for better handling and monitoring:

a. Validation Errors

- Incorrect input data
- Missing required fields
- Handled at API level with clear messages

b. Business Logic Errors

- Invalid workflow transitions (e.g., approving an already approved application)
- Rule violations (e.g., duplicate registration detected)

c. System Errors

- Database failures
- Service downtime
- External API failures

d. Integration Errors

- Lab API failures
- Payment gateway issues
- GIS service unavailability

10.1.3 Graceful Degradation

- Non-critical failures (e.g., email sending) must not block core workflows
- Fallback mechanisms:
 - Queue retries
 - Temporary caching
- Example:
 - If SMTP fails → log error + retry later

10.1.4 Retry & Recovery Mechanisms

- Implement retry logic for:
 - External API calls
 - Queue jobs
- Use:
 - Exponential backoff
 - Dead-letter queues for failed jobs

10.1.5 Exception Handling in Microservices

- Each service will:
 - Catch and handle exceptions internally
 - Avoid exposing stack traces to users
- Global exception handler (Laravel):
 - Convert exceptions into standardized responses

10.1.6 User-Friendly Error Messages

- Errors displayed to users will be:
 - Clear
 - Non-technical
- Example:
 - Instead of “SQL constraint violation”; show meaningful error messages
“A record with this Business Registration Number already exists”

10.2 Logging Strategy

10.2.1 Centralized Logging

- All services will log to a centralized logging system:
 - Cloud logging

10.2.2 Log Levels

Level	Purpose
DEBUG	Detailed debugging information
INFO	Normal operations (e.g., user login)
WARNING	Non-critical issues
ERROR	Failures affecting functionality
CRITICAL	System-wide failures

10.2.3 Structured Logging Format

Logs must be in structured JSON format:

```
{
  "timestamp": "2026-05-05T10:00:00Z",
  "service": "inspection-service",
  "level": "ERROR",
  "message": "Database connection failed",
  "trace_id": "abc123",
  "user_id": 101
}
```

10.2.4 Correlation & Traceability

- Each request will have a unique trace ID
- Passed across:
 - API Gateway
 - Microservices
 - Queue messages
- Enables end-to-end tracking of requests

10.2.5 Audit Logging

Track all critical actions:

- User logins/logout
- Application submissions
- Inspection approvals
- Complaint handling
- Enforcement actions

Stored in:

- Dedicated audit tables
- Immutable logs

10.2.6 Logging Scope

Application Logs

- API requests/responses
- Errors/exceptions

System Logs

- Server health
- Deployment logs

Security Logs

- Login attempts
- Unauthorized access

Integration Logs

- External API calls
- Failures and responses

11. Design Constraints and Assumptions

11.1 Constraints

11.1.1 Technical Constraints

- System will be developed using Laravel (PHP) as the backend framework
- No frontend framework (e.g., React, Angular); UI will use server-side rendering (Blade)
- Database will be MySQL (relational database)
- Architecture will follow microservices-based design with a shared database
- Services will be containerized using Docker
- Communication between services will use:
 - REST APIs
 - Asynchronous queues (event-driven)
- API Gateway will act as the single entry point to backend services

11.1.2 Architecture Constraints

- Microservices will be aligned with domain modules (e.g., Inspection, Complaints, Reports)
- Despite microservices, database is shared, requiring strict data governance
- Services will be independently deployable but logically interconnected
- System will support high availability (load balancer, failover servers, DB replication)
- Will support event-driven processing for notifications, emails, and background jobs

11.1.3 Integration Constraints

- System will integrate with:
 - Laboratory systems (API + document ingestion)
 - External government systems (e.g., NSW, IRD, Customs, SLSI)
 - Payment gateways (IPG)

- Email systems via SMTP
- All integrations will use secure APIs (HTTPS)
- External systems may have inconsistent formats → require adaptable interfaces

11.1.4 Data Constraints

- Will use a normalized relational schema (3NF) aligned with SRS entities
- Will maintain data integrity and consistency across services
- File uploads will support:
 - Secure storage
 - Malware scanning
- Audit logs will be persistently stored and immutable

11.2 Design Assumptions

11.2.1 Technical Assumptions

- Laravel can effectively support modular microservices implementation
- MySQL is sufficient for transactional and moderate analytical workloads
- Queue systems will be available and reliable
- Docker environment will support consistent deployments

11.2.2 Integration Assumptions

- External systems (labs, government APIs) will:
 - Provide accessible APIs or structured data formats
- GN mapping data will be accurate and maintained
- SMTP services will reliably deliver emails

11.2.3 Data Assumptions

- Data entered by users is generally accurate but requires validation

- Regulatory standards (e.g., H800 grading rules) are stable
- Historical/manual data may need digitization

11.2.4 User Assumptions

- Users have basic digital literacy
- Users will follow defined workflows and procedures

11.2.5 Operational Assumptions

- FCAU will manage:
 - System configurations (risk rules, grading criteria)